

(12) **United States Patent**
Pottorff et al.

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(54) **IMAGE ENHANCEMENT USING ONE OR MORE NEURAL NETWORKS**

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(73) Assignee: **NVIDIA CORPORATION**, Santa Clara, CA (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

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(65) **Prior Publication Data**

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Related U.S. Application Data

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(51) **Int. Cl.**
G06T 3/4046 (2024.01)
G06T 5/20 (2006.01)
(Continued)

(52) **U.S. Cl.**
CPC **G06T 3/4046** (2013.01); **G06T 5/20** (2013.01); **G06T 5/70** (2024.01); **G06T 5/73** (2024.01); **G06T 5/80** (2024.01); **G06T 15/503** (2013.01)

(58) **Field of Classification Search**

CPC G06T 3/4046; G06T 5/002; G06T 5/003; G06T 5/006; G06T 5/20; G06T 15/503; (Continued)

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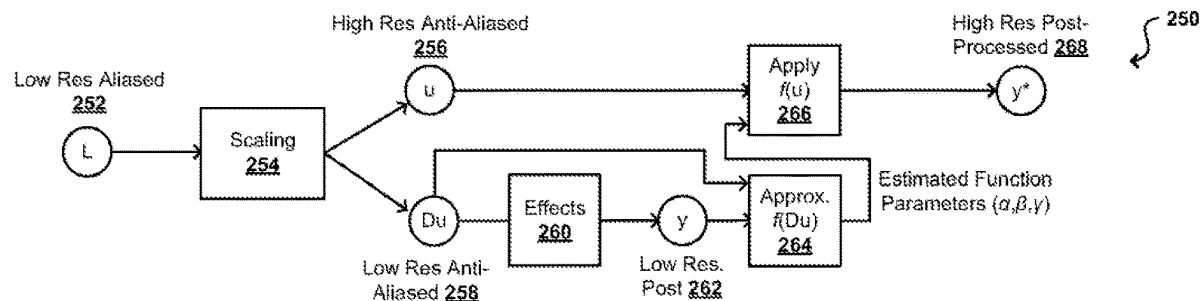
Primary Examiner — Jose L Couso

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(57) **ABSTRACT**

Apparatuses, systems, and techniques are presented to generate images with one or more visual effects applied. In at least one embodiment, one or more visual effects are applied to one or more images having a resolution that is less than a first resolution and those visual effects approximated for one or more images having a resolution that is greater than or equal to the first resolution.

17 Claims, 46 Drawing Sheets



- (51) **Int. Cl.**
G06T 5/70 (2024.01)
G06T 5/73 (2024.01)
G06T 5/80 (2024.01)
G06T 15/50 (2011.01)
- (58) **Field of Classification Search**
CPC G06T 5/00; G06T 1/20; G06T 1/60; G06T 3/40; G06T 2207/20081; G06T 2207/20084; G06T 2207/20172; G06T 3/0453; G06T 3/4092; G06T 5/001; G06T 3/4053; G06T 3/4076; G06T 5/50; G06T 2207/20016; G06T 9/002; G06N 3/0454; G06N 3/084; G06N 3/02; G06N 3/08-088; G06N 3/0445; G06N 7/00; G06N 20/00; G06K 9/6251; G06K 9/6256; G06K 9/6262; G06K 15/1873; G06K 15/1876; G06K 2215/0037; G06K 15/0057; G06K 7/1482; G06V 10/54; G06V 10/454; G06V 10/82; G06V 30/18057; H04N 19/117; H04N 19/59; H04N 19/132; H04N 19/85; H04N 21/234381
See application file for complete search history.
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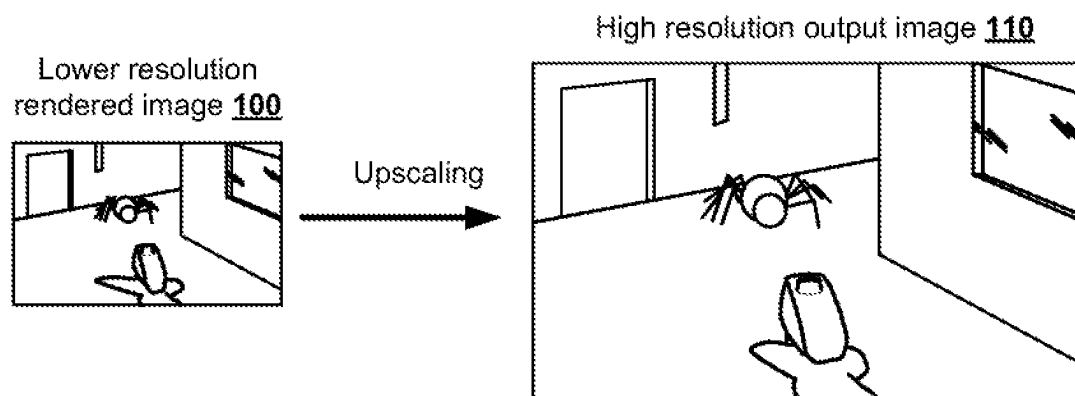


FIG. 1A

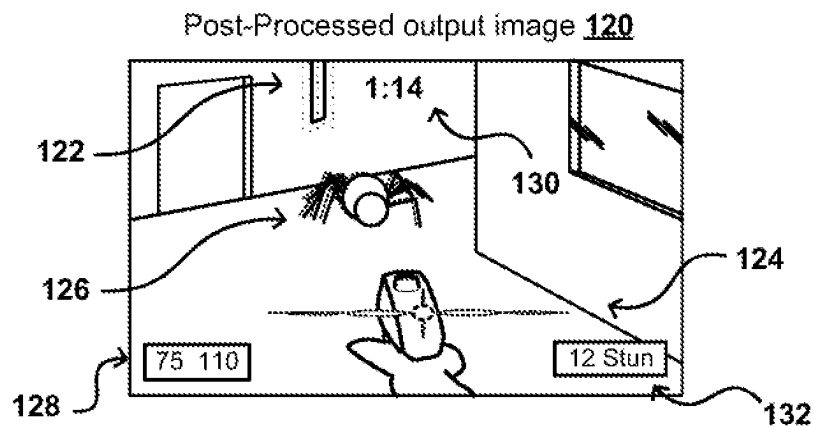


FIG. 1B

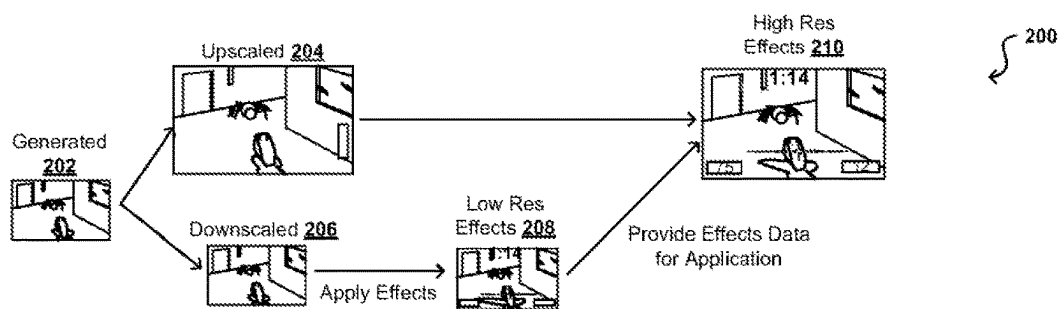


FIG. 2A

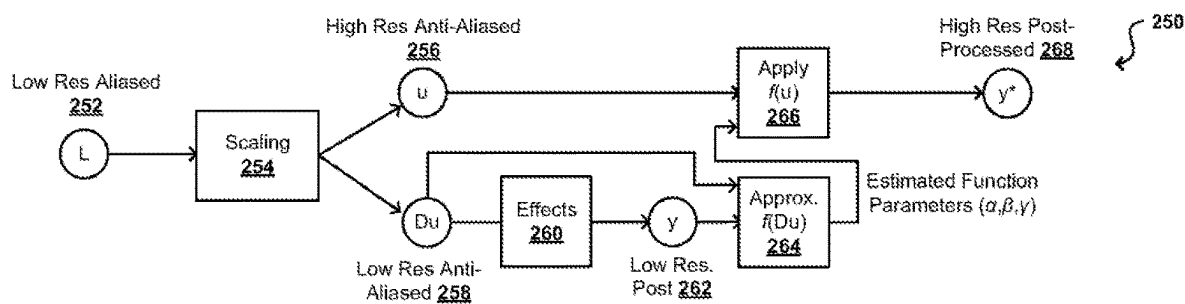


FIG. 2B

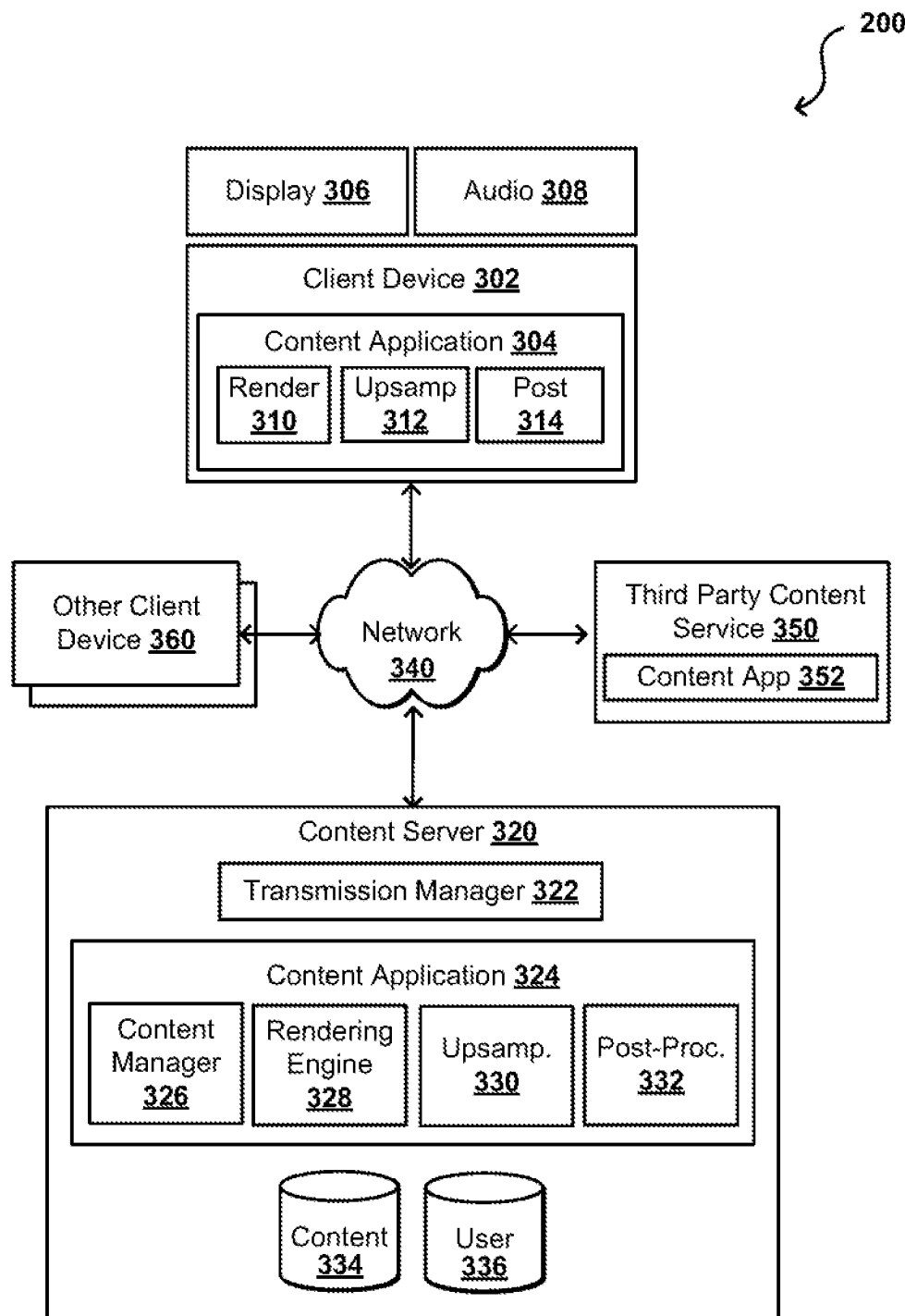


FIG. 3

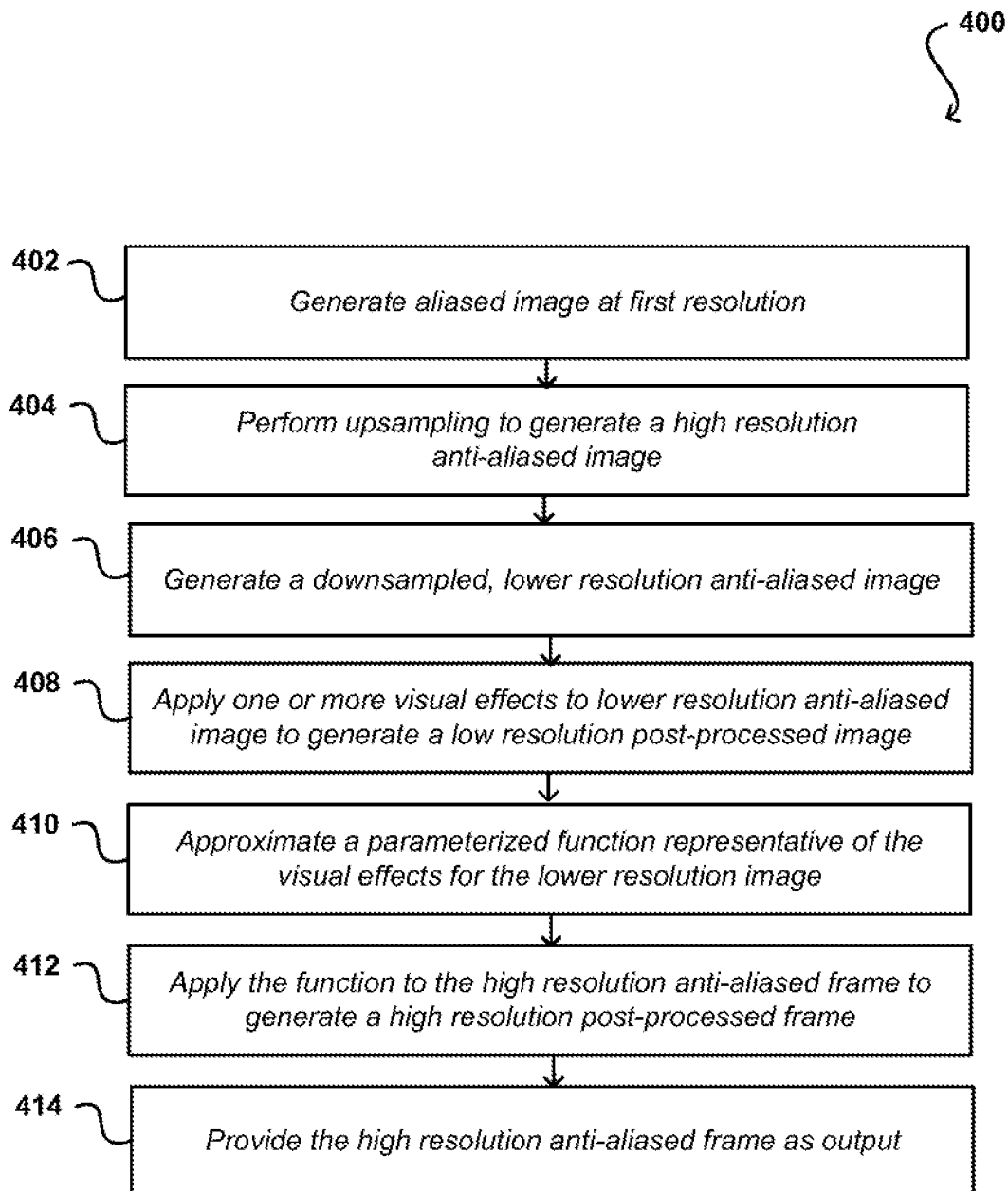


FIG. 4

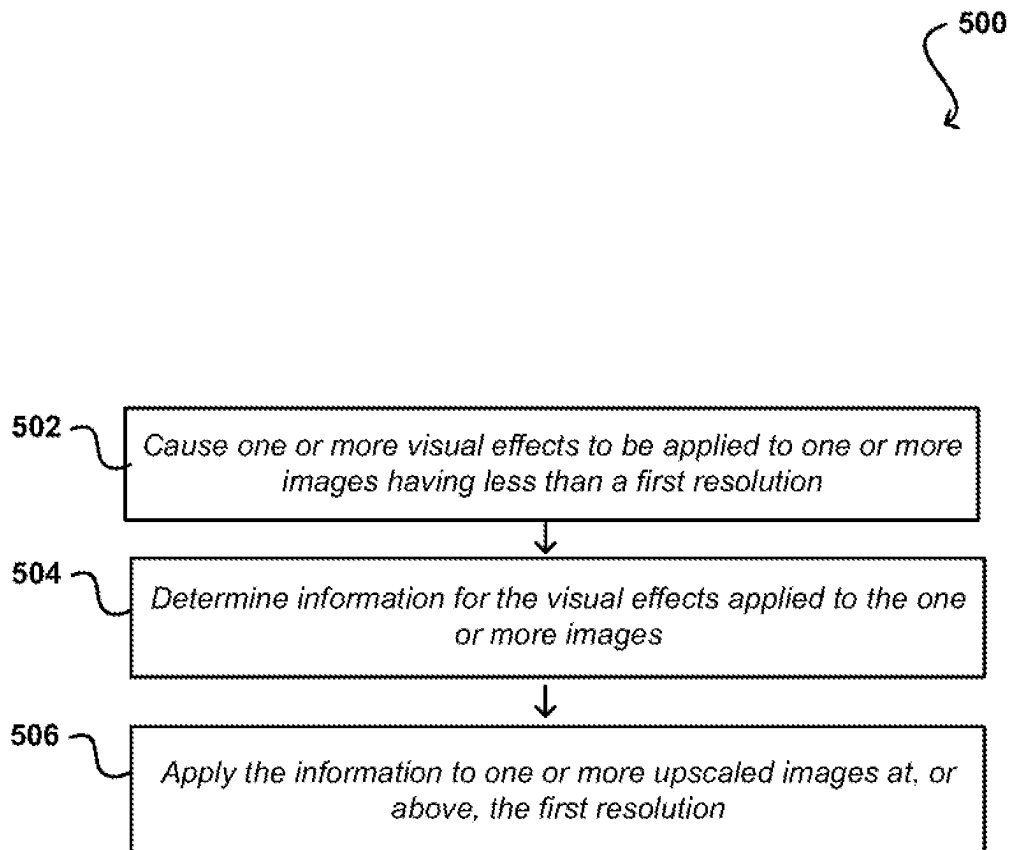


FIG. 5

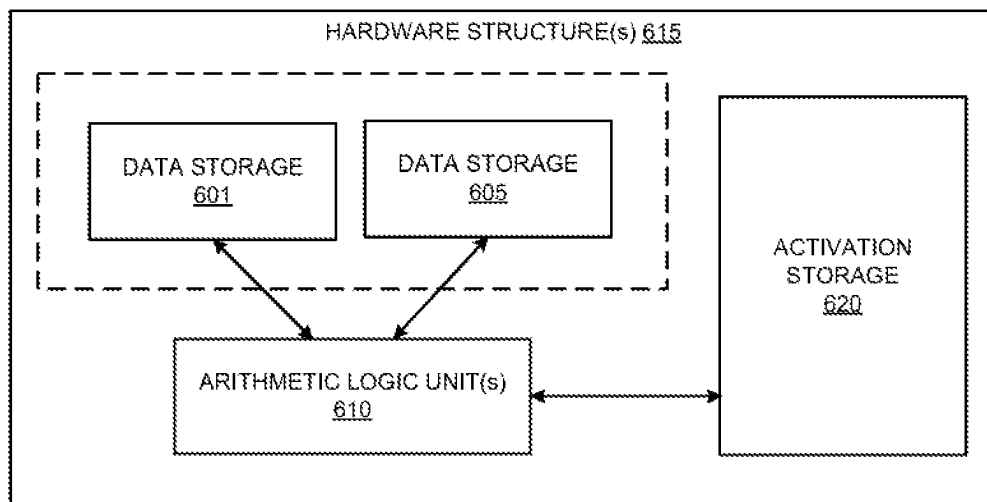


FIG. 6A

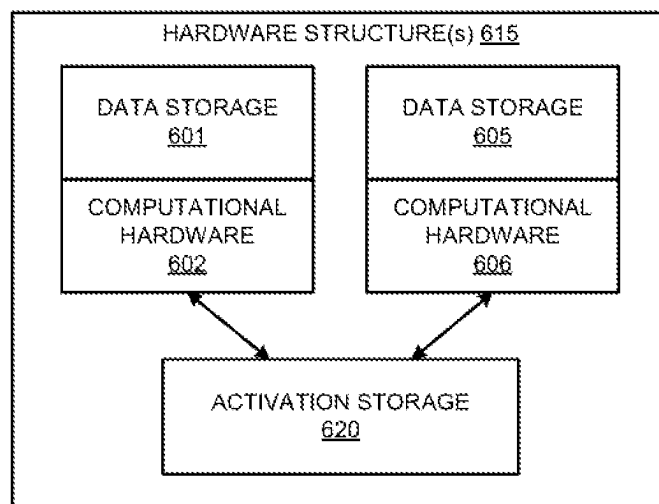


FIG. 6B

DATA CENTER
700

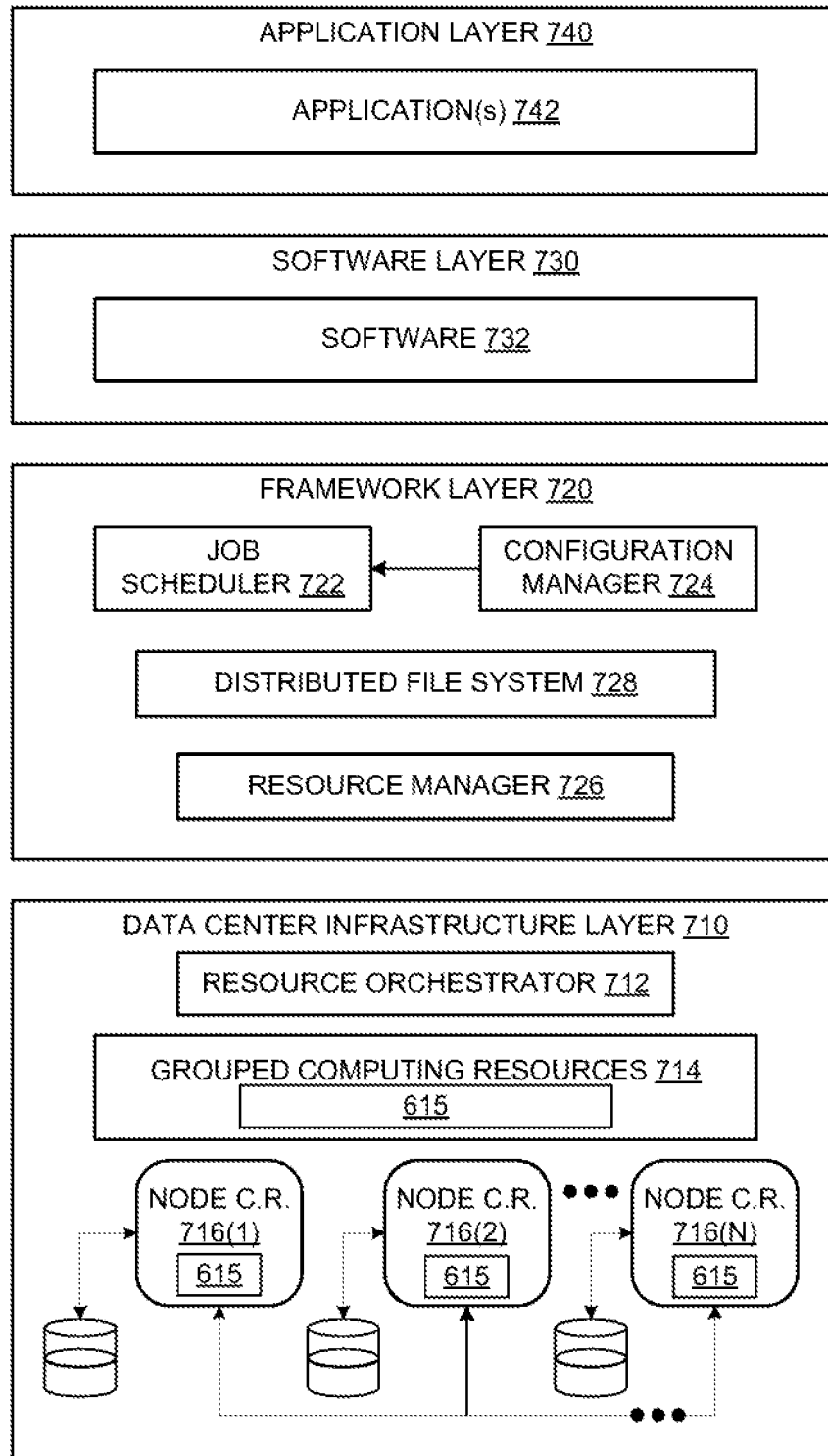


FIG. 7

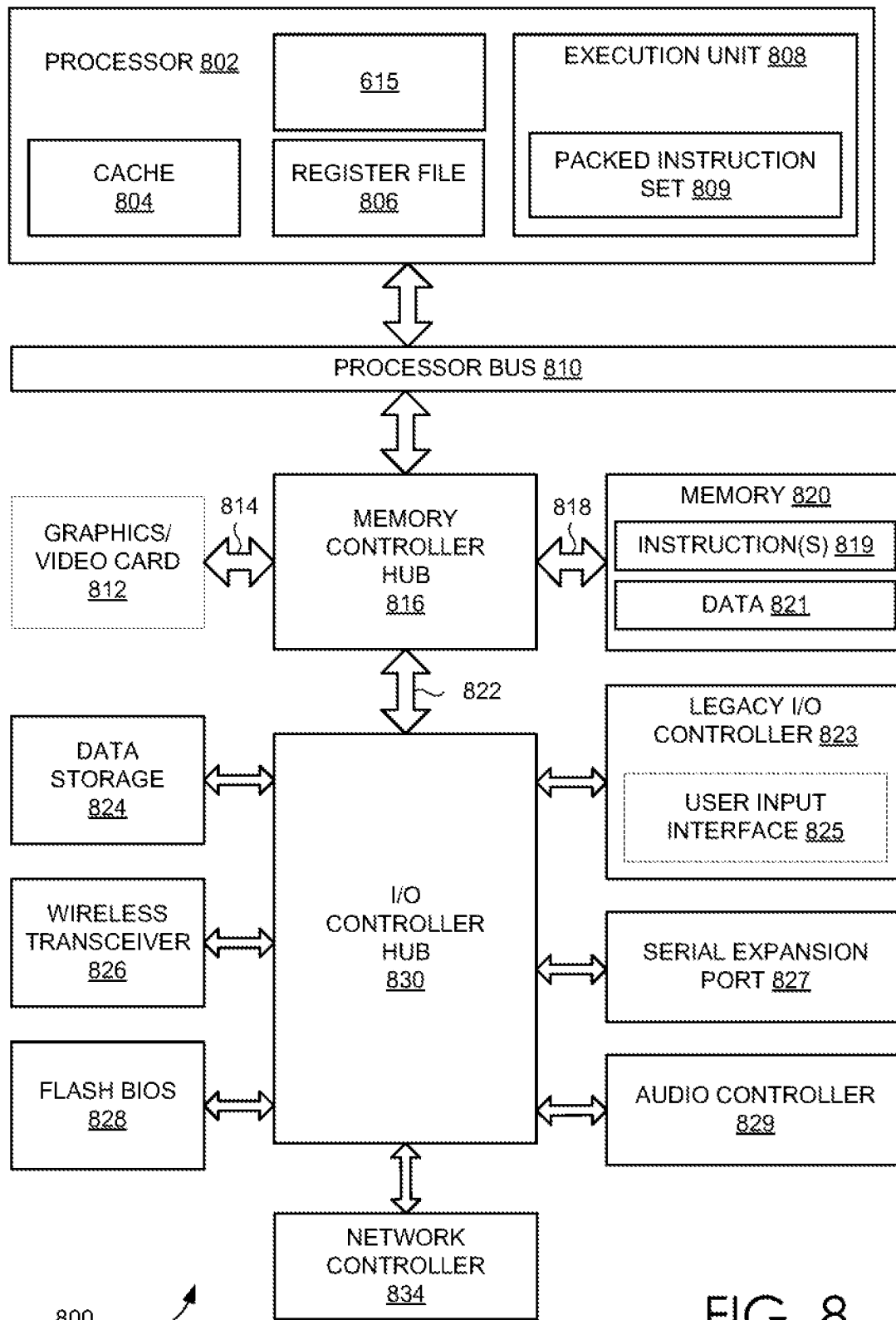
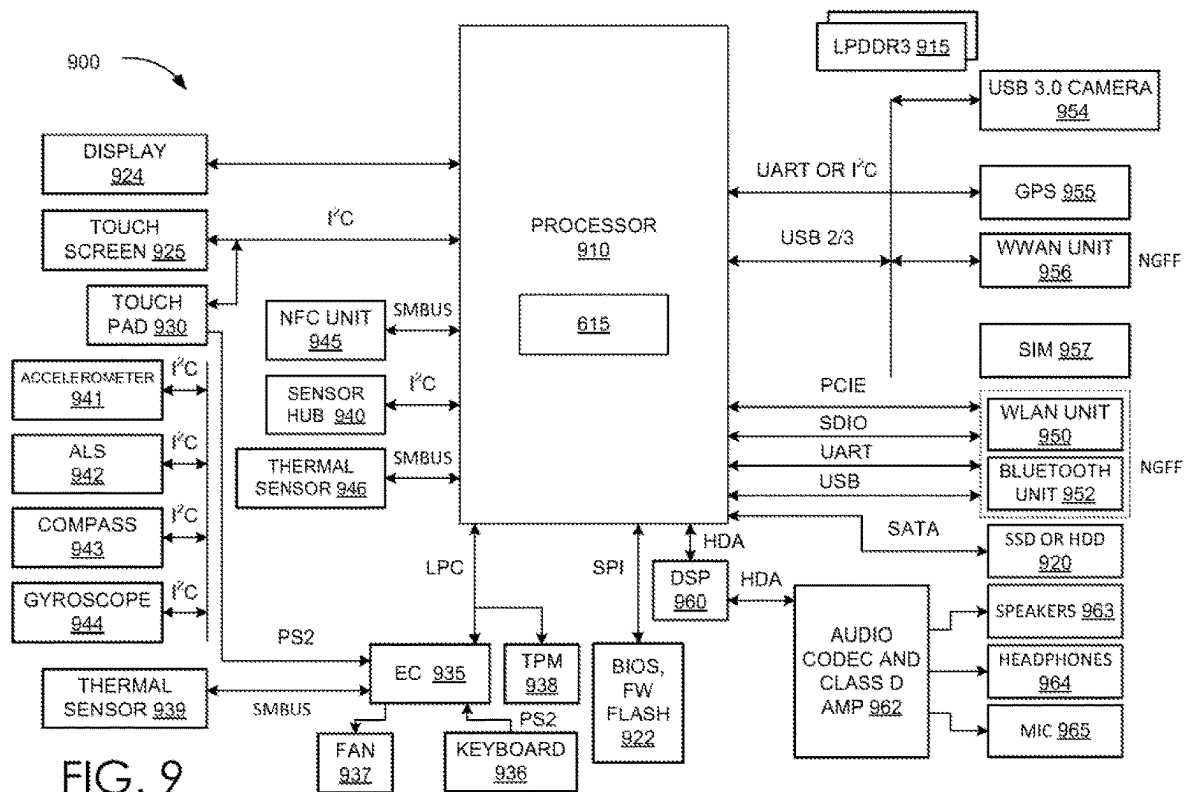


FIG. 8



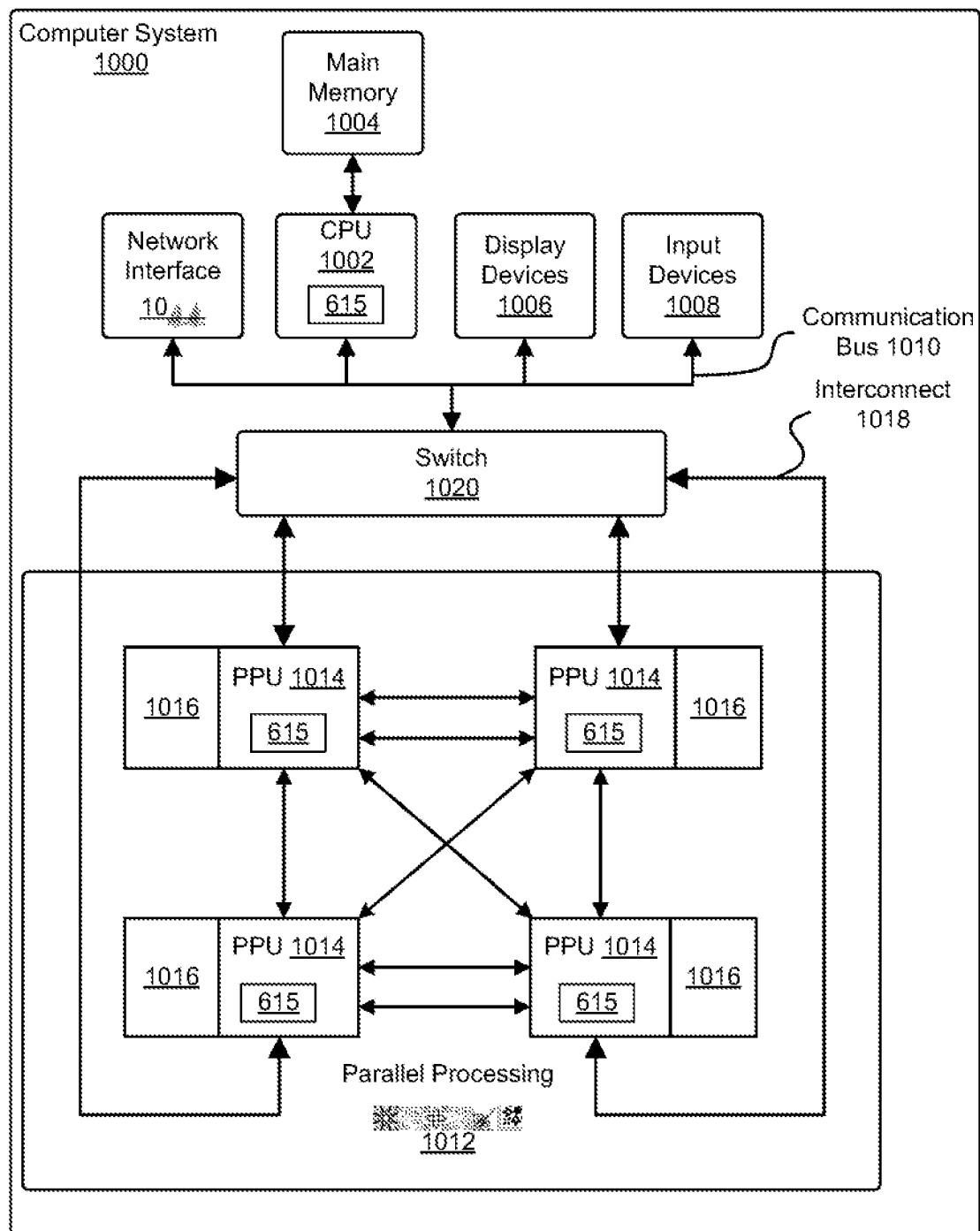


FIG. 10

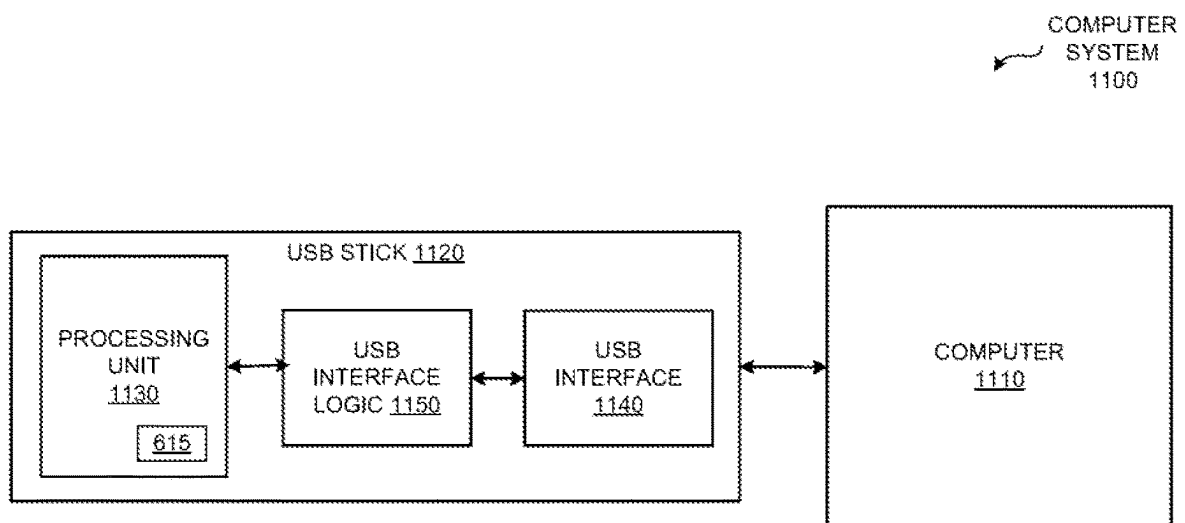


FIG. 11

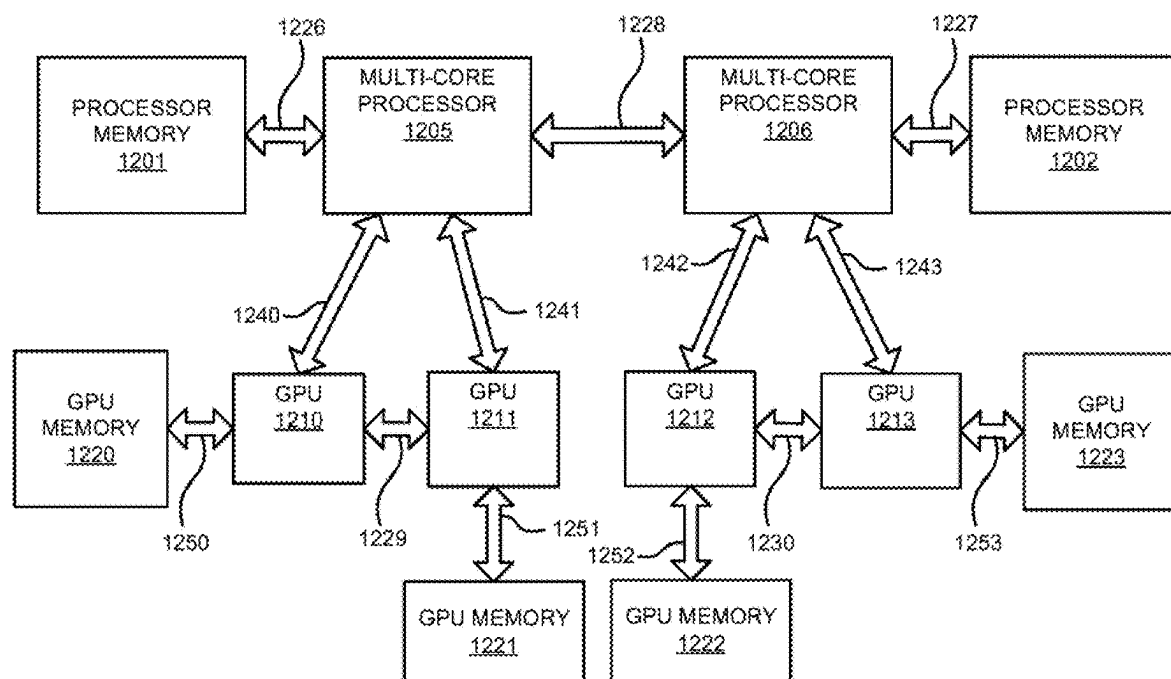


FIG. 12A

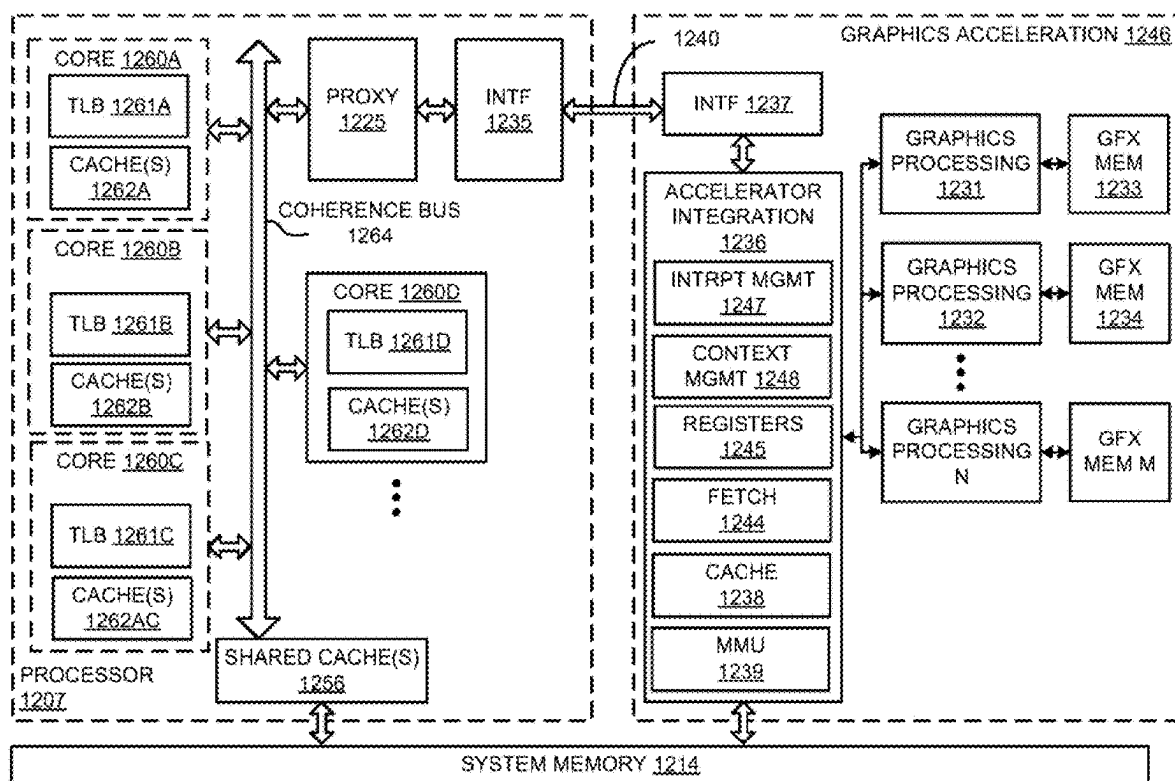


FIG. 12B

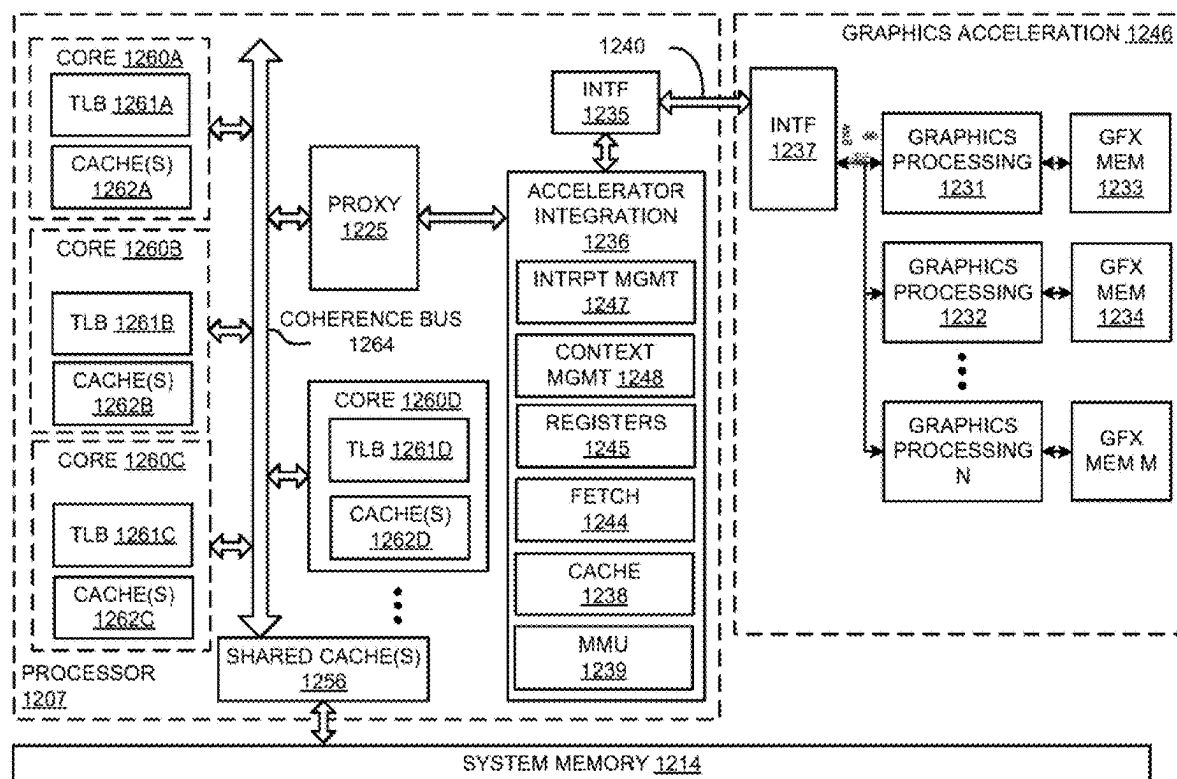
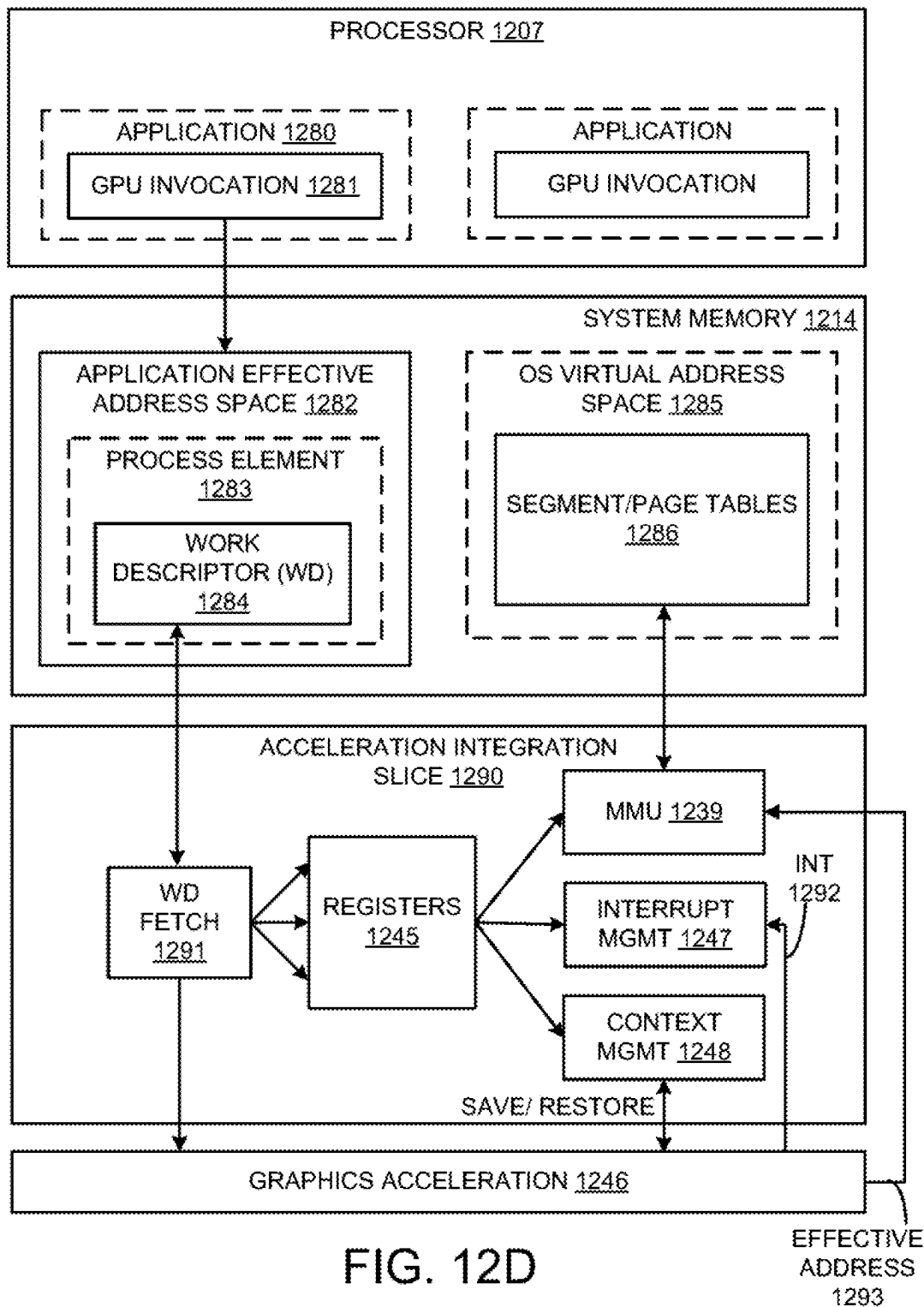
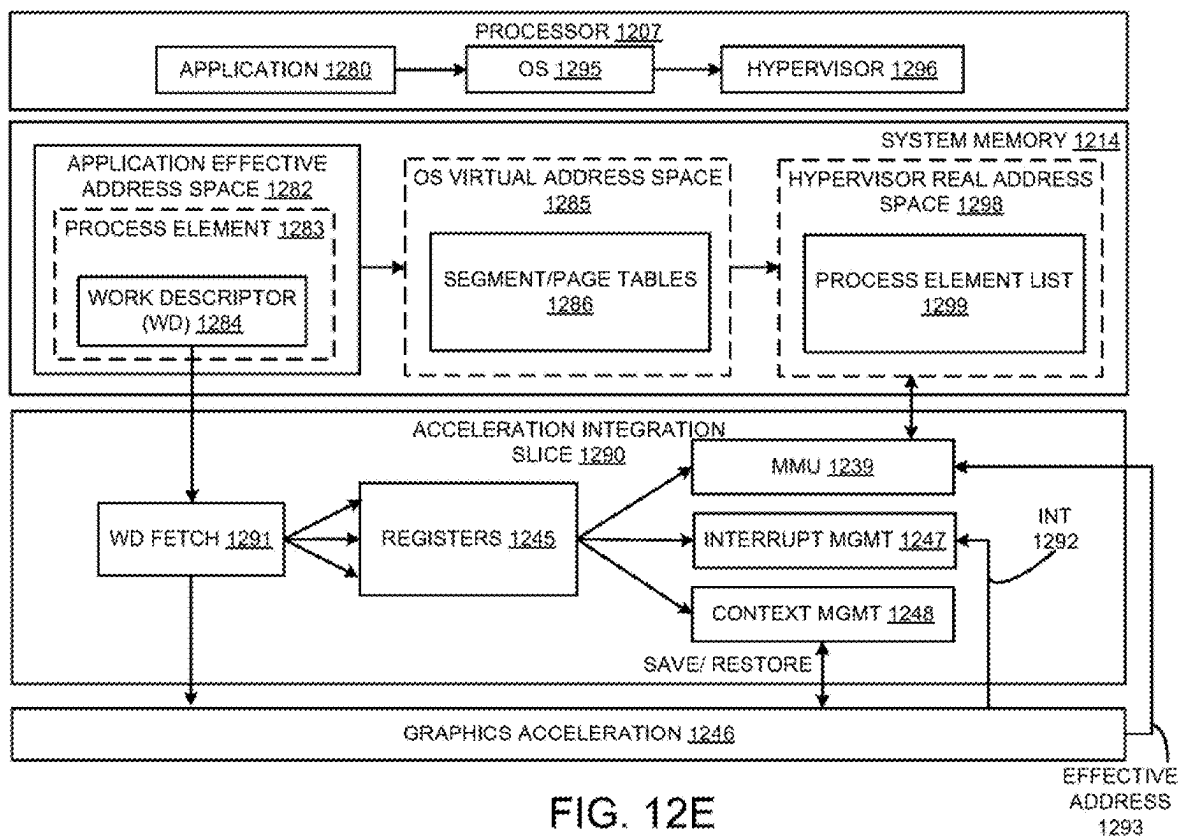


FIG. 12C





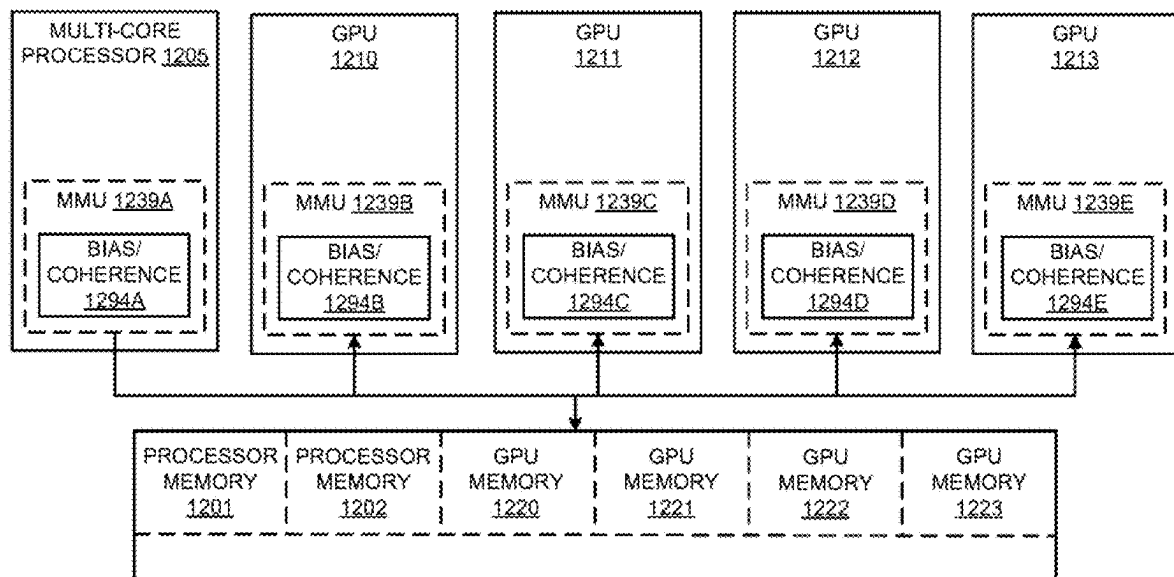


FIG. 12F

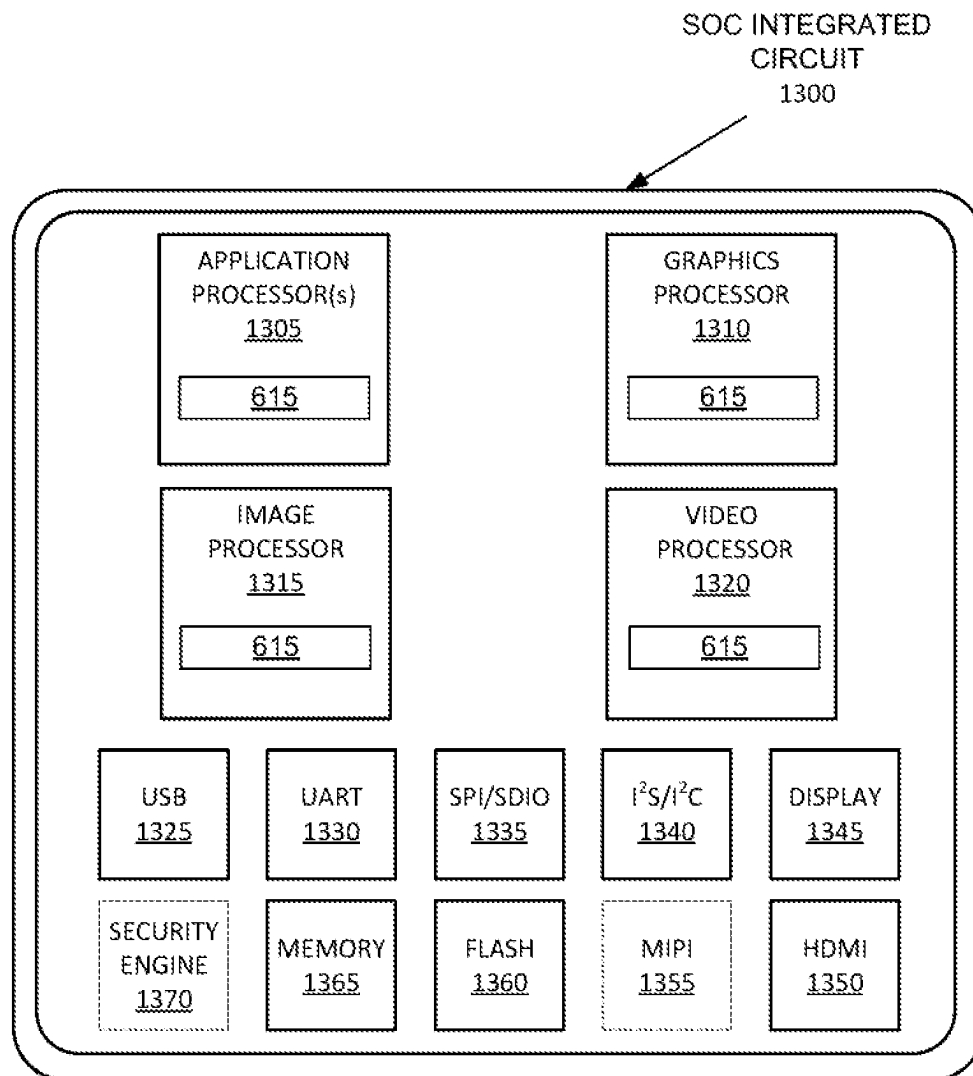


FIG. 13

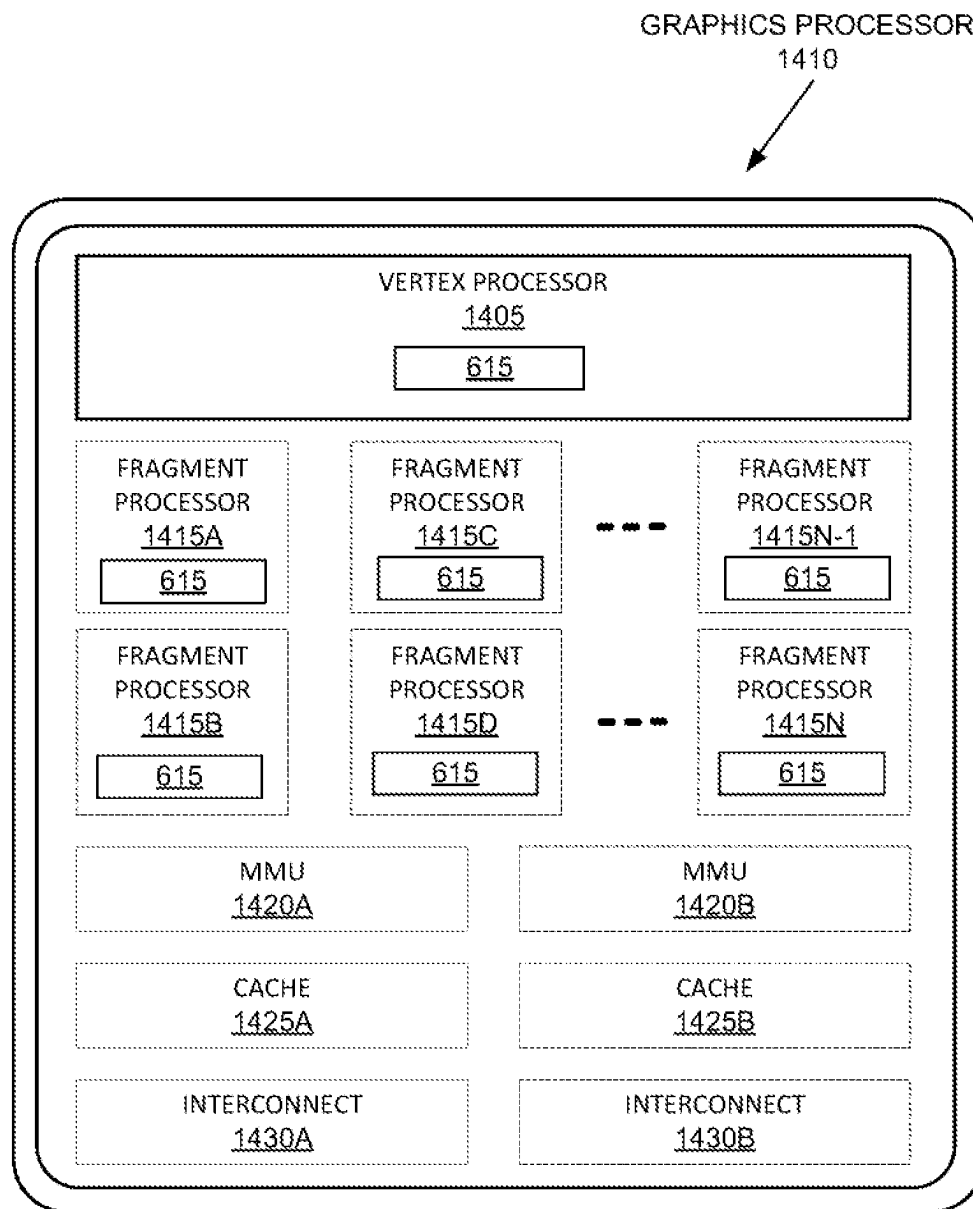


FIG. 14A

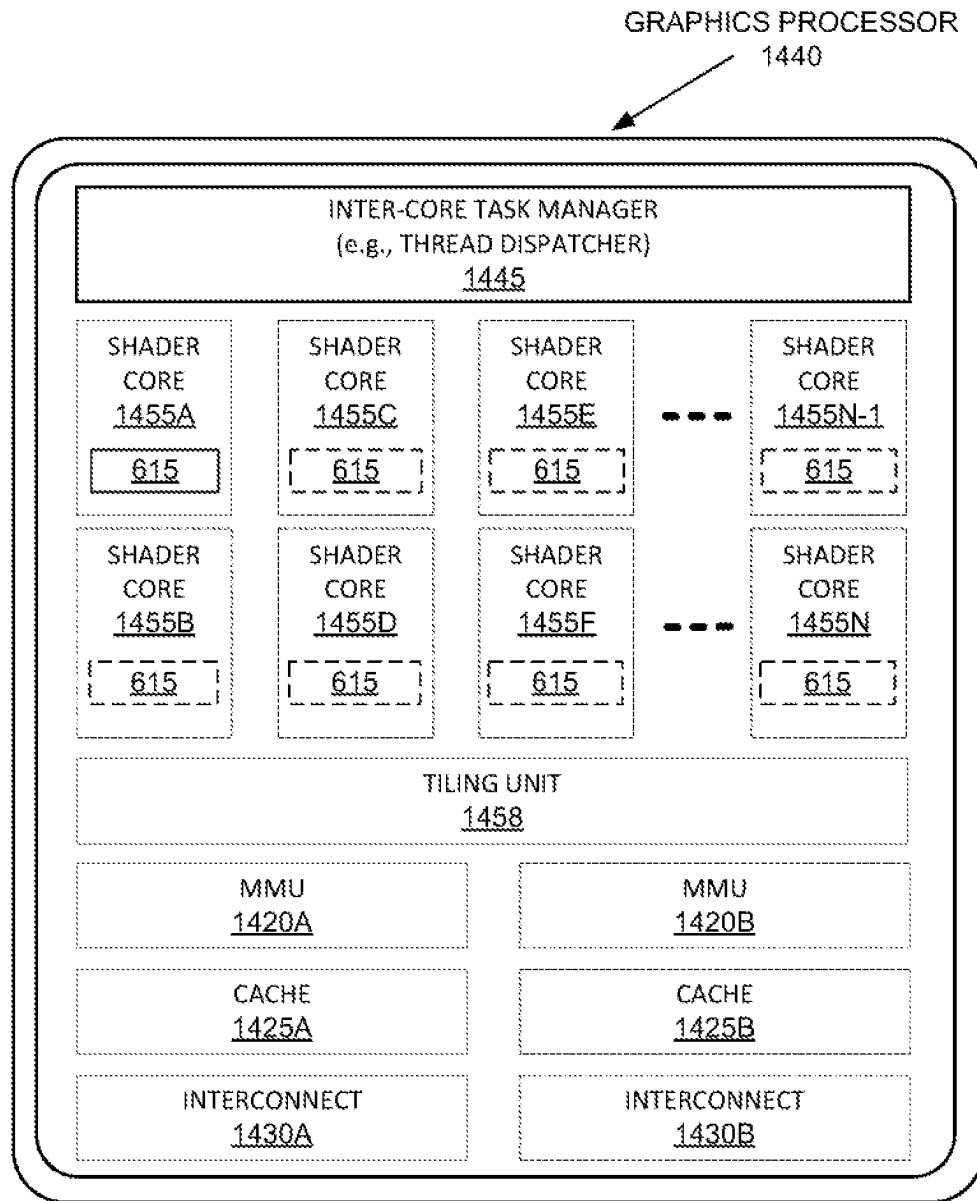


FIG. 14B

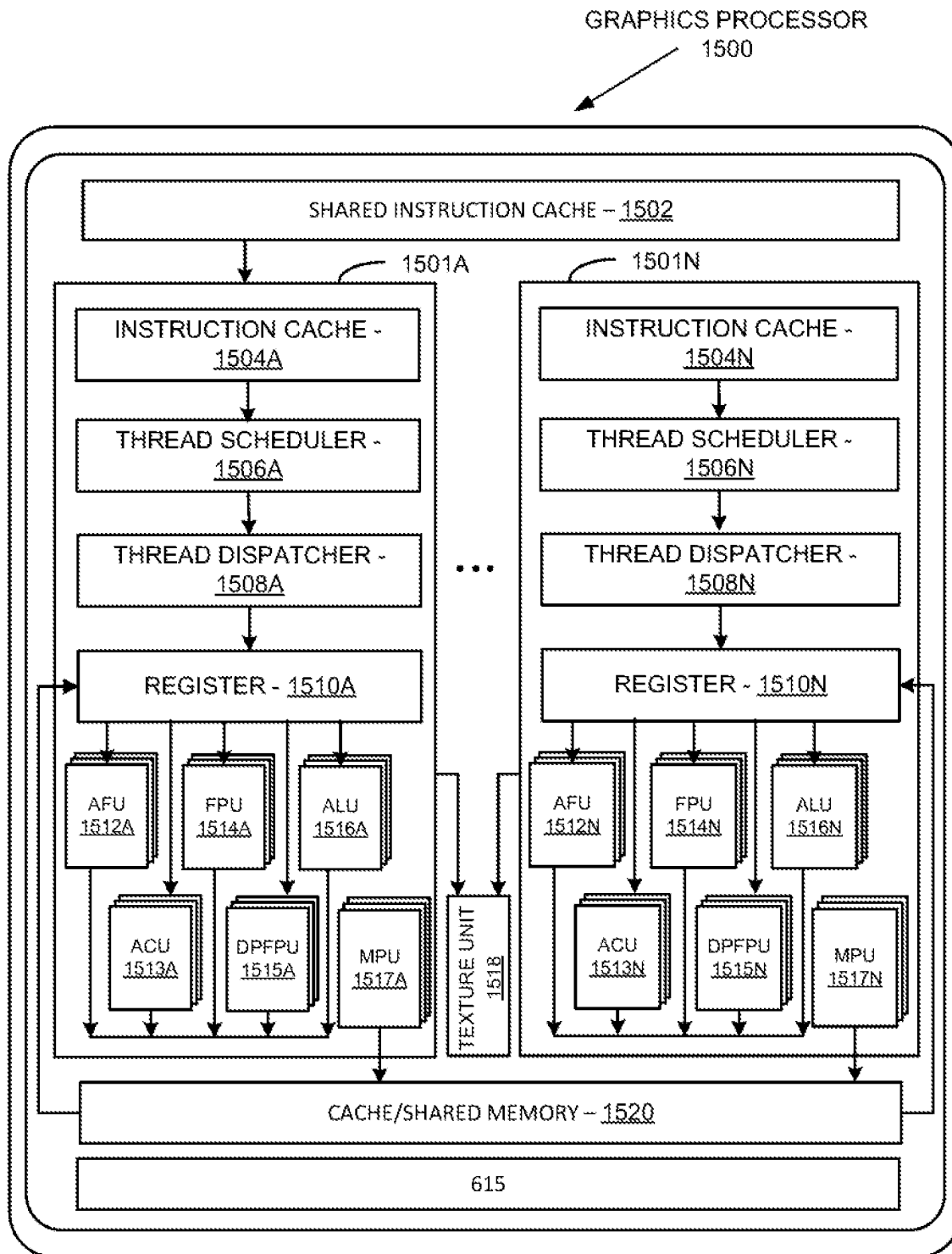


FIG. 15A

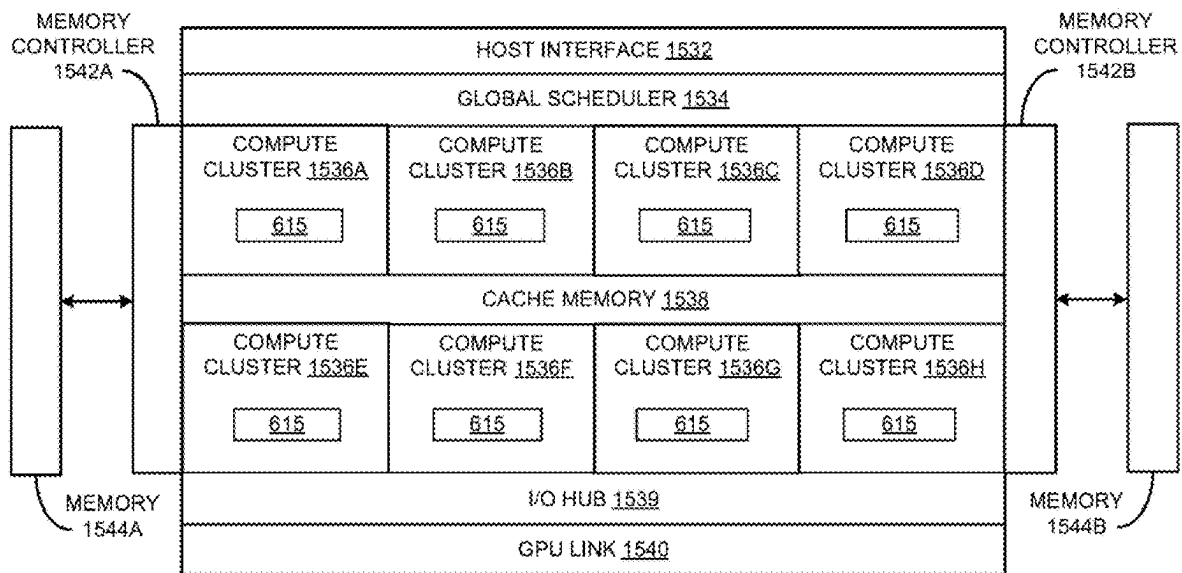


FIG. 15B

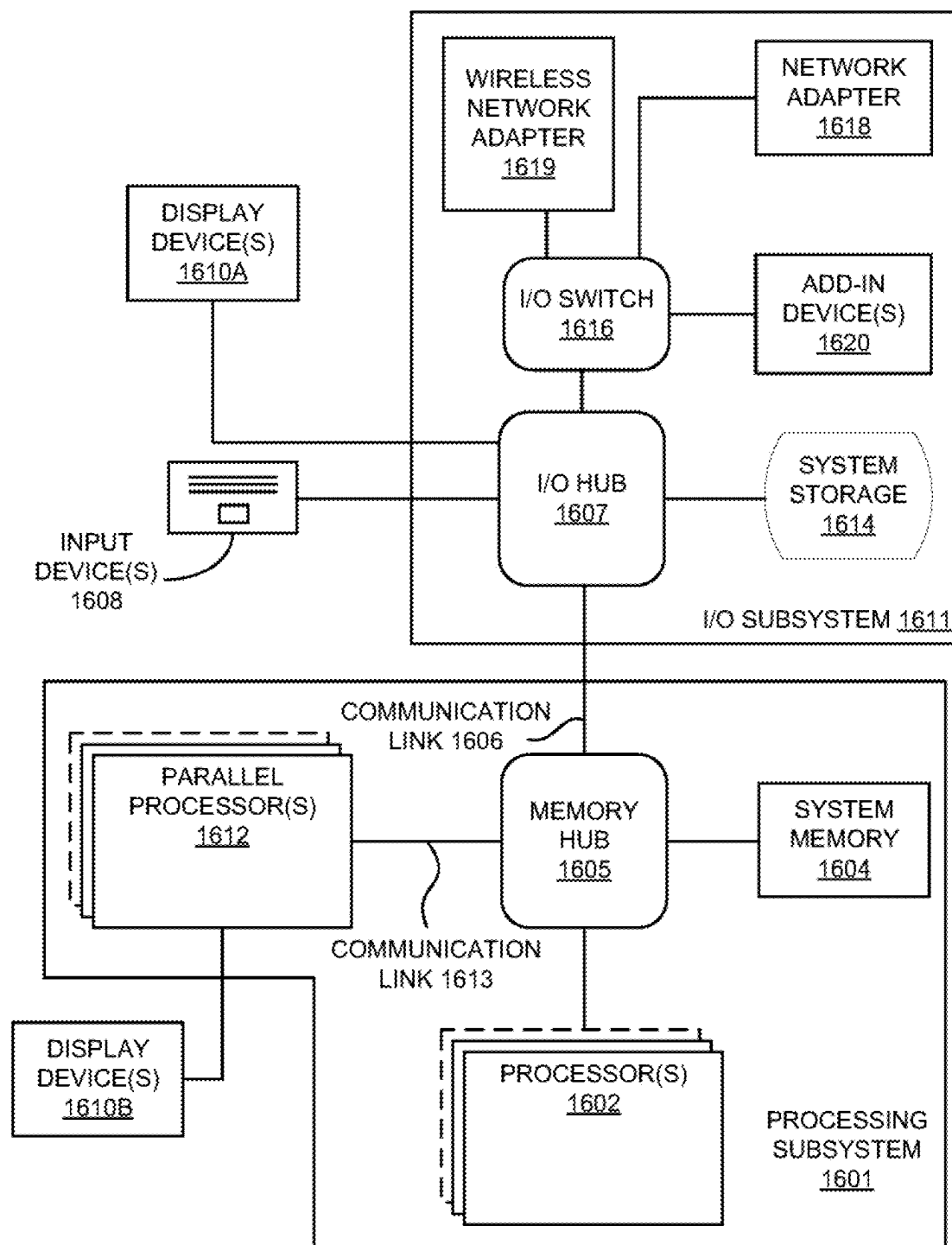


FIG. 16

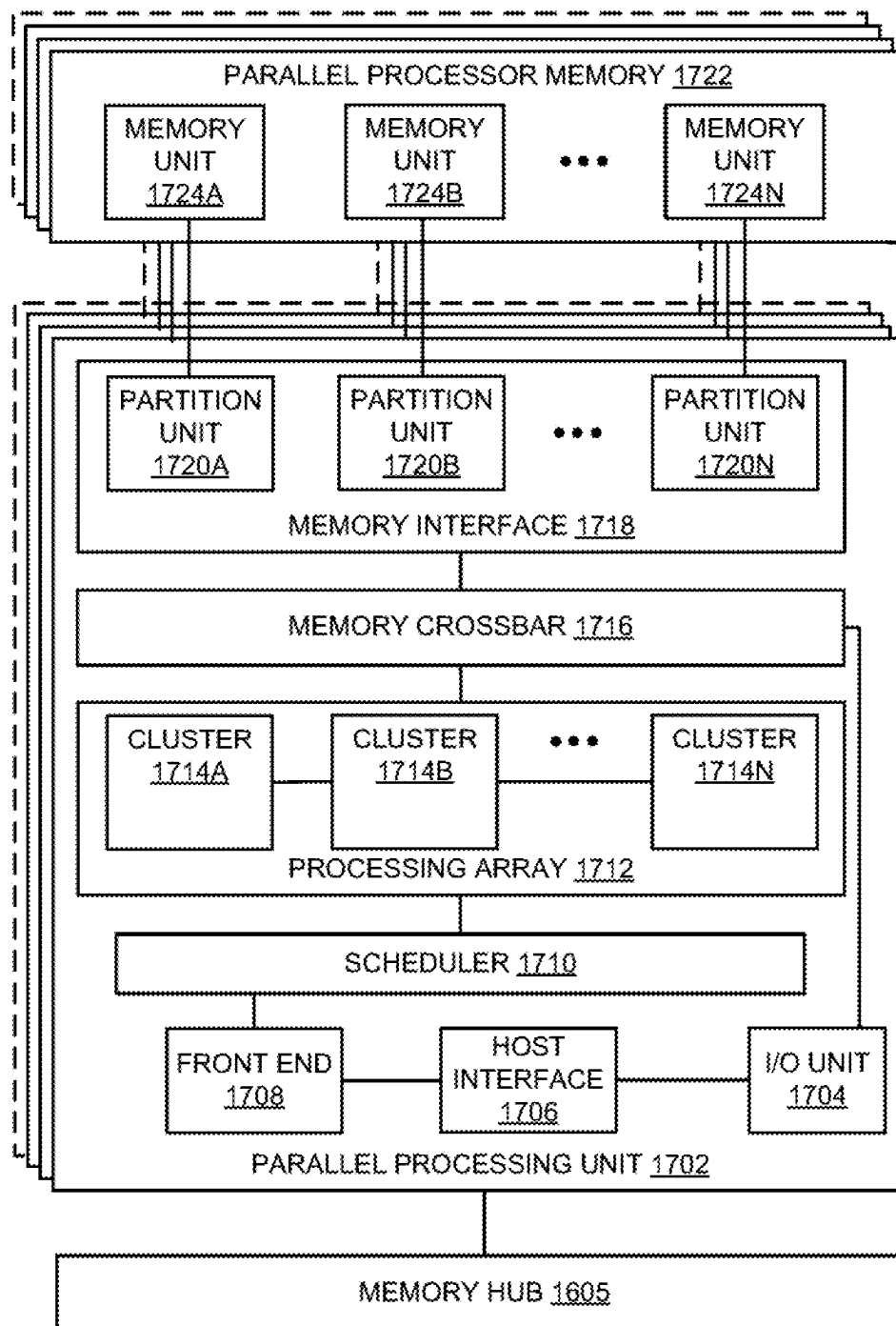


FIG. 17A

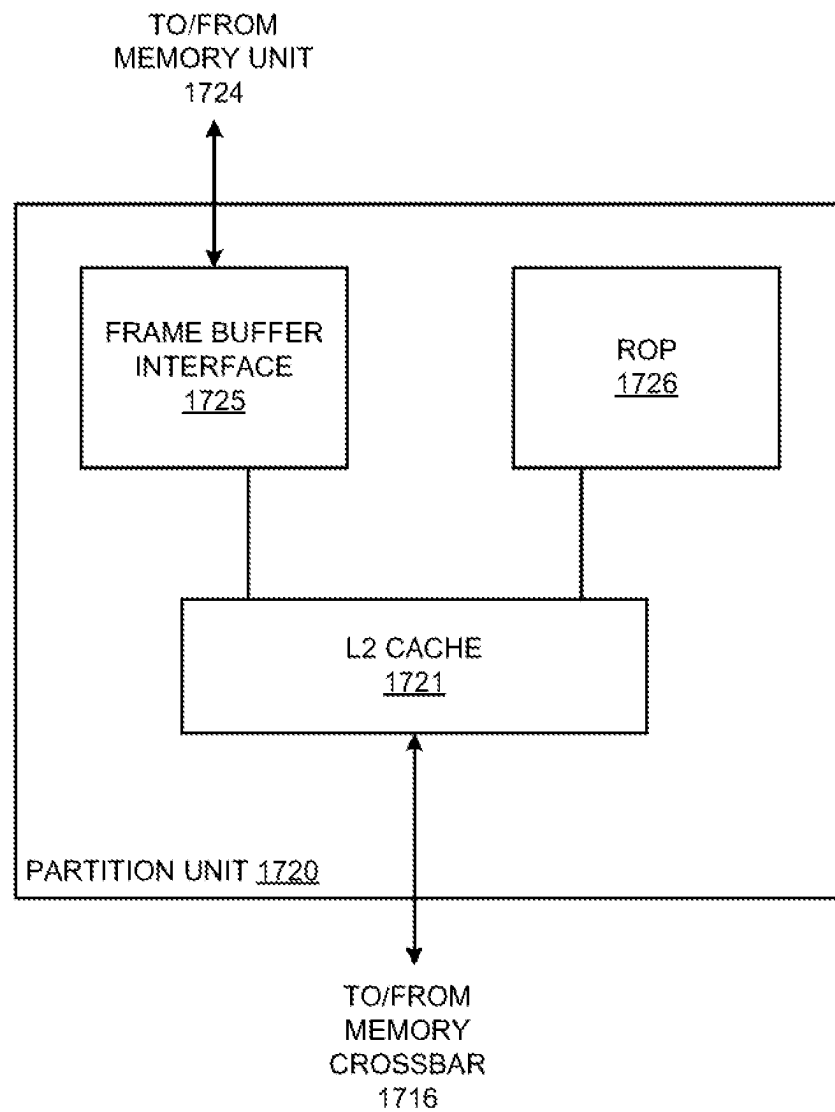


FIG. 17B

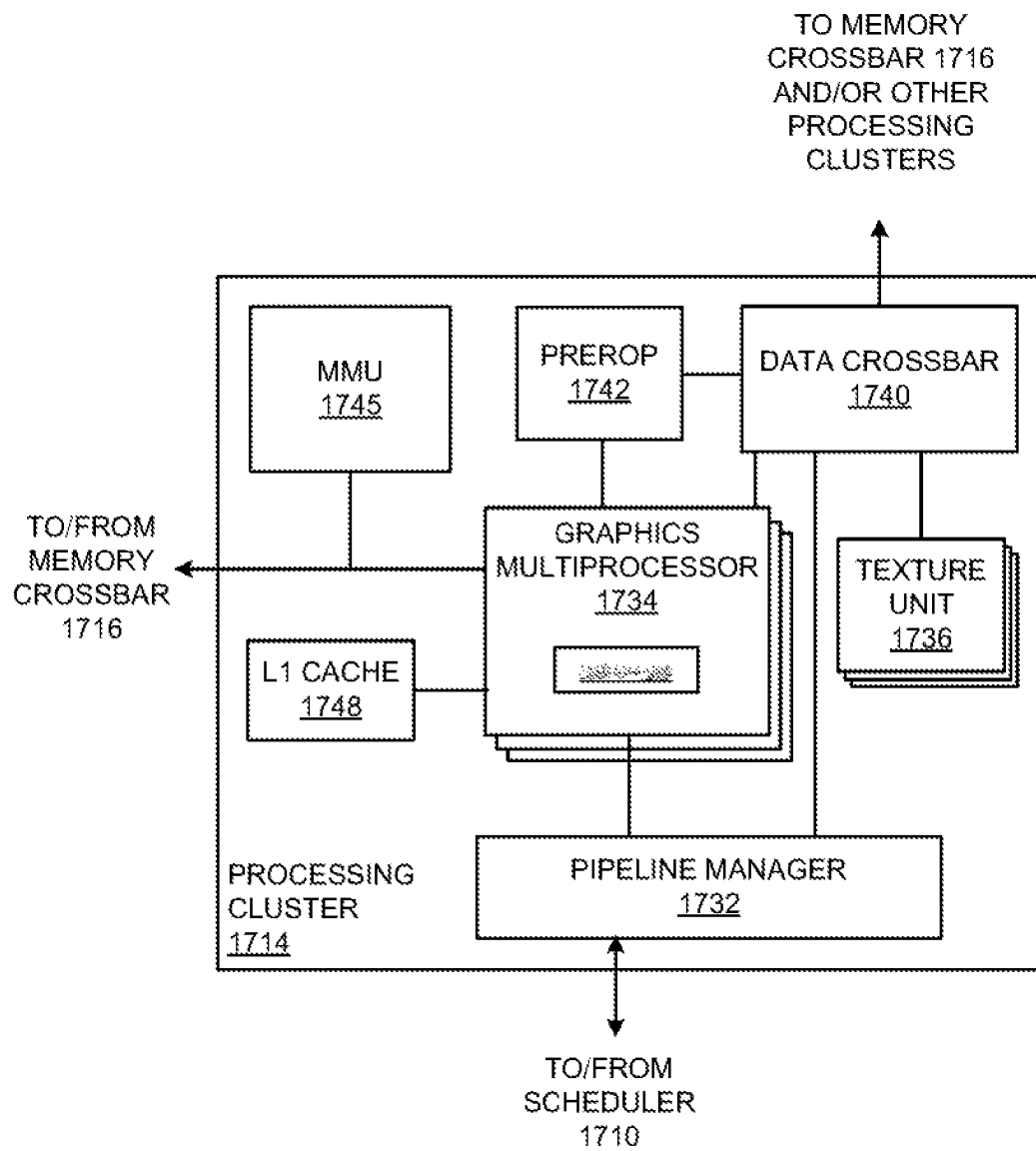


FIG. 17C

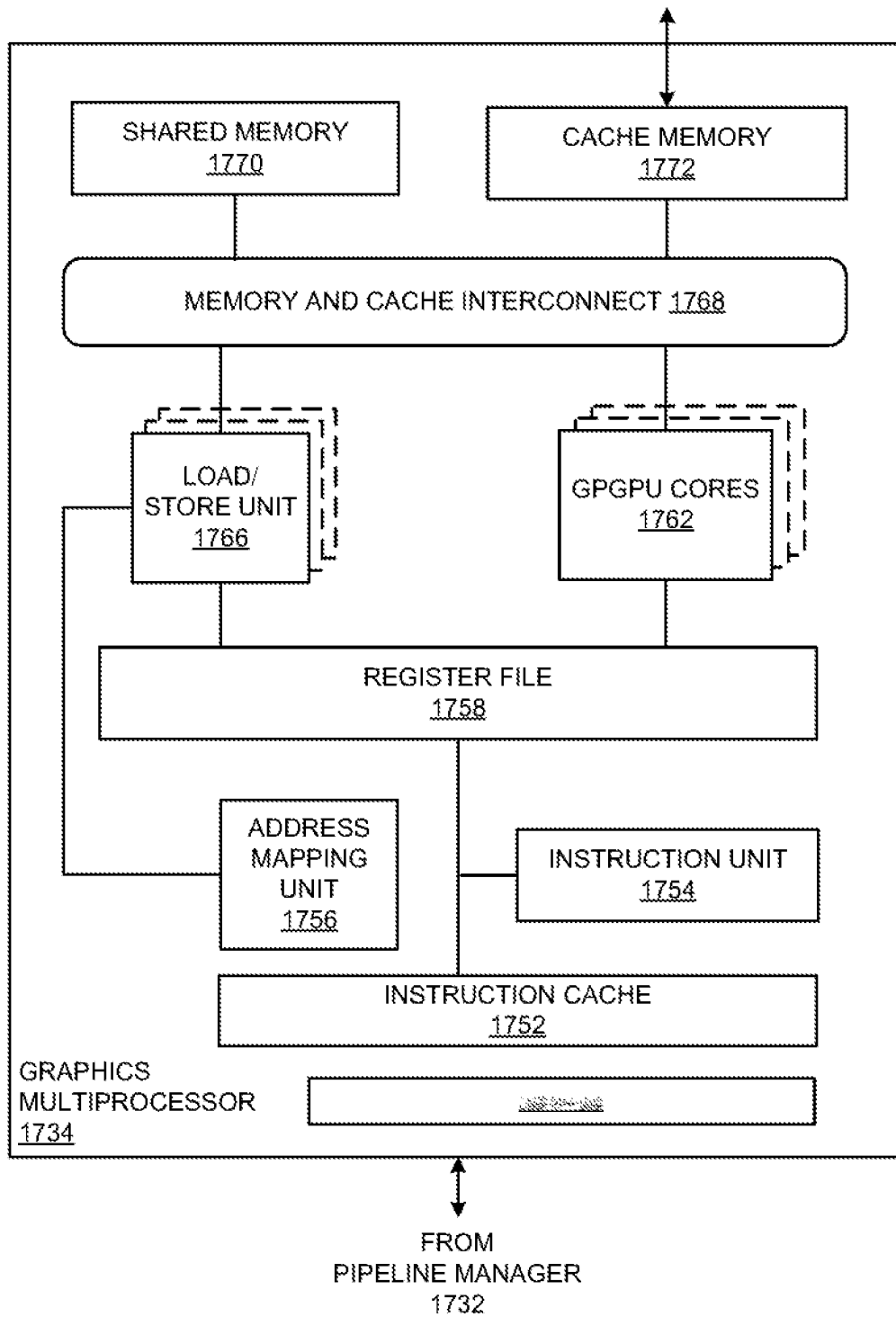


FIG. 17D

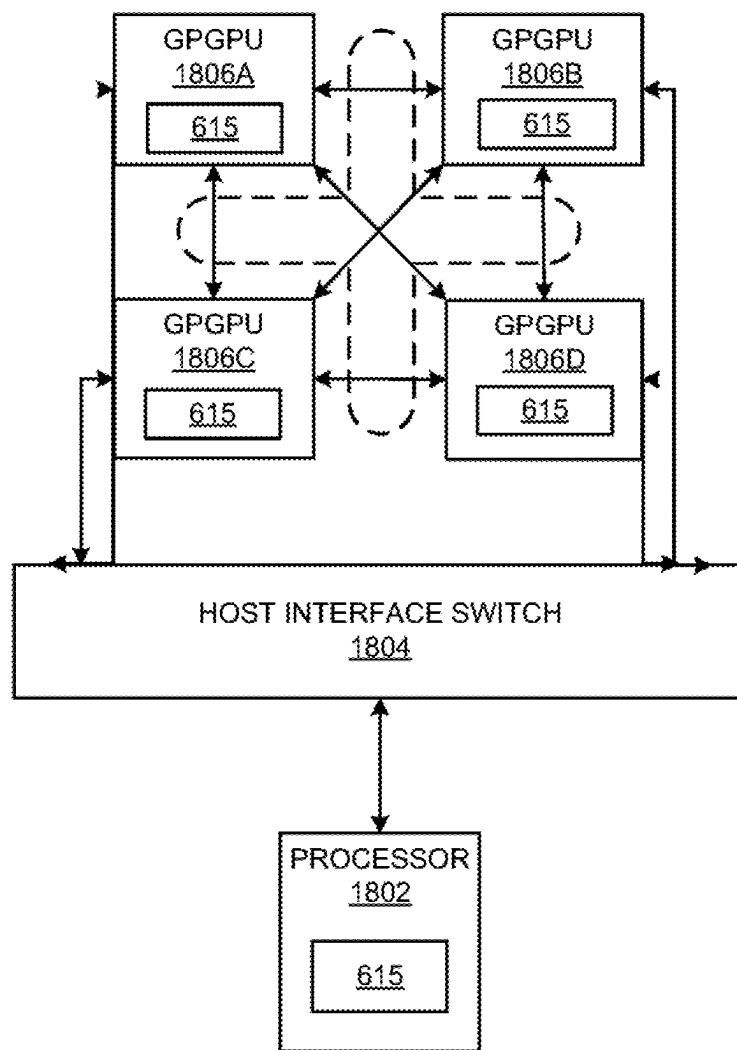


FIG. 18

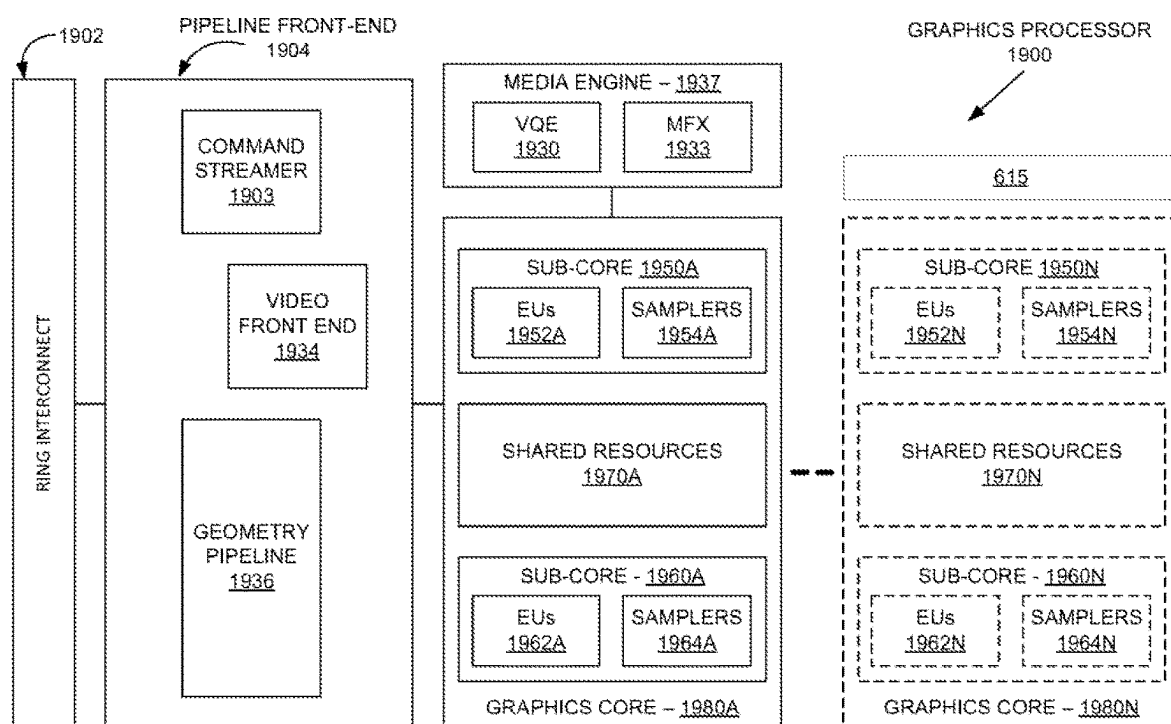


FIG. 19

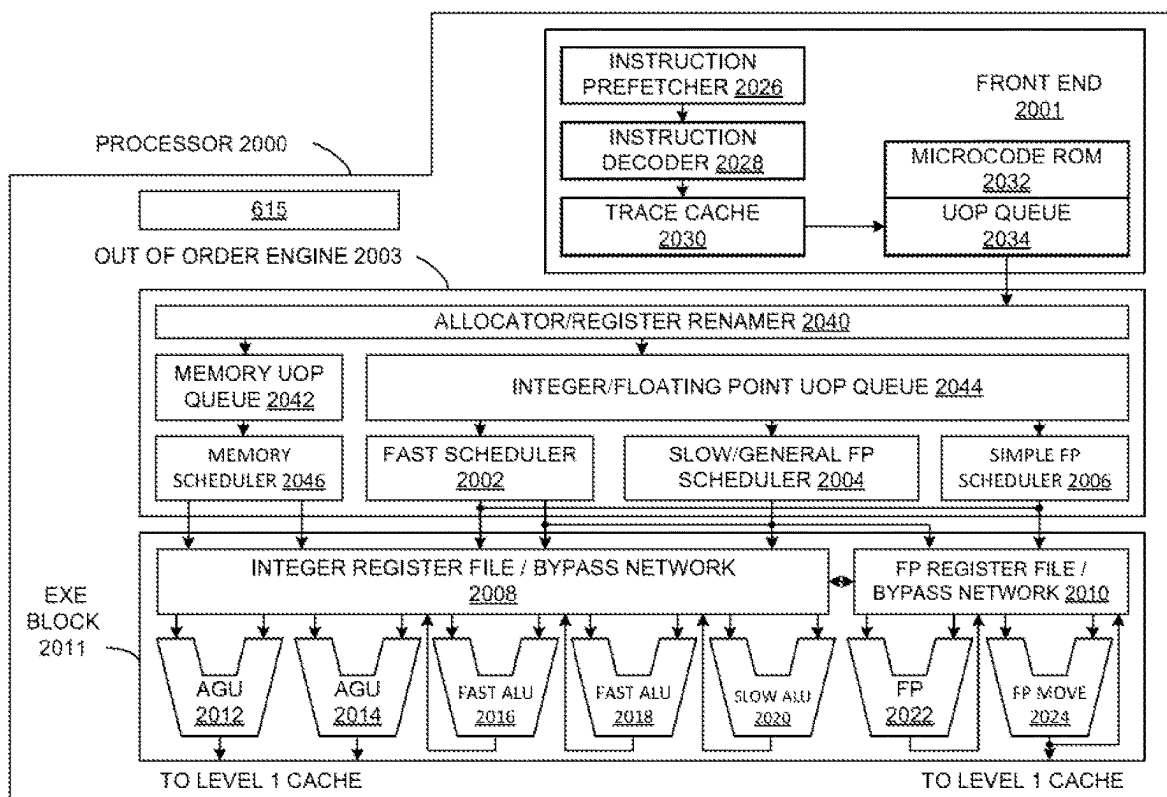


FIG. 20

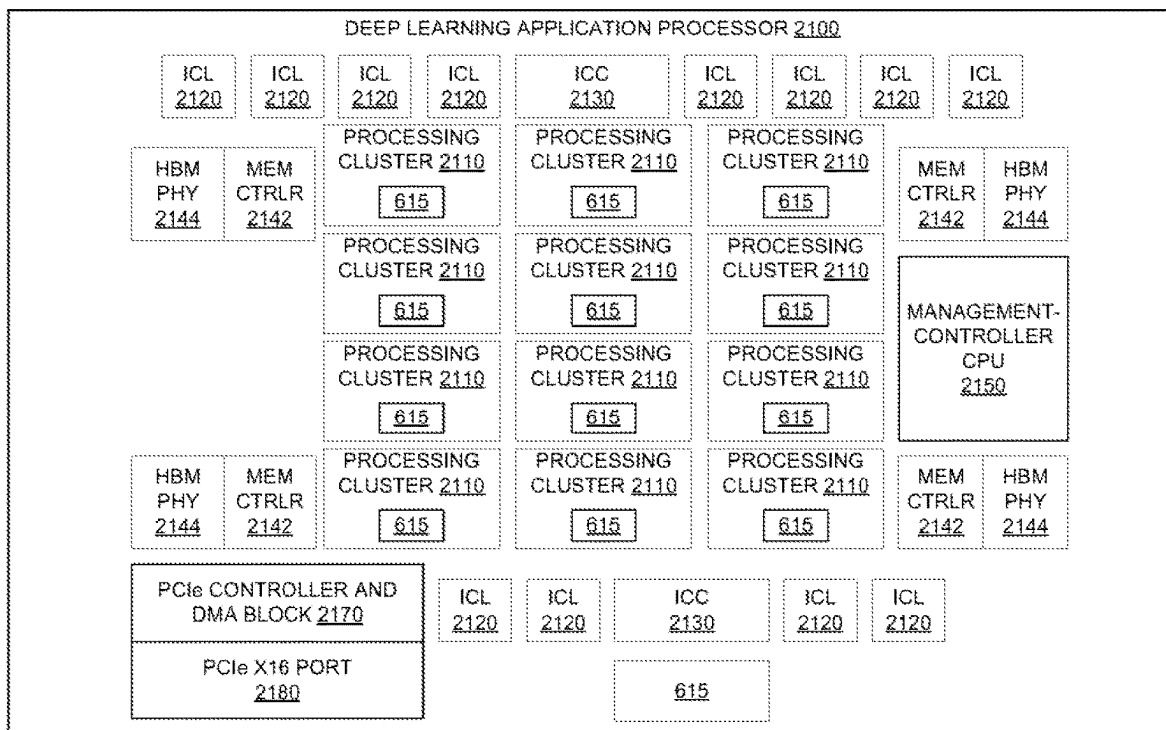
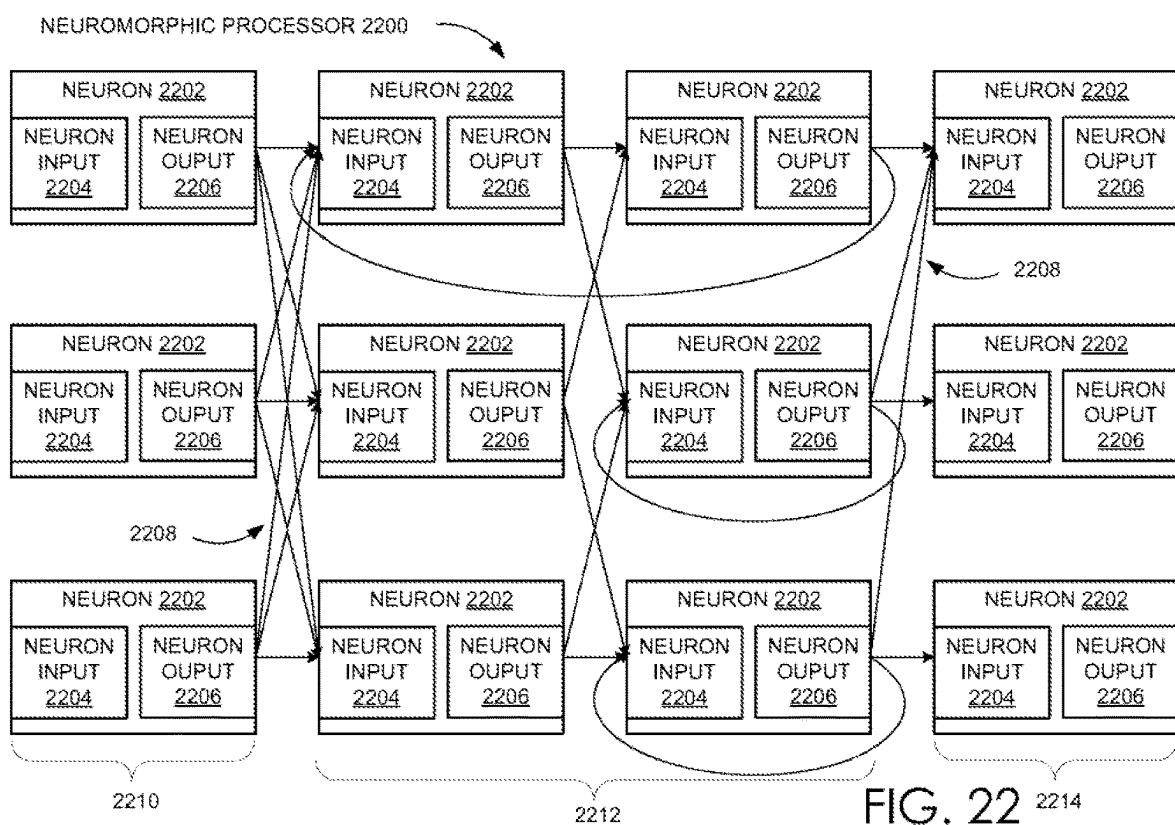


FIG. 21



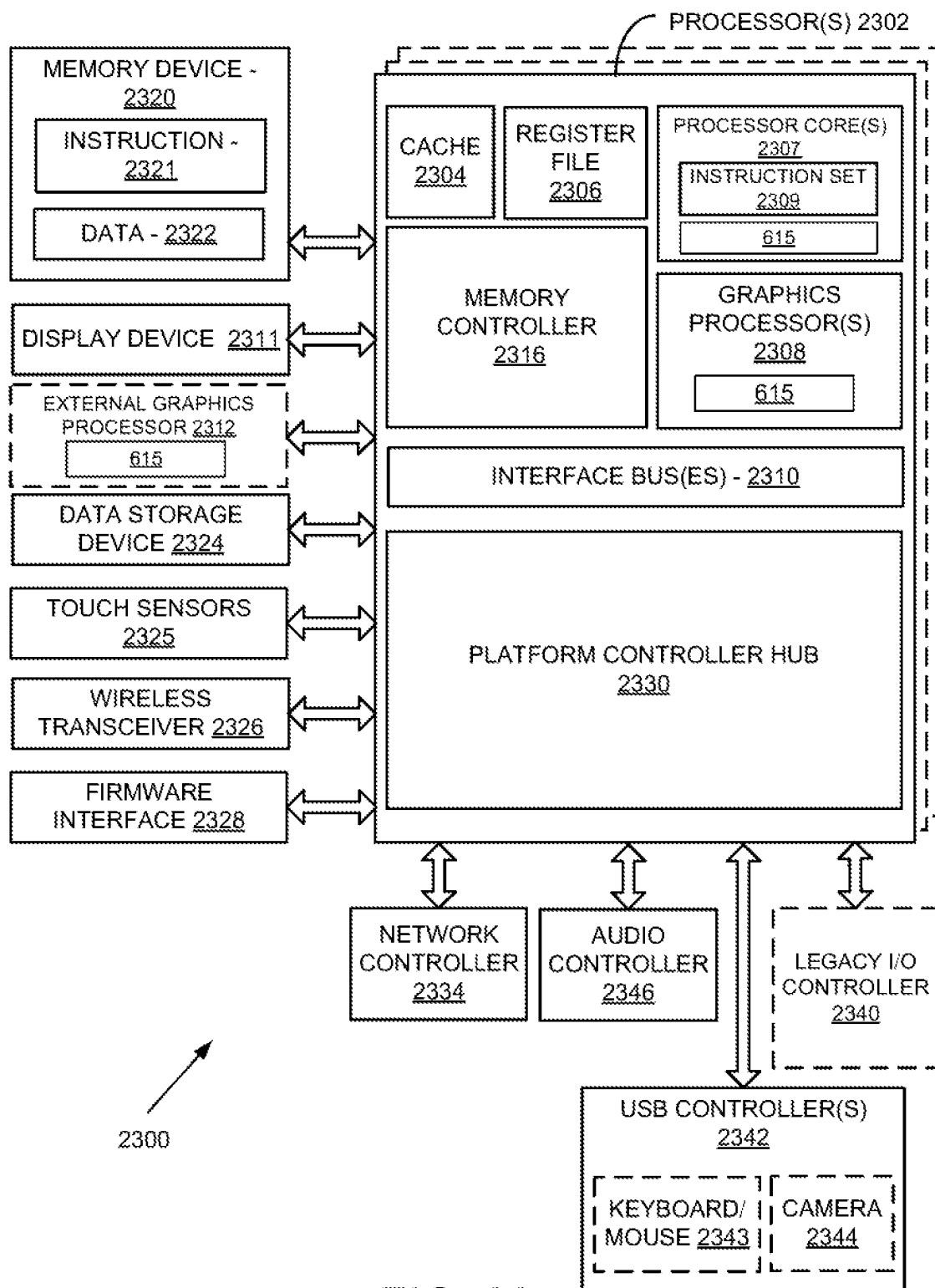


FIG. 23

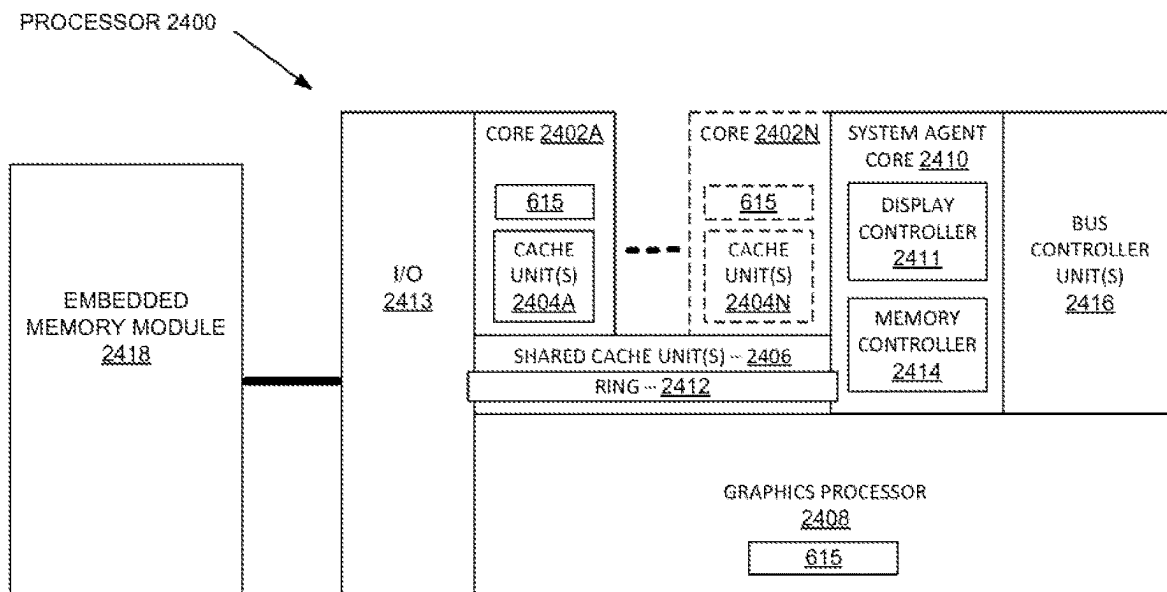
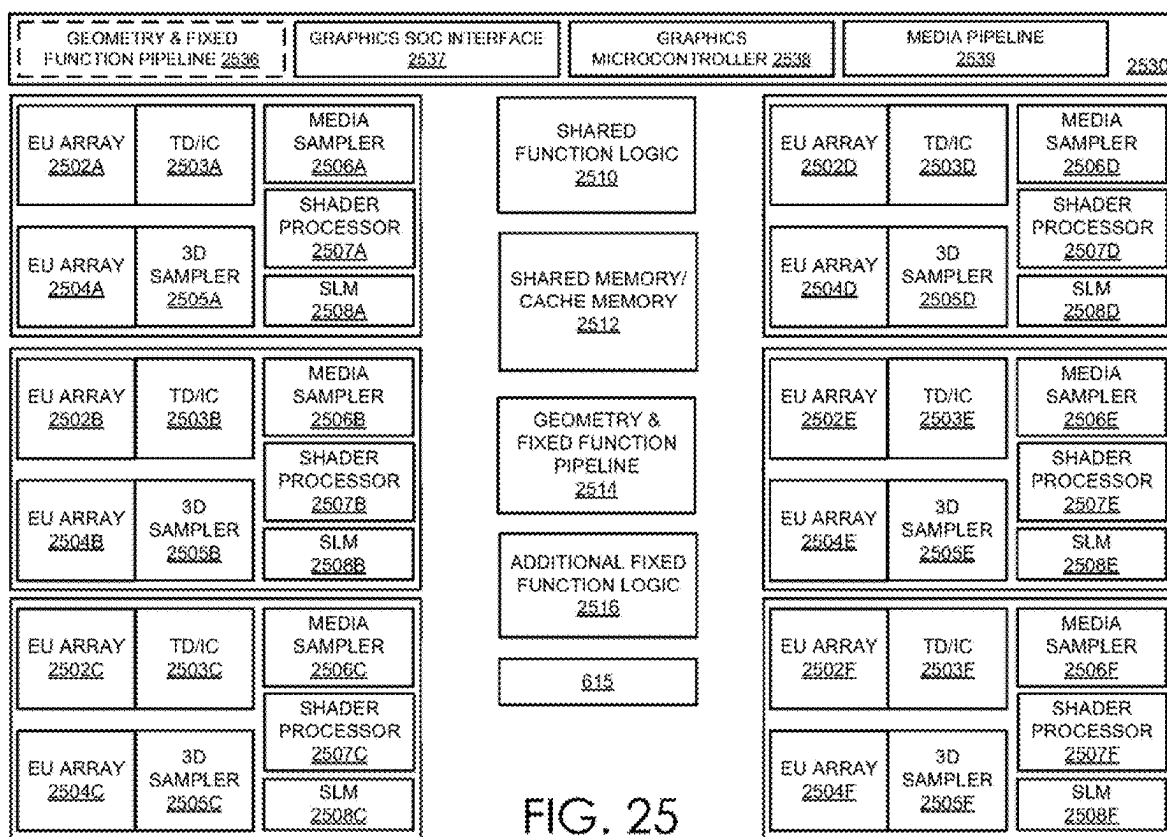


FIG. 24



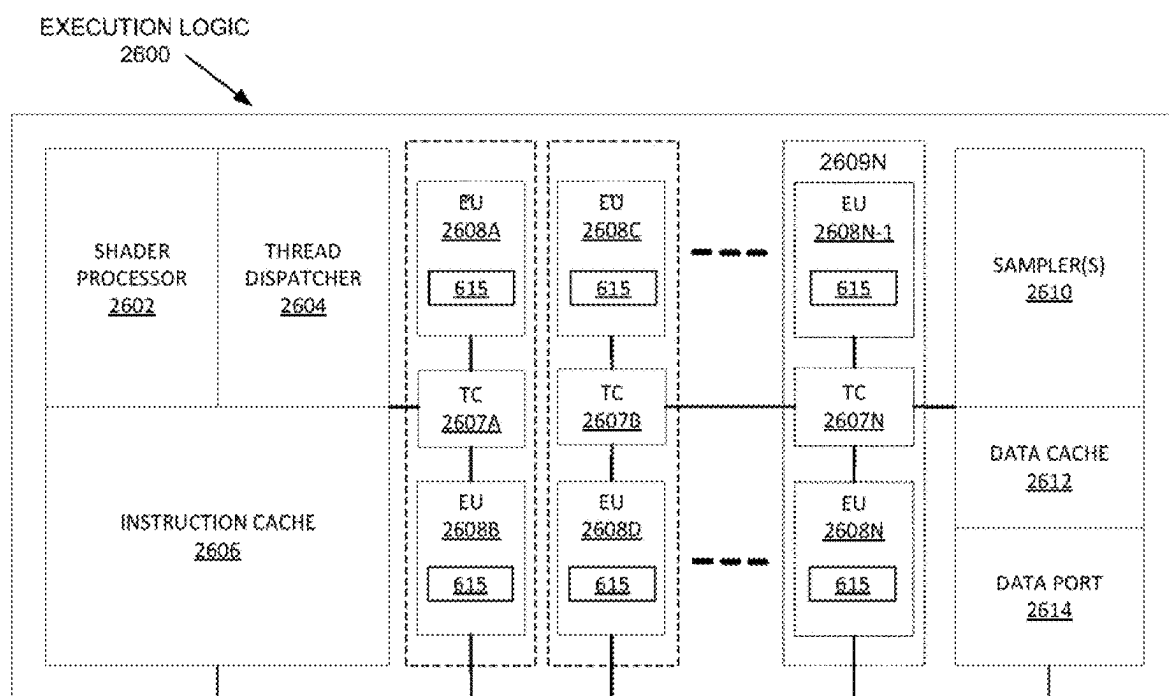


FIG. 26A

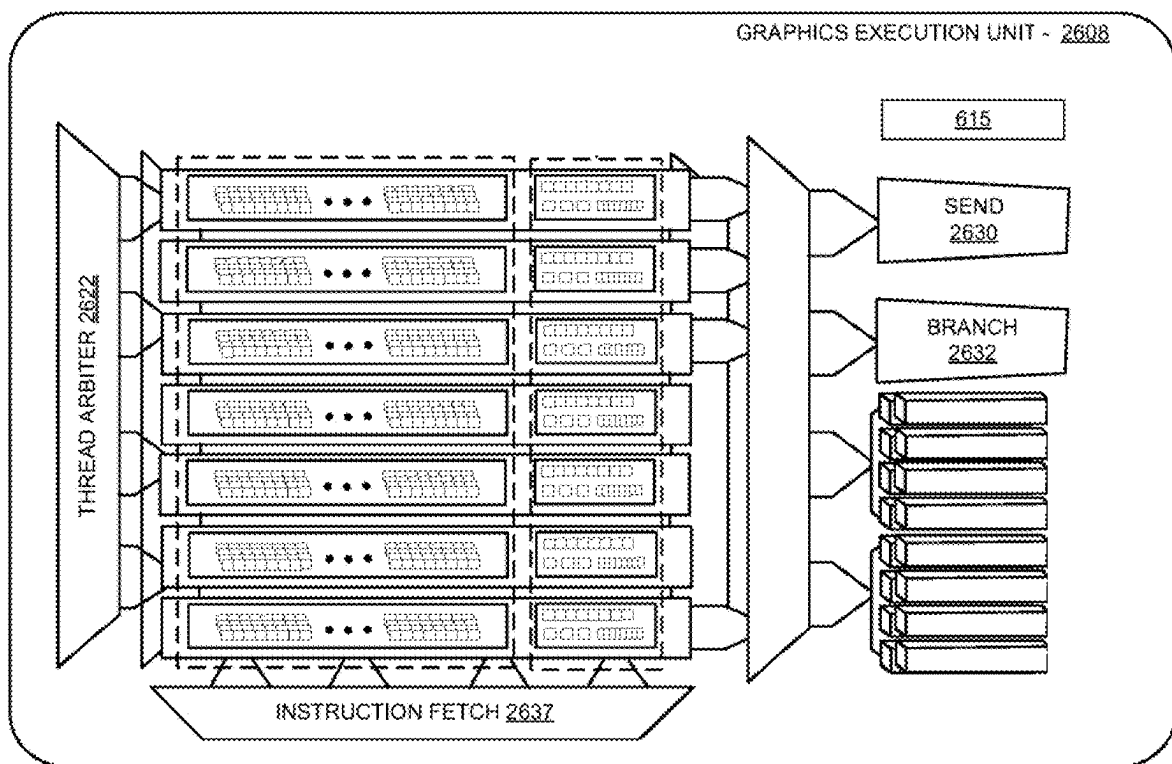


FIG. 26B

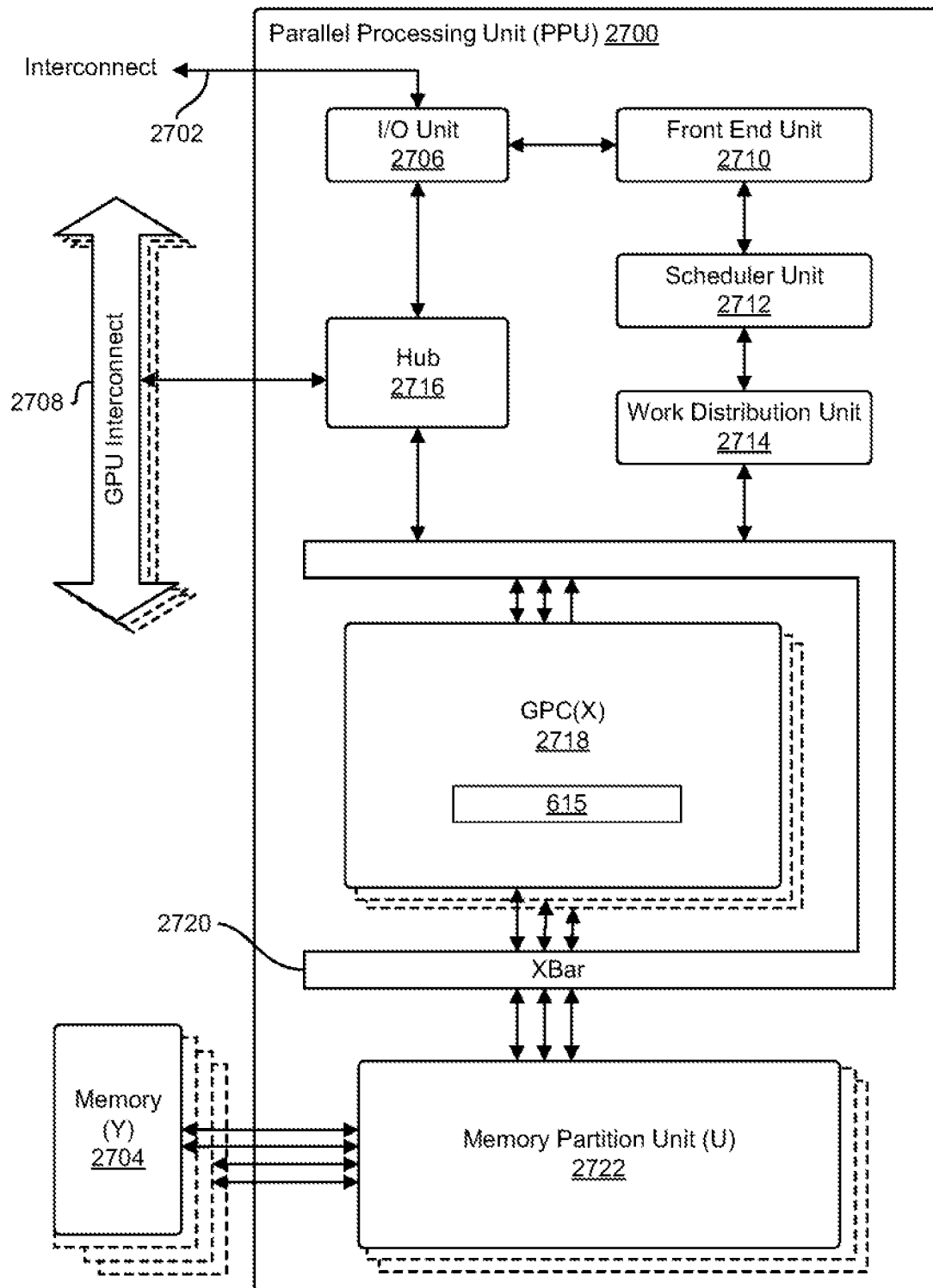


FIG. 27

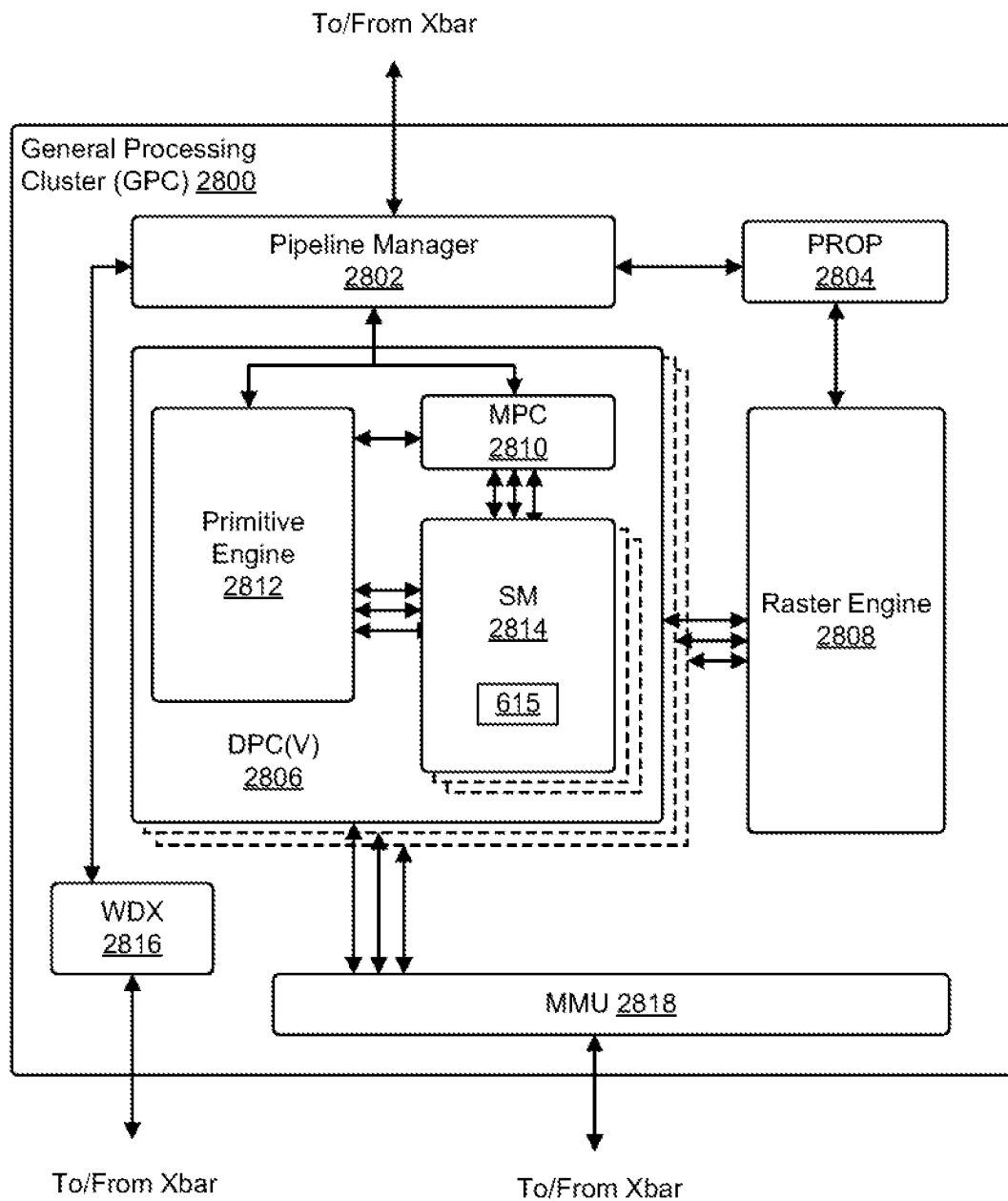


FIG. 28

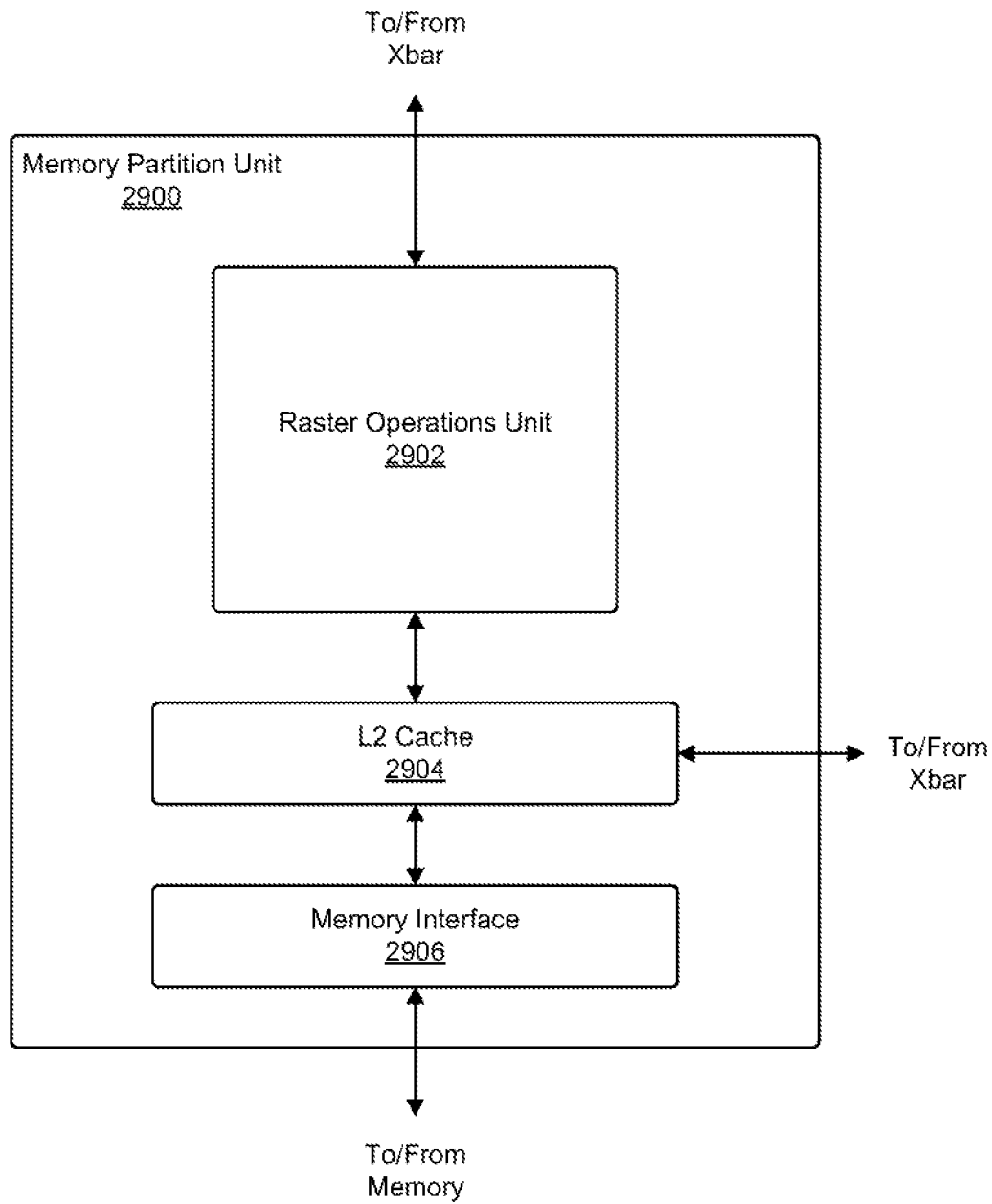


FIG. 29

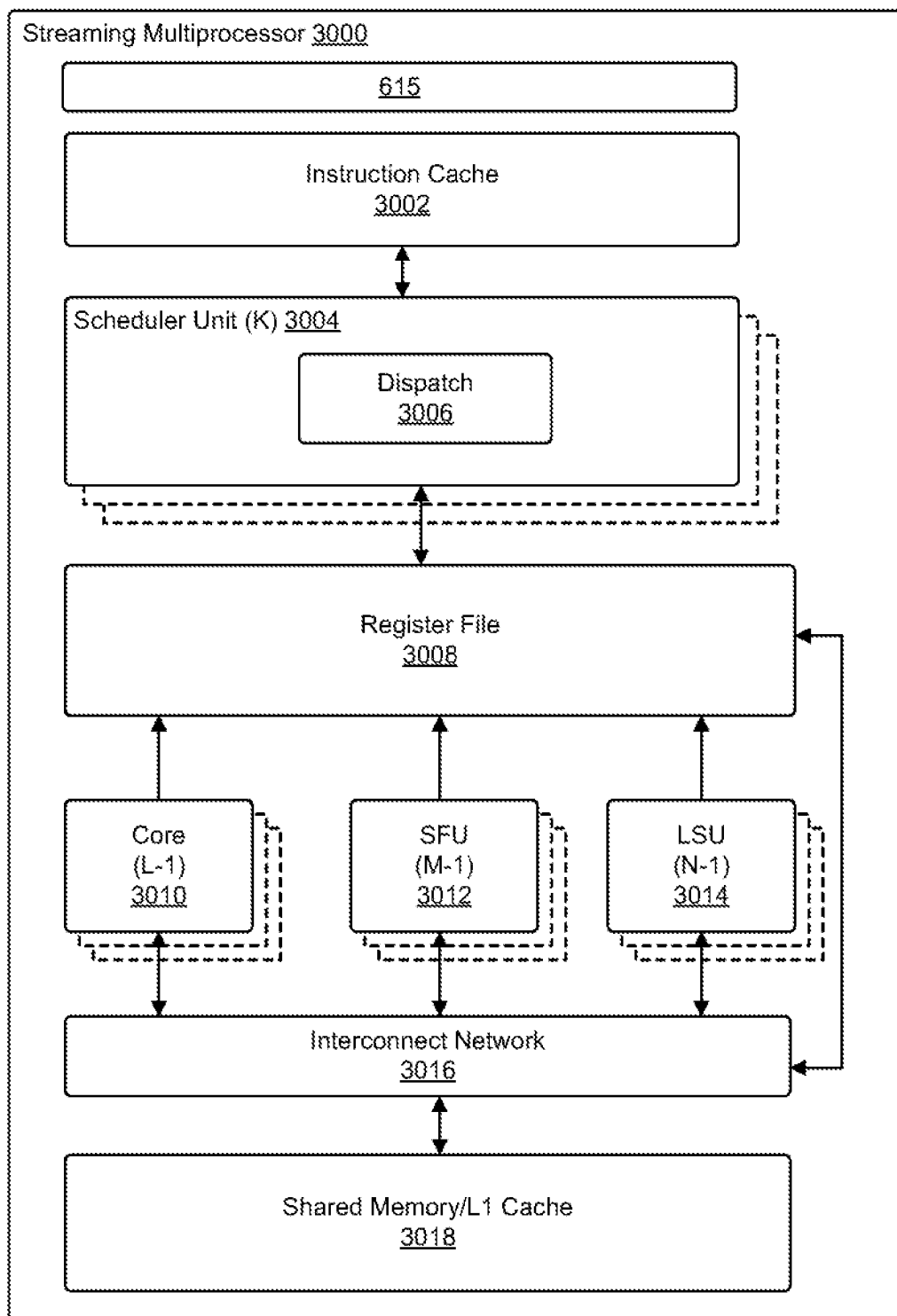


FIG. 30

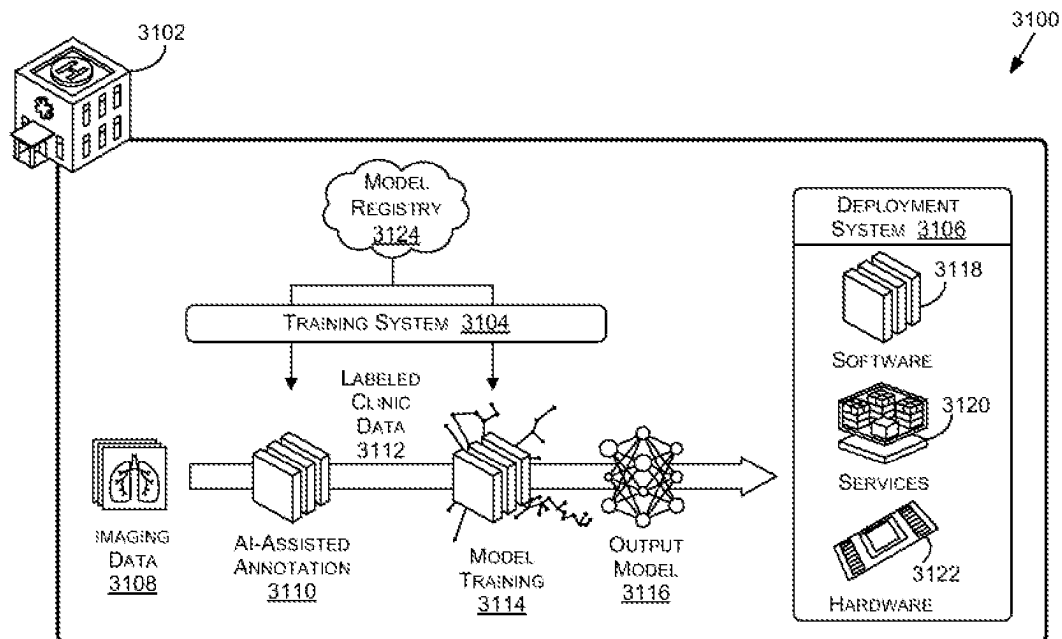


FIG. 31

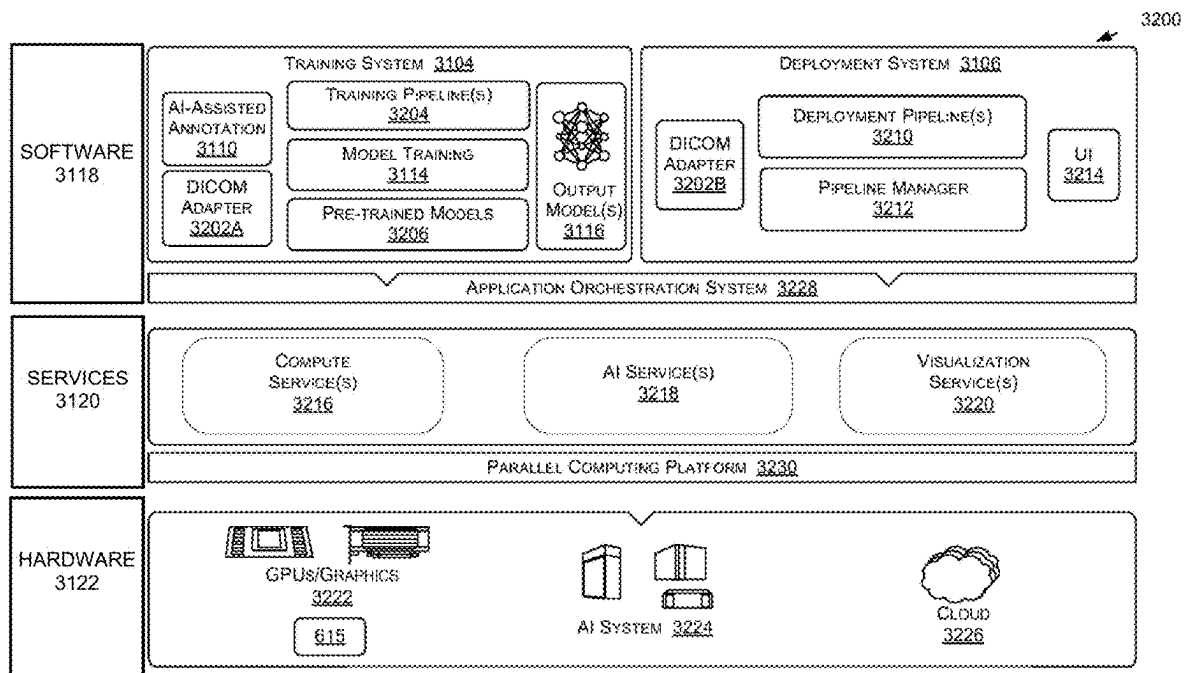


FIG. 32

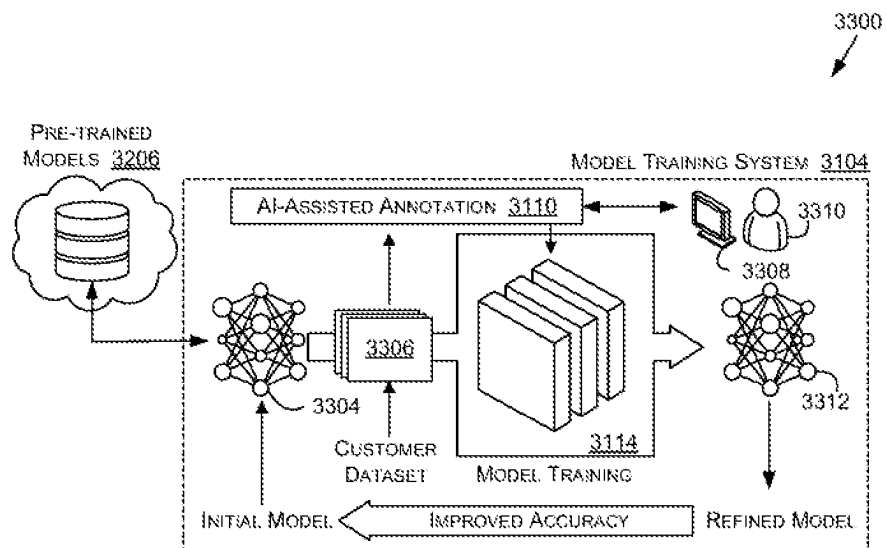


FIG. 33A

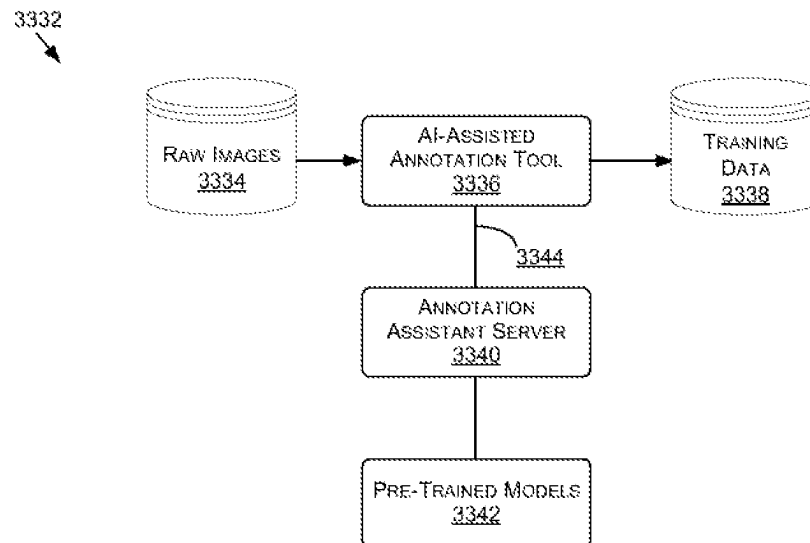


FIG. 33B

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IMAGE ENHANCEMENT USING ONE OR MORE NEURAL NETWORKS**CROSS-REFERENCE TO RELATED APPLICATIONS**

This application is a continuation of, and claims priority to, co-pending U.S. patent application Ser. No. 17/012,000, filed Sep. 3, 2020, and entitled “IMAGE ENHANCEMENT USING ONE OR MORE NEURAL NETWORKS,” which is hereby incorporated herein in its entirety and for all purposes.

FIELD

At least one embodiment pertains to processing resources used to perform and facilitate artificial intelligence. For example, at least one embodiment pertains to processors or computing systems used to train neural networks according to various novel techniques described herein.

BACKGROUND

Image and video content is increasingly being generated and displayed at higher resolutions and on higher quality displays. Approaches to generating this content at these higher resolutions often are very resource intensive, which can be problematic for devices with limited resource capacity. Further, video content is often required to be displayed with a target or minimum frame rate, and it can be difficult to generate this high resolution content at such a frame rate. Often, the quality of the resulting content is constrained by these and other limitations.

BRIEF DESCRIPTION OF THE DRAWINGS

Various embodiments in accordance with the present disclosure will be described with reference to the drawings, in which:

FIGS. 1A and 1B illustrate generated images, according to at least one embodiment;

FIGS. 2A and 2B illustrate approaches to post-processing of images, according to at least one embodiment;

FIG. 3 illustrates components of a system for providing generated image content, according to at least one embodiment;

FIG. 4 illustrates a process for enhancing an image, according to at least one embodiment;

FIG. 5 illustrates a process for applying visual effects to an image, according to at least one embodiment;

FIG. 6A illustrates inference and/or training logic, according to at least one embodiment;

FIG. 6B illustrates inference and/or training logic, according to at least one embodiment;

FIG. 7 illustrates an example data center system, according to at least one embodiment;

FIG. 8 illustrates a computer system, according to at least one embodiment;

FIG. 9 illustrates a computer system, according to at least one embodiment;

FIG. 10 illustrates a computer system, according to at least one embodiment;

FIG. 11 illustrates a computer system, according to at least one embodiment;

FIG. 12A illustrates a computer system, according to at least one embodiment;

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FIG. 12B illustrates a computer system, according to at least one embodiment;

FIG. 12C illustrates a computer system, according to at least one embodiment;

FIG. 12D illustrates a computer system, according to at least one embodiment;

FIGS. 12E and 12F illustrate a shared programming model, according to at least one embodiment;

FIG. 13 illustrates exemplary integrated circuits and associated graphics processors, according to at least one embodiment;

FIGS. 14A-14B illustrate exemplary integrated circuits and associated graphics processors, according to at least one embodiment;

FIGS. 15A-15B illustrate additional exemplary graphics processor logic, according to at least one embodiment;

FIG. 16 illustrates a computer system, according to at least one embodiment;

FIG. 17A illustrates a parallel processor, according to at least one embodiment;

FIG. 17B illustrates a partition unit, according to at least one embodiment;

FIG. 17C illustrates a processing cluster, according to at least one embodiment;

FIG. 17D illustrates a graphics multiprocessor, according to at least one embodiment;

FIG. 18 illustrates a multi-graphics processing unit (GPU) system, according to at least one embodiment;

FIG. 19 illustrates a graphics processor, according to at least one embodiment;

FIG. 20 illustrates a processor's micro-architecture, according to at least one embodiment;

FIG. 21 illustrates a deep learning application processor, according to at least one embodiment;

FIG. 22 illustrates an example neuromorphic processor, according to at least one embodiment;

FIGS. 23 and 24 illustrate at least portions of a graphics processor, according to at least one embodiment;

FIG. 25 illustrates at least portions of a graphics processor core, according to at least one embodiment;

FIGS. 26A-26B illustrate at least portions of a graphics processor core, according to at least one embodiment;

FIG. 27 illustrates a parallel processing unit (“PPU”), according to at least one embodiment;

FIG. 28 illustrates a general processing cluster (“GPC”), according to at least one embodiment;

FIG. 29 illustrates a memory partition unit of a parallel processing unit (“PPU”), according to at least one embodiment;

FIG. 30 illustrates a streaming multi-processor, according to at least one embodiment;

FIG. 31 is an example data flow diagram for an advanced computing pipeline, in accordance with at least one embodiment;

FIG. 32 is a system diagram for an example system for training, adapting, instantiating and deploying machine learning models in an advanced computing pipeline, in accordance with at least one embodiment;

FIG. 33A illustrates a data flow diagram for a process to train a machine learning model, in accordance with at least one embodiment; and

FIG. 33B is an example illustration of a client-server architecture to enhance annotation tools with pre-trained annotation models, in accordance with at least one embodiment.

DETAILED DESCRIPTION

In at least one embodiment, an image or video generation system may generate image or video content at a first

resolution, which may be of a lower resolution than one or more target output resolutions. In at least one embodiment, this can correspond to lower resolution image **100** of FIG. 1A. In at least one embodiment, this lower-resolution generation can reduce processing requirements, which can enable content to be generated faster and on devices with lesser resource (e.g., processor or memory) requirements. In at least one embodiment, this lower resolution content can then be upsampled to generate a higher resolution image **110** for output at a target resolution. In at least one embodiment, this might include going from a 1080p or 1440p resolution to a 4K or 8K resolution. In at least one embodiment, this upscaling can also accept aliased input, such as image **100**, and produce an upsampled image **110** that is also upsampled and anti-aliased. In at least one embodiment, this can be performed using a process such as Deep Learning Super Sampling (DLSS) 2.0 from NVIDIA Corporation. In at least one embodiment, this process of upscaling and antialiasing can produce a high quality, high resolution output image **110** based on a low resolution aliased image **100**, which can significantly reduce demand on a video card or other graphics processing hardware.

In at least one embodiment, at least some amount of post-processing can be used to add effects, enhancements, or augmentation to a generated image. In at least one embodiment, this can include effects and enhancements such as those illustrated in post-processed output image **120** of FIG. 1B. In at least one embodiment, post-processing can add effects such as bloom **122** around bright objects, lens flare **124**, motion blur **126**, or other effects such as may include color correction, sharpening, filtering, chromatic aberration, lens distortion, chromatic glitch, or an addition of interface elements. In at least one embodiment, post-processing can also augment an image, such as by adding graphical user interface (GUI) or heads-up display (HUD) elements, as may include health data **128**, a clock element **130**, ammunition status **132**, or other such information. In at least one embodiment, it is desired to have this additional content at a same resolution, sharpness, and image quality as other content in that image.

In at least one embodiment, this post-processing can be performed on high resolution, anti-aliased image **110**. In at least one embodiment, however, this may require more processing or resource capacity than is available, or may not be able to meet timing requirements, particularly for higher resolutions or frame rates. In at least one embodiment, post-processing could be applied on lower resolution image **100** before upscaling and anti-aliasing, but this may result in a lower quality output image than is desired, particularly for higher resolutions.

In at least one embodiment, enhancements or other visual effects can be applied to a lower resolution version of an image, and information for those visual effects then applied to a higher resolution version, as illustrated in process **200** of FIG. 2A. In at least one embodiment, an initial image **202** can be generated, such as by a rendering engine or game engine. In at least one embodiment, this initial image output by a rendering engine can have base colors set for all pixels based on determined objects or other image content for that image, as well as lighting, shading, or other effects that may be determined using ray tracing or another such process. In at least one embodiment, this can be an aliased image (or image with no anti-aliasing applied) at a resolution lower than at least one target output resolution. In at least one embodiment, this initial image is generated at a lower resolution in order to reduce memory requirements and increase rendering speed. In at least one embodiment, this

may include a frame of video game content that is rendered in real time during a session of gameplay. In at least one embodiment, this initial image **202** can have scaling applied to generate an upsampled **204** image that is higher resolution, such as for a target output resolution. In at least one embodiment, this upscaling process may involve upsampling and anti-aliasing, as performed in DLSS as mentioned above. In at least one embodiment, a lower resolution image **206** is also generated. In at least one embodiment, this lower resolution image may correspond to generated image **202**, a downsampled version of image **202**, or a downsampled version of high resolution image **204**. In at least one embodiment, an advantage to downscaling high resolution image **204** is that this image has anti-aliasing applied, as well as potentially other effects or processing. In at least one embodiment, high resolution image **204** and low resolution image **206** may be generated in parallel, with both having anti-aliasing applied, which may allow for lower resolution image **206** to be generated more quickly to reduce overall latency for post-processing.

In at least one embodiment, one or more visual effects to be applied to high resolution image **204** can instead be applied to lower resolution image **206**. In at least one embodiment, this can include any appropriate visual effect such as those mentioned or suggested herein, as may include bloom, lens flare, color adjustment, and so on. In at least one embodiment, this can result in a lower resolution, anti-aliased image **208** that has these visual effects applied. In at least one embodiment, applying these visual effects to a lower resolution image, whether applied at a per pixel level or for groups of pixels, can significantly reduce processing requirement at time versus applying these effects on high resolution image **204**. In at least one embodiment, changes to pixels of low resolution image **208** can be determined, and these changes can be applied or approximated for corresponding pixels of upsampled image **204** to produce a high resolution, anti-aliasing image **210** with these visual effects applied. In at least one embodiment, this can be done using various approaches, such as by performing pixel mapping and making corresponding adjustments, through one or more convolutions, or through use of one or more neural networks, which can take as input this visual effect data from lower resolution image **208** and upsampled image **204**, and can infer high resolution output image **210** with that effect data applied. In at least one embodiment, this can be performed using any appropriate neural network, such as a convolutional neural network (CNN) trained to generate images using provided visual effect or enhancement data.

In at least one embodiment, this visual effect data can be approximated using an enhancement function, such as may take form of an appropriate parameterized function including a linear or polynomial function. In at least one embodiment, application of one or more visual effects can result in a change individual pixels (or groups of pixels) in a lower resolution image, and these changes can be represented by parameters of that enhancement function. In at least one embodiment, an image enhancement function can take a form of:

$$f(u_{ij}) = \alpha_{ij}u_{ij} + \beta_{ij}\tilde{u}_{ij} + \gamma_{ij}$$

where α_{ij} , β_{ij} , and γ_{ij} are parameters that can be estimated using a local window, and represents a blurred version of u_{ij} . In at least one embodiment, i and j represents coordinates in two dimensions of an individual pixel. In at least one embodiment, as illustrated in view **250** of FIG. 2B, u corresponds to a high resolution image, while $\tilde{u}$ represents a downsampled (or lower resolution) version of that image. In

at least one embodiment, this image can be downsampled using any appropriate downscaling network, algorithm, or approach discussed or suggested herein. In at least one embodiment, a downsampling operation can be performed that can include various processing functions that include downsampling, and downsampling can be performed to any appropriate size, scale or resolution, which may be similar to, less than, or greater than an original rendered image. In at least one embodiment, these values can be configurable or selected based upon factors such as one or more performance goals or image characteristics. In at least one embodiment, $\tilde{u}_j$ will be a matrix rather than a value for a single pixel, which can represent a blurring of this low resolution, post-processed image. In at least one embodiment, use of this term allows for using a high resolution version of this image rather than a blurred image, enabling pixels to be borrowed from this low resolution post-processed frame when constructing higher resolution post-processed output.

In at least one embodiment, an initial low resolution aliased image (L) **252** is generated by a rendering engine. In at least one embodiment, this initial image **252** can be received from a rendering engine and stored in a buffer until time for processing. In at least one embodiment, this image can be pulled from this buffer and provided as input to a scaling module **254** or process, such as DLSS, which can generate a higher resolution, upsampled, anti-aliased image (u) **256**. In at least one embodiment, this higher resolution image can be stored temporarily to another buffer. In at least one embodiment, this scaling module **254** (or a separate module) can also generate a downsampled anti-aliased image (Du) **258** (e.g., as part of a linear downscaling process) that can represent a lower resolution (e.g., downsampled) version of (u) **256**. In at least one embodiment, this downsampled resolution is a same resolution as a resolution of originally generated image **252**, which can enable this post-processing to be performed in a game engine that may not support running post-processing at a different resolution. In at least one embodiment, one or more visual effects can be applied to this lower resolution image **258** using at least one effects module **260** or process. In at least one embodiment, this module or process may be provided by a separate system or service. In at least one embodiment, a result of this application of visual effects will be a low resolution image (y) **262** with these effects or other post-processing applied. In at least one embodiment, an enhancement approximation module **264** can take as input both low resolution image **258** and **262** as input, without and with visual effects applied, respectively. In at least one embodiment, approximation module **264** can compare these images Du and y to approximate a parameterized function $f(Du)=y$. In at least one embodiment, any changes in pixel locations between Du and y can be represented, or approximated, by a set of parameters (e.g., α , β , γ) of that function. In at least one embodiment, there may be any number of parameters in such a function, and those parameters can be determined as a best approximation of those changes to individual pixel locations in this lower resolution image. In at least one embodiment, parameter values can be determined using other approaches, such as a convolution or neural network. In at least one embodiment, other data or information representative of changes due to visual effects may be provided or utilized as well, such as pixel-specific change information or other such values.

In at least one embodiment, these estimated parameters can be provided to an application process **266** or module that will use these values to modify high resolution, anti-aliased image (u) **256** to produce a high resolution output image (y*) **268** that appears as if these visual effects were applied to

higher resolution image **256**. In at least one embodiment, this can involve plugging these estimated parameter values into a function $f(u)$ for higher resolution image **256** to cause effects-related changes to be applied to individual pixels, or groups of pixels, in this higher resolution image. In at least one embodiment, such an approach can provide for a relatively efficient and accurate estimate of a true $f(u)$ that might otherwise be applied directly to this higher resolution image. In at least one embodiment, this per-pixel function can be applied to any or all pixels of higher resolution image **256**. In at least one embodiment, a quality of this approximation may depend at least in part upon a difference in resolution between lower resolution image (y) **262** and higher resolution image (y*) **268**, as going from 1440p to 4K may provide more accurate and crisp visual effects or image enhancements than going from 1080p to 8 k resolution. In at least one embodiment, this produces an inferred, approximated, or estimated version of what this higher resolution image would have looked like if these visual effects had been applied directly to this higher resolution version. In at least one embodiment, this higher resolution, anti-aliased image **268**, with approximated visual effects applied, can then be provided as output at a target resolution. In at least one embodiment, there may be multiple output images at different resolutions or aspect ratios, and estimated parameters from one or more lower resolution images can be used to approximate these effects onto one or more of these higher resolution output images.

In at least one embodiment, there may be different approximating functions utilized. In at least one embodiment, an approximating function can assume that each pixel has a dedicated polynomial function, such as may result in three parameters per pixel. In at least one embodiment, these parameters would change over a spatial extent of an image. In at least one embodiment, this can be used to approximate pixel differences for post-processing effects that operate at a pixel level. In at least one embodiment, some post-processing effects such as blur may operate based on a sliding pixel window, such as where an actual computation may involve averaging a 3x3 window of pixels. In at least one embodiment, a pixel-wise approximation may not provide for this, as it may not have enough inputs to approximate that function, so a different function or approach may be utilized that is not pixel-specific. In at least one embodiment, f can allow for additional complexity to handle such situations. In at least one embodiment, a convolution or neural network can be used for this approximation to allow for this additional complexity. In at least one embodiment, additional steps might be taken to reduce complexity, such as to only perform approximations for a single color channel, such as luma, to which humans are more sensitive. In at least one embodiment, such a reduction in complexity can also reduce processing time.

In at least one embodiment, at least some of this image or video content may be provided or presented locally on a client device **302** as illustrated in FIG. 3. In at least one embodiment, at least a portion of this content may be provided by a content server **320**, such as a game server or provider system, across at least one wired or wireless network **340**. In at least one embodiment, content to be presented may include various types of content, as may include video game, virtual reality (VR), augmented reality (AR), image, textual, audio, haptic, or video content. In at least one embodiment, client device **302** may include or comprise a device such as a desktop computer, notebook

computer, gaming console, smart phone, tablet computer, VR headset, AR goggles, a wearable computer, or a smart television.

In at least one embodiment, client device **302** can generate content for a session, such as a gaming session or video viewing session, using components of a content application **304** on client device **302** and data stored locally on that client device. In at least one embodiment, a content application **324** (e.g., a gaming or streaming media application) executing on content server **320** can initiate a session associated with at least client device **302**, as may utilize a session manager and user data stored in a user database **334**, and can cause content **332** to be determined by a content manager **326** and rendered using a rendering engine **328**, if needed for this type of content or platform, and transmitted to client device **302** using an appropriate transmission manager **322** to send by download, streaming, or another such transmission channel. In at least one embodiment, client device **302** receiving this content can provide this content to a corresponding content application **304**, which may also or alternatively include a rendering engine **310** for rendering at least some of this content for presentation via client device **302**, such as video content through a display **306** and audio, such as sounds and music, through at least one audio playback device **308**, such as speakers or headphones. In at least one embodiment, at least some of this content may already be stored on, rendered on, or accessible to client device **302** such that transmission over network **340** is not required for at least that portion of content, such as where that content may have been previously downloaded or stored locally on a hard drive or optical disk. In at least one embodiment, a transmission mechanism such as data streaming can be used to transfer this content from server **320**, or content database **334**, to client device **302**. In at least one embodiment, at least a portion of this content can be obtained or streamed from another source, such as a third party content service **350** that may also include a content application **352** for generating or providing content. In at least one embodiment, portions of this functionality can be performed using multiple computing devices, or multiple processors within one or more computing devices, such as may include a combination of CPUs and GPUs.

In at least one embodiment, content application **324** includes a content manager **326** that can determine or analyze content before this content is transmitted to client device **302**. In at least one embodiment, content manager **326** can also include, or work with, other components that are able to generate, modify, or enhance content to be provided. In at least one embodiment, this can include a rendering engine **328** for rendering content, such as aliased content at a first resolution. In at least one embodiment, an upsampling or scaling component **330** can generate at least one additional version of this image at a different resolution, higher or lower, and can perform at least some processing such as anti-aliasing. In at least one embodiment, a post-processing component **332** can perform post-process on one or more of those images, such as to apply visual effects, enhancements, or augmentations as discussed herein. In at least one embodiment, content manager **326** can then select an image or video frame of an appropriate resolution to send to client device **302**. In at least one embodiment, a content application **304** on client device **302** may also include components such as a rendering engine **310**, upsampling module **312**, and post-processing module **314**, such that any or all of this functionality can additionally, or alternatively, be performed on client device **302**. In at least one embodiment, a content application **352** on a third party content

service system **350** can also include such functionality. In at least one embodiment, locations where at least some of this functionality is performed may be configurable, or may depend upon factors such as a type of client device **302** or availability of a network connection with appropriate bandwidth, among other such factors. In at least one embodiment, an upsampling module **330** or post-processing module **332** may include one or more neural networks for performing or assisting in this functionality, where those neural networks can also be provided, or made available, to other client devices **360**, such as for download or streaming from a media source storing a copy of that image or video content. In at least one embodiment, this may include transmitting images of game content for a multiplayer game, where different client devices may display that content at different resolutions.

In at least one embodiment, a process **400** for generating image or video content can be performed as illustrated in FIG. **4**. In at least one embodiment, an image is generated **402** at a first resolution, which may or may not have any anti-aliasing or other processing applied, as may be produced by a game engine. In at least one embodiment, this first resolution may be lower than a target output resolution, and thus will be referred to as “low” or “lower” resolution although that should not be interpreted as specifying a specific resolution unless otherwise specifically stated. In at least one embodiment, upsampling can be performed **404** to generate a high resolution, anti-aliased image, where “high” resolution in this instance refers to a resolution that is higher than this first resolution. In at least one embodiment, multiple higher resolution versions can be generated. In at least one embodiment, at least one downsampled, lower resolution version of this anti-aliased image can also be generated **406**, either in parallel or based on one of these higher resolution anti-aliased images that are generated. In at least one embodiment, one or more visual effects, augmentations, or enhancements can be applied **408** to this lower resolution, anti-aliased image to generate a low resolution, post-processed image that includes these effects, augmentations, or enhancements. In at least one embodiment, changes due to these effects, augmentations, or enhancements can be approximated **410** using a parameterized function for coordinates of that lower resolution image. In at least one embodiment, these parameters can then be used with a function for a higher resolution anti-aliased image, where that function can be applied **412** to that image in order to determine changes to individual pixel locations, or groups of pixel locations, corresponding to these effects, augmentations, or enhancements that were applied to this lower resolution image. In at least one embodiment, this can result in at least one higher resolution image that appears as if this post-processing was applied to this higher resolution version. In at least one embodiment, this high resolution anti-aliased image or video frame can then be provided **414** as output, such as for display by a client device configured for display or presentation at that resolution.

In at least one embodiment, a process for applying visual effects to an image can be performed **500** as illustrated in FIG. **5**. In at least one embodiment, one or more visual effects can be caused **502** to be applied to one or more images having less than a first resolution. In at least one

embodiment, information for these visual effects that were applied to these one or more images can be determined 504, as may relate to parameters of an enhancement function. In at least one embodiment, this information can be applied 506 to one or more upscaled images at, or above, this first resolution, in order to generate one or more images having at least this first resolution and that include an approximation of these visual effects applied to one or more images at less than this first resolution.

In at least one embodiment, some post-processing may be applied directly to a higher resolution image after upsampling and anti-aliasing. In at least one embodiment, this may include addition of text or user interface elements that are to be provided as an overlay overtop of objects in an image, such as on top of game content. In at least one embodiment, adding such augmentation or enhancement does not require significant resource capacity, such that this content may be added directly to this higher resolution image to ensure crispness and high quality of that particular content. In at least one embodiment, other visual effects such as bloom 20 and blur can be approximated using data from visual effects applied to a lower resolution version, and this additional content can be applied to that higher resolution image after this approximation. In at least one embodiment, an image may correspond to a sequence of images or frames of video, 25 and an approximated function can be applied to each image or frame of that sequence. In at least one embodiment, application of this function can involve upsampling of these parameters of this function, with this “upsampled” function being used to enhance a higher resolution image.

In at least one embodiment, there may be different types or versions of post-processing applied to an image. In at least one embodiment, a post-processing version operates at a pixel level. In at least one embodiment, input to a true post-processing function involves color data for a single pixel, with output being a single pixel color. In at least one embodiment, this version might be used for effects such as tone-mapping or converting color to grayscale. In at least one embodiment, a pixel-wise approach can be used because a different approach might result in destroying some information, and thus detail, in a higher resolution image. In at least one embodiment, another version can analyze small spatial windows of information for effects such as bloom or blur. In at least one embodiment, rendering a neon light will result in pixels with a higher level of brightness, which can result in some glow around that neon that is referred to as bloom. In at least one embodiment, applying bloom can result in blurring bright pixel values across nearby pixels in a semi-transparent fashion by approximating this glow within a small window. In at least one embodiment, another 50 version of post-processing can replace a color completely, such as for additional of GUI overlays or a heads-up display. In at least one embodiment, a set of convolutions could be utilized and a determination made as to which convolution best approximates a given input/output pair, which could 55 allow for functionality such as application of an actual blurring kernel.

In at least one embodiment, there may be an approximation function or equation for each pixel in a downsampled, anti-aliased frame. In at least one embodiment, if there are a million pixels with a million equations, each with a number (e.g., 3) of unknown parameters, then there will be more unknowns than equations which can be challenging to solve. In at least one embodiment, an assumption can be made that these unknowns (e.g., alpha, beta, and gamma) are 65 shared across a relatively small window, such as a 2x2 or 8x8 pixel region. In at least one embodiment, this can result

in fewer unknowns than equations, which can allow for simplified approximation. In at least one embodiment, linear regression can be used to determine values for these unknowns. In at least one embodiment, these determined values can then be upsampled to obtain different alpha, beta, and gamma (or other parameter) values all pixels of a higher resolution image, such that this function can be applied at this higher resolution.

In at least one embodiment, an approximation function f is a linear function that can be solved using a process such as matrix inversion. In at least one embodiment, a matrix can be constructed and an inverse of that matrix solved for in a shader. In at least one embodiment, linear regression can be used for a linear function, while a different estimation technique can be used for a non-linear function. In at least one embodiment, f could be a neural network in its functional form, where a technique such as gradient descent can be used to determine values for parameters off. In at least one embodiment, other techniques can be used as well, as may include Gauss-Newton optimization. In at least one embodiment, fast Fourier transforms (FFTs) can be used to estimate phase shifts in terms of color or displacement in space, in order to estimate part of a transform that approximates f , such as may provide improved performance for groups or windows of pixels. In at least one embodiment, historical data can also be used, where pixel values from a prior frame can be used as starting points for estimating parameters for a current frame. In at least one embodiment, this can result in smaller search windows for these parameters, and may result in a somewhat higher resolution approximating function. In at least one embodiment, data for a prior three frames could be used, which would allow for a same number of equations as unknowns, if these equations each have three unknowns, allowing for different parameter values for each pixel and higher accuracy results. In at least one embodiment, a recurrent neural network (RNN) could be used to learn this information over multiple frames. In at least one embodiment, such an RNN can utilize motion vectors to warp historical data, in a way similar to a warping process used in DLSS. In at least one embodiment, these motion vectors can be vectors using in upsampling, or may be generated by yet another deep learning network, such as an interpolation or optical flow network. In at least one embodiment, Cholesky decomposition can be used to estimate these parameters for overlapping or non-overlapping pixel windows. In at least one embodiment, a function may not be specific to color-to-color transform, but may take in a coordinate and output a color, such as may be appropriate for text or other such content.

In at least one embodiment, noise or other variations may be added to a downsampled image in order to provide for more accurate approximation. In at least one embodiment, knowledge of added noise can help to more quickly and accurately learn information about a blurring kernel used. In at least one embodiment, since this lower resolution post-processed image is never displayed to an end user, that image can be modified as appropriate to attempt to generate a better approximation for corresponding high resolution output.

60 Inference and Training Logic

FIG. 6A illustrates inference and/or training logic 615 used to perform inferencing and/or training operations associated with one or more embodiments. Details regarding inference and/or training logic 615 are provided below in conjunction with FIGS. 6A and/or 6B.

In at least one embodiment, inference and/or training logic 615 may include, without limitation, code and/or data

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storage **601** to store forward and/or output weight and/or input/output data, and/or other parameters to configure neurons or layers of a neural network trained and/or used for inferencing in aspects of one or more embodiments. In at least one embodiment, training logic **615** may include, or be coupled to code and/or data storage **601** to store graph code or other software to control timing and/or order, in which weight and/or other parameter information is to be loaded to configure, logic, including integer and/or floating point units (collectively, arithmetic logic units (ALUs)). In at least one embodiment, code, such as graph code, loads weight or other parameter information into processor ALUs based on architecture of a neural network to which this code corresponds. In at least one embodiment, code and/or data storage **601** stores weight parameters and/or input/output data of each layer of a neural network trained or used in conjunction with one or more embodiments during forward propagation of input/output data and/or weight parameters during training and/or inferencing using aspects of one or more embodiments. In at least one embodiment, any portion of code and/or data storage **601** may be included with other on-chip or off-chip data storage, including a processor's L1, L2, or L3 cache or system memory.

In at least one embodiment, any portion of code and/or data storage **601** may be internal or external to one or more processors or other hardware logic devices or circuits. In at least one embodiment, code and/or data storage **601** may be cache memory, dynamic randomly addressable memory ("DRAM"), static randomly addressable memory ("SRAM"), non-volatile memory (e.g., Flash memory), or other storage. In at least one embodiment, choice of whether code and/or data storage **601** is internal or external to a processor, for example, or comprised of DRAM, SRAM, Flash or some other storage type may depend on available storage on-chip versus off-chip, latency requirements of training and/or inferencing functions being performed, batch size of data used in inferencing and/or training of a neural network, or some combination of these factors.

In at least one embodiment, inference and/or training logic **615** may include, without limitation, a code and/or data storage **605** to store backward and/or output weight and/or input/output data corresponding to neurons or layers of a neural network trained and/or used for inferencing in aspects of one or more embodiments. In at least one embodiment, code and/or data storage **605** stores weight parameters and/or input/output data of each layer of a neural network trained or used in conjunction with one or more embodiments during backward propagation of input/output data and/or weight parameters during training and/or inferencing using aspects of one or more embodiments. In at least one embodiment, training logic **615** may include, or be coupled to code and/or data storage **605** to store graph code or other software to control timing and/or order, in which weight and/or other parameter information is to be loaded to configure, logic, including integer and/or floating point units (collectively, arithmetic logic units (ALUs)). In at least one embodiment, code, such as graph code, loads weight or other parameter information into processor ALUs based on an architecture of a neural network to which this code corresponds. In at least one embodiment, any portion of code and/or data storage **605** may be included with other on-chip or off-chip data storage, including a processor's L1, L2, or L3 cache or system memory. In at least one embodiment, any portion of code and/or data storage **605** may be internal or external to one or more processors or other hardware logic devices or circuits. In at least one embodiment, code and/or data storage **605** may be cache memory, DRAM,

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SRAM, non-volatile memory (e.g., Flash memory), or other storage. In at least one embodiment, choice of whether code and/or data storage **605** is internal or external to a processor, for example, or comprised of DRAM, SRAM, Flash or some other storage type may depend on available storage on-chip versus off-chip, latency requirements of training and/or inferencing functions being performed, batch size of data used in inferencing and/or training of a neural network, or some combination of these factors.

In at least one embodiment, code and/or data storage **601** and code and/or data storage **605** may be separate storage structures. In at least one embodiment, code and/or data storage **601** and code and/or data storage **605** may be same storage structure. In at least one embodiment, code and/or data storage **601** and code and/or data storage **605** may be partially same storage structure and partially separate storage structures. In at least one embodiment, any portion of code and/or data storage **601** and code and/or data storage **605** may be included with other on-chip or off-chip data storage, including a processor's L1, L2, or L3 cache or system memory.

In at least one embodiment, inference and/or training logic **615** may include, without limitation, one or more arithmetic logic unit(s) ("ALU(s)") **610**, including integer and/or floating point units, to perform logical and/or mathematical operations based, at least in part on, or indicated by, training and/or inference code (e.g., graph code), a result of which may produce activations (e.g., output values from layers or neurons within a neural network) stored in an activation storage **620** that are functions of input/output and/or weight parameter data stored in code and/or data storage **601** and/or code and/or data storage **605**. In at least one embodiment, activations stored in activation storage **620** are generated according to linear algebraic and or matrix-based mathematics performed by ALU(s) **610** in response to performing instructions or other code, wherein weight values stored in code and/or data storage **605** and/or code and/or data storage **601** are used as operands along with other values, such as bias values, gradient information, momentum values, or other parameters or hyperparameters, any or all of which may be stored in code and/or data storage **605** or code and/or data storage **601** or another storage on or off-chip.

In at least one embodiment, ALU(s) **610** are included within one or more processors or other hardware logic devices or circuits, whereas in another embodiment, ALU(s) **610** may be external to a processor or other hardware logic device or circuit that uses them (e.g., a co-processor). In at least one embodiment, ALUs **610** may be included within a processor's execution units or otherwise within a bank of ALUs accessible by a processor's execution units either within same processor or distributed between different processors of different types (e.g., central processing units, graphics processing units, fixed function units, etc.). In at least one embodiment, code and/or data storage **601**, code and/or data storage **605**, and activation storage **620** may be on same processor or other hardware logic device or circuit, whereas in another embodiment, they may be in different processors or other hardware logic devices or circuits, or some combination of same and different processors or other hardware logic devices or circuits. In at least one embodiment, any portion of activation storage **620** may be included with other on-chip or off-chip data storage, including a processor's L1, L2, or L3 cache or system memory. Furthermore, inferencing and/or training code may be stored with other code accessible to a processor or other hardware

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logic or circuit and fetched and/or processed using a processor's fetch, decode, scheduling, execution, retirement and/or other logical circuits.

In at least one embodiment, activation storage **620** may be cache memory, DRAM, SRAM, non-volatile memory (e.g., Flash memory), or other storage. In at least one embodiment, activation storage **620** may be completely or partially within or external to one or more processors or other logical circuits. In at least one embodiment, choice of whether activation storage **620** is internal or external to a processor, for example, or comprised of DRAM, SRAM, Flash or some other storage type may depend on available storage on-chip versus off-chip, latency requirements of training and/or inferencing functions being performed, batch size of data used in inferencing and/or training of a neural network, or some combination of these factors. In at least one embodiment, inference and/or training logic **615** illustrated in FIG. **6A** may be used in conjunction with an application-specific integrated circuit ("ASIC"), such as Tensorflow® Processing Unit from Google, an inference processing unit (IPU) from Graphcore™, or a Nervana® (e.g., "Lake Crest") processor from Intel Corp. In at least one embodiment, inference and/or training logic **615** illustrated in FIG. **6A** may be used in conjunction with central processing unit ("CPU") hardware, graphics processing unit ("GPU") hardware or other hardware, such as field programmable gate arrays ("FPGAs").

FIG. **6B** illustrates inference and/or training logic **615**, according to at least one or more embodiments. In at least one embodiment, inference and/or training logic **615** may include, without limitation, hardware logic in which computational resources are dedicated or otherwise exclusively used in conjunction with weight values or other information corresponding to one or more layers of neurons within a neural network. In at least one embodiment, inference and/or training logic **615** illustrated in FIG. **6B** may be used in conjunction with an application-specific integrated circuit (ASIC), such as Tensorflow® Processing Unit from Google, an inference processing unit (IPU) from Graphcore™, or a Nervana® (e.g., "Lake Crest") processor from Intel Corp. In at least one embodiment, inference and/or training logic **615** illustrated in FIG. **6B** may be used in conjunction with central processing unit (CPU) hardware, graphics processing unit (GPU) hardware or other hardware, such as field programmable gate arrays (FPGAs). In at least one embodiment, inference and/or training logic **615** includes, without limitation, code and/or data storage **601** and code and/or data storage **605**, which may be used to store code (e.g., graph code), weight values and/or other information, including bias values, gradient information, momentum values, and/or other parameter or hyperparameter information. In at least one embodiment illustrated in FIG. **6B**, each of code and/or data storage **601** and code and/or data storage **605** is associated with a dedicated computational resource, such as computational hardware **602** and computational hardware **606**, respectively. In at least one embodiment, each of computational hardware **602** and computational hardware **606** comprises one or more ALUs that perform mathematical functions, such as linear algebraic functions, only on information stored in code and/or data storage **601** and code and/or data storage **605**, respectively, result of which is stored in activation storage **620**.

In at least one embodiment, each of code and/or data storage **601** and **605** and corresponding computational hardware **602** and **606**, respectively, correspond to different layers of a neural network, such that resulting activation from one "storage/computational pair **601/602**" of code

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and/or data storage **601** and computational hardware **602** is provided as an input to "storage/computational pair **605/606**" of code and/or data storage **605** and computational hardware **606**, in order to mirror conceptual organization of a neural network. In at least one embodiment, each of storage/computational pairs **601/602** and **605/606** may correspond to more than one neural network layer. In at least one embodiment, additional storage/computation pairs (not shown) subsequent to or in parallel with storage computation pairs **601/602** and **605/606** may be included in inference and/or training logic **615**.

Data Center

FIG. **7** illustrates an example data center **700**, in which at least one embodiment may be used. In at least one embodiment, data center **700** includes a data center infrastructure layer **710**, a framework layer **720**, a software layer **730**, and an application layer **740**.

In at least one embodiment, as shown in FIG. **7**, data center infrastructure layer **710** may include a resource orchestrator **712**, grouped computing resources **714**, and node computing resources ("node C.R.s") **716(1)-716(N)**, where "N" represents any whole, positive integer. In at least one embodiment, node C.R.s **716(1)-716(N)** may include, but are not limited to, any number of central processing units ("CPUs") or other processors (including accelerators, field programmable gate arrays (FPGAs), graphics processors, etc.), memory devices (e.g., dynamic read-only memory), storage devices (e.g., solid state or disk drives), network input/output ("NW I/O") devices, network switches, virtual machines ("VMs"), power modules, and cooling modules, etc. In at least one embodiment, one or more node C.R.s from among node C.R.s **716(1)-716(N)** may be a server having one or more of above-mentioned computing resources.

In at least one embodiment, grouped computing resources **714** may include separate groupings of node C.R.s housed within one or more racks (not shown), or many racks housed in data centers at various geographical locations (also not shown). Separate groupings of node C.R.s within grouped computing resources **714** may include grouped compute, network, memory or storage resources that may be configured or allocated to support one or more workloads. In at least one embodiment, several node C.R.s including CPUs or processors may grouped within one or more racks to provide compute resources to support one or more workloads. In at least one embodiment, one or more racks may also include any number of power modules, cooling modules, and network switches, in any combination.

In at least one embodiment, resource orchestrator **712** may configure or otherwise control one or more node C.R.s **716(1)-716(N)** and/or grouped computing resources **714**. In at least one embodiment, resource orchestrator **712** may include a software design infrastructure ("SDI") management entity for data center **700**. In at least one embodiment, resource orchestrator may include hardware, software or some combination thereof.

In at least one embodiment, as shown in FIG. **7**, framework layer **720** includes a job scheduler **722**, a configuration manager **724**, a resource manager **726** and a distributed file system **728**. In at least one embodiment, framework layer **720** may include a framework to support software **732** of software layer **730** and/or one or more application(s) **742** of application layer **740**. In at least one embodiment, software **732** or application(s) **742** may respectively include web-based service software or applications, such as those provided by Amazon Web Services, Google Cloud and Microsoft Azure. In at least one embodiment, framework

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layer 720 may be, but is not limited to, a type of free and open-source software web application framework such as Apache Spark™ (hereinafter “Spark”) that may utilize distributed file system 728 for large-scale data processing (e.g., “big data”). In at least one embodiment, job scheduler 722 may include a Spark driver to facilitate scheduling of workloads supported by various layers of data center 700. In at least one embodiment, configuration manager 724 may be capable of configuring different layers such as software layer 730 and framework layer 720 including Spark and distributed file system 728 for supporting large-scale data processing. In at least one embodiment, resource manager 726 may be capable of managing clustered or grouped computing resources mapped to or allocated for support of distributed file system 728 and job scheduler 722. In at least one embodiment, clustered or grouped computing resources may include grouped computing resource 714 at data center infrastructure layer 710. In at least one embodiment, resource manager 726 may coordinate with resource orchestrator 712 to manage these mapped or allocated computing resources.

In at least one embodiment, software 732 included in software layer 730 may include software used by at least portions of node C.R.s 716(1)-716(N), grouped computing resources 714, and/or distributed file system 728 of framework layer 720. one or more types of software may include, but are not limited to, Internet web page search software, e-mail virus scan software, database software, and streaming video content software.

In at least one embodiment, application(s) 742 included in application layer 740 may include one or more types of applications used by at least portions of node C.R.s 716(1)-716(N), grouped computing resources 714, and/or distributed file system 728 of framework layer 720. One or more types of applications may include, but are not limited to, any number of a genomics application, a cognitive compute, and a machine learning application, including training or inferencing software, machine learning framework software (e.g., PyTorch, TensorFlow, Caffe, etc.) or other machine learning applications used in conjunction with one or more embodiments.

In at least one embodiment, any of configuration manager 724, resource manager 726, and resource orchestrator 712 may implement any number and type of self-modifying actions based on any amount and type of data acquired in any technically feasible fashion. In at least one embodiment, self-modifying actions may relieve a data center operator of data center 700 from making possibly bad configuration decisions and possibly avoiding underutilized and/or poor performing portions of a data center.

In at least one embodiment, data center 700 may include tools, services, software or other resources to train one or more machine learning models or predict or infer information using one or more machine learning models according to one or more embodiments described herein. For example, in at least one embodiment, a machine learning model may be trained by calculating weight parameters according to a neural network architecture using software and computing resources described above with respect to data center 700. In at least one embodiment, trained machine learning models corresponding to one or more neural networks may be used to infer or predict information using resources described above with respect to data center 700 by using weight parameters calculated through one or more training techniques described herein.

In at least one embodiment, data center may use CPUs, application-specific integrated circuits (ASICs), GPUs,

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FPGAs, or other hardware to perform training and/or inferencing using above-described resources. Moreover, one or more software and/or hardware resources described above may be configured as a service to allow users to train or performing inferencing of information, such as image recognition, speech recognition, or other artificial intelligence services.

Inference and/or training logic 615 are used to perform inferencing and/or training operations associated with one or more embodiments. Details regarding inference and/or training logic 615 are provided below in conjunction with FIGS. 6A and/or 6B. In at least one embodiment, inference and/or training logic 615 may be used in system FIG. 7 for inferencing or predicting operations based, at least in part, on weight parameters calculated using neural network training operations, neural network functions and/or architectures, or neural network use cases described herein.

Inference and/or training logic 615 are used to perform inferencing and/or training operations associated with one or more embodiments. In at least one embodiment, this logic can be used with components of these figures to generate one or more images with visual effects approximated from similar effects applied to one or more lower resolution versions of those images.

Computer Systems

FIG. 8 is a block diagram illustrating an exemplary computer system, which may be a system with interconnected devices and components, a system-on-a-chip (SOC) or some combination thereof 800 formed with a processor that may include execution units to execute an instruction, according to at least one embodiment. In at least one embodiment, computer system 800 may include, without limitation, a component, such as a processor 802 to employ execution units including logic to perform algorithms for process data, in accordance with present disclosure, such as in embodiment described herein. In at least one embodiment, computer system 800 may include processors, such as PENTIUM® Processor family, Xeon™, Itanium®, XScale™ and/or StrongARM™, Intel® Core™, or Intel® Nervana™ microprocessors available from Intel Corporation of Santa Clara, California, although other systems (including PCs having other microprocessors, engineering workstations, set-top boxes and like) may also be used. In at least one embodiment, computer system 800 may execute a version of WINDOWS® operating system available from Microsoft Corporation of Redmond, Wash., although other operating systems (UNIX and Linux for example), embedded software, and/or graphical user interfaces, may also be used.

Embodiments may be used in other devices such as handheld devices and embedded applications. Some examples of handheld devices include cellular phones, Internet Protocol devices, digital cameras, personal digital assistants (“PDAs”), and handheld PCs. In at least one embodiment, embedded applications may include a microcontroller, a digital signal processor (“DSP”), system on a chip, network computers (“NetPCs”), set-top boxes, network hubs, wide area network (“WAN”) switches, or any other system that may perform one or more instructions in accordance with at least one embodiment.

In at least one embodiment, computer system 800 may include, without limitation, processor 802 that may include, without limitation, one or more execution units 808 to perform machine learning model training and/or inferencing according to techniques described herein. In at least one embodiment, computer system 800 is a single processor desktop or server system, but in another embodiment com-

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puter system **800** may be a multiprocessor system. In at least one embodiment, processor **802** may include, without limitation, a complex instruction set computer (“CISC”) microprocessor, a reduced instruction set computing (“RISC”) microprocessor, a very long instruction word (“VLIW”) microprocessor, a processor implementing a combination of instruction sets, or any other processor device, such as a digital signal processor, for example. In at least one embodiment, processor **802** may be coupled to a processor bus **810** that may transmit data signals between processor **802** and other components in computer system **800**.

In at least one embodiment, processor **802** may include, without limitation, a Level 1 (“L1”) internal cache memory (“cache”) **804**. In at least one embodiment, processor **802** may have a single internal cache or multiple levels of internal cache. In at least one embodiment, cache memory may reside external to processor **802**. Other embodiments may also include a combination of both internal and external caches depending on particular implementation and needs. In at least one embodiment, register file **806** may store different types of data in various registers including, without limitation, integer registers, floating point registers, status registers, and instruction pointer register.

In at least one embodiment, execution unit **808**, including, without limitation, logic to perform integer and floating point operations, also resides in processor **802**. In at least one embodiment, processor **802** may also include a microcode (“ucode”) read only memory (“ROM”) that stores microcode for certain macro instructions. In at least one embodiment, execution unit **808** may include logic to handle a packed instruction set **809**. In at least one embodiment, by including packed instruction set **809** in an instruction set of a general-purpose processor **802**, along with associated circuitry to execute instructions, operations used by many multimedia applications may be performed using packed data in a general-purpose processor **802**. In one or more embodiments, many multimedia applications may be accelerated and executed more efficiently by using full width of a processor’s data bus for performing operations on packed data, which may eliminate need to transfer smaller units of data across processor’s data bus to perform one or more operations one data element at a time.

In at least one embodiment, execution unit **808** may also be used in microcontrollers, embedded processors, graphics devices, DSPs, and other types of logic circuits. In at least one embodiment, computer system **800** may include, without limitation, a memory **820**. In at least one embodiment, memory **820** may be implemented as a Dynamic Random Access Memory (“DRAM”) device, a Static Random Access Memory (“SRAM”) device, flash memory device, or other memory device. In at least one embodiment, memory **820** may store instruction(s) **819** and/or data **821** represented by data signals that may be executed by processor **802**.

In at least one embodiment, system logic chip may be coupled to processor bus **810** and memory **820**. In at least one embodiment, system logic chip may include, without limitation, a memory controller hub (“MCH”) **816**, and processor **802** may communicate with MCH **816** via processor bus **810**. In at least one embodiment, MCH **816** may provide a high bandwidth memory path **818** to memory **820** for instruction and data storage and for storage of graphics commands, data and textures. In at least one embodiment, MCH **816** may direct data signals between processor **802**, memory **820**, and other components in computer system **800** and to bridge data signals between processor bus **810**, memory **820**, and a system I/O **822**. In at least one embodiment, system logic chip may provide a graphics port for

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coupling to a graphics controller. In at least one embodiment, MCH **816** may be coupled to memory **820** through a high bandwidth memory path **818** and graphics/video card **812** may be coupled to MCH **816** through an Accelerated Graphics Port (“AGP”) interconnect **814**.

In at least one embodiment, computer system **800** may use system I/O **822** that is a proprietary hub interface bus to couple MCH **816** to I/O controller hub (“ICH”) **830**. In at least one embodiment, ICH **830** may provide direct connections to some I/O devices via a local I/O bus. In at least one embodiment, local I/O bus may include, without limitation, a high-speed I/O bus for connecting peripherals to memory **820**, chipset, and processor **802**. Examples may include, without limitation, an audio controller **829**, a firmware hub (“flash BIOS”) **828**, a wireless transceiver **826**, a data storage **824**, a legacy I/O controller **823** containing user input and keyboard interfaces **825**, a serial expansion port **827**, such as Universal Serial Bus (“USB”), and a network controller **834**. data storage **824** may comprise a hard disk drive, a floppy disk drive, a CD-ROM device, a flash memory device, or other mass storage device.

In at least one embodiment, FIG. **8** illustrates a system, which includes interconnected hardware devices or “chips”, whereas in other embodiments, FIG. **8** may illustrate an exemplary System on a Chip (“SoC”). In at least one embodiment, devices illustrated in FIG. **8** may be interconnected with proprietary interconnects, standardized interconnects (e.g., PCIe) or some combination thereof. In at least one embodiment, one or more components of computer system **800** are interconnected using compute express link (CXL) interconnects.

Inference and/or training logic **615** are used to perform inferencing and/or training operations associated with one or more embodiments. Details regarding inference and/or training logic **615** are provided below in conjunction with FIGS. **6A** and/or **6B**. In at least one embodiment, inference and/or training logic **615** may be used in system FIG. **8** for inferencing or predicting operations based, at least in part, on weight parameters calculated using neural network training operations, neural network functions and/or architectures, or neural network use cases described herein.

Inference and/or training logic **615** are used to perform inferencing and/or training operations associated with one or more embodiments. In at least one embodiment, this logic can be used with components of these figures to generate one or more images with visual effects approximated from similar effects applied to one or more lower resolution versions of those images.

FIG. **9** is a block diagram illustrating an electronic device **900** for utilizing a processor **910**, according to at least one embodiment. In at least one embodiment, electronic device **900** may be, for example and without limitation, a notebook, a tower server, a rack server, a blade server, a laptop, a desktop, a tablet, a mobile device, a phone, an embedded computer, or any other suitable electronic device.

In at least one embodiment, system **900** may include, without limitation, processor **910** communicatively coupled to any suitable number or kind of components, peripherals, modules, or devices. In at least one embodiment, processor **910** coupled using a bus or interface, such as a 1° C. bus, a System Management Bus (“SMBus”), a Low Pin Count (LPC) bus, a Serial Peripheral Interface (“SPI”), a High Definition Audio (“HDA”) bus, a Serial Advance Technology Attachment (“SATA”) bus, a Universal Serial Bus (“USB”) (versions 1, 2, 3), or a Universal Asynchronous Receiver/Transmitter (“UART”) bus. In at least one embodiment, FIG. **9** illustrates a system, which includes intercon-

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nected hardware devices or “chips”, whereas in other embodiments, FIG. 9 may illustrate an exemplary System on a Chip (“SoC”). In at least one embodiment, devices illustrated in FIG. 9 may be interconnected with proprietary interconnects, standardized interconnects (e.g., PCIe) or some combination thereof. In at least one embodiment, one or more components of FIG. 9 are interconnected using compute express link (CXL) interconnects.

In at least one embodiment, FIG. 9 may include a display 924, a touch screen 925, a touch pad 930, a Near Field Communications unit (“NFC”) 945, a sensor hub 940, a thermal sensor 946, an Express Chipset (“EC”) 935, a Trusted Platform Module (“TPM”) 938, BIOS/firmware/flash memory (“BIOS, FW Flash”) 922, a DSP 960, a drive 920 such as a Solid State Disk (“SSD”) or a Hard Disk Drive (“HDD”), a wireless local area network unit (“WLAN”) 950, a Bluetooth unit 952, a Wireless Wide Area Network unit (“WWAN”) 956, a Global Positioning System (GPS) 955, a camera (“USB 3.0 camera”) 954 such as a USB 3.0 camera, and/or a Low Power Double Data Rate (“LPDDR”) memory unit (“LPDDR3”) 915 implemented in, for example, LPDDR3 standard. These components may each be implemented in any suitable manner.

In at least one embodiment, other components may be communicatively coupled to processor 910 through components discussed above. In at least one embodiment, an accelerometer 941, Ambient Light Sensor (“ALS”) 942, compass 943, and a gyroscope 944 may be communicatively coupled to sensor hub 940. In at least one embodiment, thermal sensor 939, a fan 937, a keyboard 946, and a touch pad 930 may be communicatively coupled to EC 935. In at least one embodiment, speaker 963, headphones 964, and microphone (“mic”) 965 may be communicatively coupled to an audio unit (“audio codec and class d amp”) 962, which may in turn be communicatively coupled to DSP 960. In at least one embodiment, audio unit 964 may include, for example and without limitation, an audio coder/decoder (“codec”) and a class D amplifier. In at least one embodiment, SIM card (“SIM”) 957 may be communicatively coupled to WWAN unit 956. In at least one embodiment, components such as WLAN unit 950 and Bluetooth unit 952, as well as WWAN unit 956 may be implemented in a Next Generation Form Factor (“NGFF”).

Inference and/or training logic 615 are used to perform inferencing and/or training operations associated with one or more embodiments. Details regarding inference and/or training logic 615 are provided below in conjunction with FIGS. 6A and/or 6B. In at least one embodiment, inference and/or training logic 615 may be used in system FIG. 9 for inferencing or predicting operations based, at least in part, on weight parameters calculated using neural network training operations, neural network functions and/or architectures, or neural network use cases described herein.

Inference and/or training logic 615 are used to perform inferencing and/or training operations associated with one or more embodiments. In at least one embodiment, this logic can be used with components of these figures to generate one or more images with visual effects approximated from similar effects applied to one or more lower resolution versions of those images.

FIG. 10 illustrates a computer system 1000, according to at least one embodiment. In at least one embodiment, computer system 1000 is configured to implement various processes and methods described throughout this disclosure.

In at least one embodiment, computer system 1000 comprises, without limitation, at least one central processing unit (“CPU”) 1002 that is connected to a communication bus

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1010 implemented using any suitable protocol, such as PCI (“Peripheral Component Interconnect”), peripheral component interconnect express (“PCI-Express”), AGP (“Accelerated Graphics Port”), HyperTransport, or any other bus or point-to-point communication protocol(s). In at least one embodiment, computer system 1000 includes, without limitation, a main memory 1004 and control logic (e.g., implemented as hardware, software, or a combination thereof) and data are stored in main memory 1004 which may take form of random access memory (“RAM”). In at least one embodiment, a network interface subsystem (“network interface”) 1022 provides an interface to other computing devices and networks for receiving data from and transmitting data to other systems from computer system 1000.

In at least one embodiment, computer system 1000, in at least one embodiment, includes, without limitation, input devices 1008, parallel processing system 1012, and display devices 1006 which can be implemented using a conventional cathode ray tube (“CRT”), liquid crystal display (“LCD”), light emitting diode (“LED”), plasma display, or other suitable display technologies. In at least one embodiment, user input is received from input devices 1008 such as keyboard, mouse, touchpad, microphone, and more. In at least one embodiment, each of foregoing modules can be situated on a single semiconductor platform to form a processing system.

Inference and/or training logic 615 are used to perform inferencing and/or training operations associated with one or more embodiments. Details regarding inference and/or training logic 615 are provided below in conjunction with FIGS. 6A and/or 6B. In at least one embodiment, inference and/or training logic 615 may be used in system FIG. 10 for inferencing or predicting operations based, at least in part, on weight parameters calculated using neural network training operations, neural network functions and/or architectures, or neural network use cases described herein.

Inference and/or training logic 615 are used to perform inferencing and/or training operations associated with one or more embodiments. In at least one embodiment, this logic can be used with components of these figures to generate one or more images with visual effects approximated from similar effects applied to one or more lower resolution versions of those images.

FIG. 11 illustrates a computer system 1100, according to at least one embodiment. In at least one embodiment, computer system 1100 includes, without limitation, a computer 1110 and a USB stick 1120. In at least one embodiment, computer 1110 may include, without limitation, any number and type of processor(s) (not shown) and a memory (not shown). In at least one embodiment, computer 1110 includes, without limitation, a server, a cloud instance, a laptop, and a desktop computer.

In at least one embodiment, USB stick 1120 includes, without limitation, a processing unit 1130, a USB interface 1140, and USB interface logic 1150. In at least one embodiment, processing unit 1130 may be any instruction execution system, apparatus, or device capable of executing instructions. In at least one embodiment, processing unit 1130 may include, without limitation, any number and type of processing cores (not shown). In at least one embodiment, processing core 1130 comprises an application specific integrated circuit (“ASIC”) that is optimized to perform any amount and type of operations associated with machine learning. For instance, in at least one embodiment, processing core 1130 is a tensor processing unit (“TPU”) that is optimized to perform machine learning inference operations. In at least one embodiment, processing core 1130 is a vision

processing unit (“VPU”) that is optimized to perform machine vision and machine learning inference operations.

In at least one embodiment, USB interface **1140** may be any type of USB connector or USB socket. For instance, in at least one embodiment, USB interface **1140** is a USB 3.0 Type-C socket for data and power. In at least one embodiment, USB interface **1140** is a USB 3.0 Type-A connector. In at least one embodiment, USB interface logic **1150** may include any amount and type of logic that enables processing unit **1130** to interface with or devices (e.g., computer **1110**) via USB connector **1140**.

Inference and/or training logic **615** are used to perform inferencing and/or training operations associated with one or more embodiments. Details regarding inference and/or training logic **615** are provided below in conjunction with FIGS. **6A** and/or **6B**. In at least one embodiment, inference and/or training logic **615** may be used in system FIG. **11** for inferencing or predicting operations based, at least in part, on weight parameters calculated using neural network training operations, neural network functions and/or architectures, or neural network use cases described herein.

Inference and/or training logic **615** are used to perform inferencing and/or training operations associated with one or more embodiments. In at least one embodiment, this logic can be used with components of these figures to generate one or more images with visual effects approximated from similar effects applied to one or more lower resolution versions of those images.

FIG. **12A** illustrates an exemplary architecture in which a plurality of GPUs **1210-1213** is communicatively coupled to a plurality of multi-core processors **1205-1206** over high-speed links **1240-1243** (e.g., buses, point-to-point interconnects, etc.). In one embodiment, high-speed links **1240-1243** support a communication throughput of 4 GB/s, 30 GB/s, 80 GB/s or higher. Various interconnect protocols may be used including, but not limited to, PCIe 4.0 or 5.0 and NVLink 2.0.

In addition, and in one embodiment, two or more of GPUs **1210-1213** are interconnected over high-speed links **1229-1230**, which may be implemented using same or different protocols/links than those used for high-speed links **1240-1243**. Similarly, two or more of multi-core processors **1205-1206** may be connected over high speed link **1228** which may be symmetric multi-processor (SMP) buses operating at 20 GB/s, 30 GB/s, 120 GB/s or higher. Alternatively, all communication between various system components shown in FIG. **12A** may be accomplished using same protocols/links (e.g., over a common interconnection fabric).

In one embodiment, each multi-core processor **1205-1206** is communicatively coupled to a processor memory **1201-1202**, via memory interconnects **1226-1227**, respectively, and each GPU **1210-1213** is communicatively coupled to GPU memory **1220-1223** over GPU memory interconnects **1250-1253**, respectively. Memory interconnects **1226-1227** and **1250-1253** may utilize same or different memory access technologies. By way of example, and not limitation, processor memories **1201-1202** and GPU memories **1220-1223** may be volatile memories such as dynamic random access memories (DRAMs) (including stacked DRAMs), Graphics DDR SDRAM (GDDR) (e.g., GDDR5, GDDR6), or High Bandwidth Memory (HBM) and/or may be non-volatile memories such as 3D(Point or Nano-Ram. In one embodiment, some portion of processor memories **1201-1202** may be volatile memory and another portion may be non-volatile memory (e.g., using a two-level memory (2LM) hierarchy).

As described below, although various processors **1205-1206** and GPUs **1210-1213** may be physically coupled to a

particular memory **1201-1202**, **1220-1223**, respectively, a unified memory architecture may be implemented in which a same virtual system address space (also referred to as “effective address” space) is distributed among various physical memories. For example, processor memories **1201-1202** may each comprise 64 GB of system memory address space and GPU memories **1220-1223** may each comprise 32 GB of system memory address space (resulting in a total of 256 GB addressable memory in this example).

FIG. **12B** illustrates additional details for an interconnection between a multi-core processor **1207** and a graphics acceleration module **1246** in accordance with one exemplary embodiment. Graphics acceleration module **1246** may include one or more GPU chips integrated on a line card which is coupled to processor **1207** via high-speed link **1240**. Alternatively, graphics acceleration module **1246** may be integrated on a same package or chip as processor **1207**.

In at least one embodiment, illustrated processor **1207** includes a plurality of cores **1260A-1260D**, each with a translation lookaside buffer **1261A-1261D** and one or more caches **1262A-1262D**. In at least one embodiment, cores **1260A-1260D** may include various other components for executing instructions and processing data which are not illustrated. Caches **1262A-1262D** may comprise level 1 (L1) and level 2 (L2) caches. In addition, one or more shared caches **1256** may be included in caches **1262A-1262D** and shared by sets of cores **1260A-1260D**. For example, one embodiment of processor **1207** includes 24 cores, each with its own L1 cache, twelve shared L2 caches, and twelve shared L3 caches. In this embodiment, one or more L2 and L3 caches are shared by two adjacent cores. Processor **1207** and graphics acceleration module **1246** connect with system memory **1214**, which may include processor memories **1201-1202** of FIG. **12A**.

Coherency is maintained for data and instructions stored in various caches **1262A-1262D**, **1256** and system memory **1214** via inter-core communication over a coherence bus **1264**. For example, each cache may have cache coherency logic/circuitry associated therewith to communicate to over coherence bus **1264** in response to detected reads or writes to particular cache lines. In one implementation, a cache snooping protocol is implemented over coherence bus **1264** to snoop cache accesses.

In one embodiment, a proxy circuit **1225** communicatively couples graphics acceleration module **1246** to coherence bus **1264**, allowing graphics acceleration module **1246** to participate in a cache coherence protocol as a peer of cores **1260A-1260D**. In particular, an interface **1235** provides connectivity to proxy circuit **1225** over high-speed link **1240** (e.g., a PCIe bus, NVLink, etc.) and an interface **1237** connects graphics acceleration module **1246** to link **1240**.

In one implementation, an accelerator integration circuit **1236** provides cache management, memory access, context management, and interrupt management services on behalf of a plurality of graphics processing engines **1231**, **1232**, N of graphics acceleration module **1246**. Graphics processing engines **1231**, **1232**, N may each comprise a separate graphics processing unit (GPU). Alternatively, graphics processing engines **1231**, **1232**, N may comprise different types of graphics processing engines within a GPU such as graphics execution units, media processing engines (e.g., video encoders/decoders), samplers, and blit engines. In at least one embodiment, graphics acceleration module **1246** may be a GPU with a plurality of graphics processing

engines **1231-1232**, N or graphics processing engines **1231-1232**, N may be individual GPUs integrated on a common package, line card, or chip.

In one embodiment, accelerator integration circuit **1236** includes a memory management unit (MMU) **1239** for performing various memory management functions such as virtual-to-physical memory translations (also referred to as effective-to-real memory translations) and memory access protocols for accessing system memory **1214**. MMU **1239** may also include a translation lookaside buffer (TLB) (not shown) for caching virtual/effective to physical/real address translations. In one implementation, a cache **1238** stores commands and data for efficient access by graphics processing engines **1231-1232**, N. In one embodiment, data stored in cache **1238** and graphics memories **1233-1234**, M is kept coherent with core caches **1262A-1262D**, **1256**, and system memory **1214**. As mentioned above, this may be accomplished via proxy circuit **1225** on behalf of cache **1238** and memories **1233-1234**, M (e.g., sending updates to cache **1238** related to modifications/accesses of cache lines on processor caches **1262A-1262D**, **1256**, and receiving updates from cache **1238**).

A set of registers **1245** store context data for threads executed by graphics processing engines **1231-1232**, N and a context management circuit **1248** manages thread contexts. For example, context management circuit **1248** may perform save and restore operations to save and restore contexts of various threads during contexts switches (e.g., where a first thread is saved and a second thread is stored so that a second thread can be executed by a graphics processing engine). For example, on a context switch, context management circuit **1248** may store current register values to a designated region in memory (e.g., identified by a context pointer). It may then restore register values when returning to a context. In one embodiment, an interrupt management circuit **1247** receives and processes interrupts received from system devices.

In one implementation, virtual/effective addresses from a graphics processing engine **1231** are translated to real/physical addresses in system memory **1214** by MMU **1239**. One embodiment of accelerator integration circuit **1236** supports multiple (e.g., 4, 8, 16) graphics accelerator modules **1246** and/or other accelerator devices. Graphics accelerator module **1246** may be dedicated to a single application executed on processor **1207** or may be shared between multiple applications. In one embodiment, a virtualized graphics execution environment is presented in which resources of graphics processing engines **1231-1232**, N are shared with multiple applications or virtual machines (VMs). In at least one embodiment, resources may be subdivided into “slices” which are allocated to different VMs and/or applications based on processing requirements and priorities associated with VMs and/or applications.

In at least one embodiment, accelerator integration circuit **1236** performs as a bridge to a system for graphics acceleration module **1246** and provides address translation and system memory cache services. In addition, accelerator integration circuit **1236** may provide virtualization facilities for a host processor to manage virtualization of graphics processing engines **1231-1232**, N, interrupts, and memory management.

Because hardware resources of graphics processing engines **1231-1232**, N are mapped explicitly to a real address space seen by host processor **1207**, any host processor can address these resources directly using an effective address value. One function of accelerator integration circuit

1236, in one embodiment, is physical separation of graphics processing engines **1231-1232**, N so that they appear to a system as independent units.

In at least one embodiment, one or more graphics memories **1233-1234**, M are coupled to each of graphics processing engines **1231-1232**, N, respectively. Graphics memories **1233-1234**, M store instructions and data being processed by each of graphics processing engines **1231-1232**, N. Graphics memories **1233-1234**, M may be volatile memories such as DRAMs (including stacked DRAMs), GDDR memory (e.g., GDDR5, GDDR6), or HBM, and/or may be non-volatile memories such as 3D XPoint or Nano-Ram.

In one embodiment, to reduce data traffic over link **1240**, biasing techniques are used to ensure that data stored in graphics memories **1233-1234**, M is data which will be used most frequently by graphics processing engines **1231-1232**, N and preferably not used by cores **1260A-1260D** (at least not frequently). Similarly, a biasing mechanism attempts to keep data needed by cores (and preferably not graphics processing engines **1231-1232**, N) within caches **1262A-1262D**, **1256** of cores and system memory **1214**.

FIG. **12C** illustrates another exemplary embodiment in which accelerator integration circuit **1236** is integrated within processor **1207**. In at least this embodiment, graphics processing engines **1231-1232**, N communicate directly over high-speed link **1240** to accelerator integration circuit **1236** via interface **1237** and interface **1235** (which, again, may be utilize any form of bus or interface protocol). Accelerator integration circuit **1236** may perform same operations as those described with respect to FIG. **12B**, but potentially at a higher throughput given its close proximity to coherence bus **1264** and caches **1262A-1262D**, **1256**. At least one embodiment supports different programming models including a dedicated-process programming model (no graphics acceleration module virtualization) and shared programming models (with virtualization), which may include programming models which are controlled by accelerator integration circuit **1236** and programming models which are controlled by graphics acceleration module **1246**.

In at least one embodiment, graphics processing engines **1231-1232**, N are dedicated to a single application or process under a single operating system. In at least one embodiment, a single application can funnel other application requests to graphics processing engines **1231-1232**, N, providing virtualization within a VM/partition.

In at least one embodiment, graphics processing engines **1231-1232**, N, may be shared by multiple VM/application partitions. In at least one embodiment, shared models may use a system hypervisor to virtualize graphics processing engines **1231-1232**, N to allow access by each operating system. For single-partition systems without a hypervisor, graphics processing engines **1231-1232**, N are owned by an operating system. In at least one embodiment, an operating system can virtualize graphics processing engines **1231-1232**, N to provide access to each process or application.

In at least one embodiment, graphics acceleration module **1246** or an individual graphics processing engine **1231-1232**, N selects a process element using a process handle. In at least one embodiment, process elements are stored in system memory **1214** and are addressable using an effective address to real address translation techniques described herein. In at least one embodiment, a process handle may be an implementation-specific value provided to a host process when registering its context with graphics processing engine **1231-1232**, N (that is, calling system software to add a process element to a process element linked list). In at least

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one embodiment, a lower 16-bits of a process handle may be an offset of a process element within a process element linked list.

FIG. 12D illustrates an exemplary accelerator integration slice 1290. As used herein, a “slice” comprises a specified portion of processing resources of accelerator integration circuit 1236. Application effective address space 1282 within system memory 1214 stores process elements 1283. In one embodiment, process elements 1283 are stored in response to GPU invocations 1281 from applications 1280 executed on processor 1207. A process element 1283 contains process state for corresponding application 1280. A work descriptor (WD) 1284 contained in process element 1283 can be a single job requested by an application or may contain a pointer to a queue of jobs. In at least one embodiment, WD 1284 is a pointer to a job request queue in an application’s address space 1282.

Graphics acceleration module 1246 and/or individual graphics processing engines 1231-1232, N can be shared by all or a subset of processes in a system. In at least one embodiment, an infrastructure for setting up process state and sending a WD 1284 to a graphics acceleration module 1246 to start a job in a virtualized environment may be included.

In at least one embodiment, a dedicated-process programming model is implementation-specific. In this model, a single process owns graphics acceleration module 1246 or an individual graphics processing engine 1231. Because graphics acceleration module 1246 is owned by a single process, a hypervisor initializes accelerator integration circuit 1236 for an owning partition and an operating system initializes accelerator integration circuit 1236 for an owning process when graphics acceleration module 1246 is assigned.

In operation, a WD fetch unit 1291 in accelerator integration slice 1290 fetches next WD 1284 which includes an indication of work to be done by one or more graphics processing engines of graphics acceleration module 1246. Data from WD 1284 may be stored in registers 1245 and used by MMU 1239, interrupt management circuit 1247, and/or context management circuit 1248 as illustrated. For example, one embodiment of MMU 1239 includes segment/page walk circuitry for accessing segment/page tables 1286 within OS virtual address space 1285. Interrupt management circuit 1247 may process interrupt events 1292 received from graphics acceleration module 1246. When performing graphics operations, an effective address 1293 generated by a graphics processing engine 1231-1232, N is translated to a real address by MMU 1239.

In one embodiment, a same set of registers 1245 are duplicated for each graphics processing engine 1231-1232, N and/or graphics acceleration module 1246 and may be initialized by a hypervisor or operating system. Each of these duplicated registers may be included in an accelerator integration slice 1290. Exemplary registers that may be initialized by a hypervisor are shown in Table 1.

TABLE 1

Hypervisor Initialized Registers	
1	Slice Control Register
2	Real Address (RA) Scheduled Processes Area Pointer
3	Authority Mask Override Register
4	Interrupt Vector Table Entry Offset
5	Interrupt Vector Table Entry Limit
6	State Register
7	Logical Partition ID

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TABLE 1-continued

Hypervisor Initialized Registers	
8	Real address (RA) Hypervisor Accelerator Utilization Record Pointer
9	Storage Description Register

Exemplary registers that may be initialized by an operating system are shown in Table 2.

TABLE 2

Operating System Initialized Registers	
1	Process and Thread Identification
2	Effective Address (EA) Context Save/Restore Pointer
3	Virtual Address (VA) Accelerator Utilization Record Pointer
4	Virtual Address (VA) Storage Segment Table Pointer
5	Authority Mask
6	Work descriptor

In one embodiment, each WD 1284 is specific to a particular graphics acceleration module 1246 and/or graphics processing engines 1231-1232, N. It contains all information required by a graphics processing engine 1231-1232, N to do work or it can be a pointer to a memory location where an application has set up a command queue of work to be completed.

FIG. 12E illustrates additional details for one exemplary embodiment of a shared model. This embodiment includes a hypervisor real address space 1298 in which a process element list 1299 is stored. Hypervisor real address space 1298 is accessible via a hypervisor 1296 which virtualizes graphics acceleration module engines for operating system 1295.

In at least one embodiment, shared programming models allow for all or a subset of processes from all or a subset of partitions in a system to use a graphics acceleration module 1246. There are two programming models where graphics acceleration module 1246 is shared by multiple processes and partitions: time-sliced shared and graphics-directed shared.

In this model, system hypervisor 1296 owns graphics acceleration module 1246 and makes its function available to all operating systems 1295. For a graphics acceleration module 1246 to support virtualization by system hypervisor 1296, graphics acceleration module 1246 may adhere to the following: 1) An application’s job request must be autonomous (that is, state does not need to be maintained between jobs), or graphics acceleration module 1246 must provide a context save and restore mechanism. 2) An application’s job request is guaranteed by graphics acceleration module 1246 to complete in a specified amount of time, including any translation faults, or graphics acceleration module 1246 provides an ability to preempt processing of a job. 3) Graphics acceleration module 1246 must be guaranteed fairness between processes when operating in a directed shared programming model.

In at least one embodiment, application 1280 is required to make an operating system 1295 system call with a graphics acceleration module 1246 type, a work descriptor (WD), an authority mask register (AMR) value, and a context save/restore area pointer (CSRP). In at least one embodiment, graphics acceleration module 1246 type describes a targeted acceleration function for a system call. In at least one embodiment, graphics acceleration module 1246 type may be a system-specific value. In at least one embodiment, WD is formatted specifically for graphics

acceleration module **1246** and can be in a form of a graphics acceleration module **1246** command, an effective address pointer to a user-defined structure, an effective address pointer to a queue of commands, or any other data structure to describe work to be done by graphics acceleration module **1246**. In one embodiment, an AMR value is an AMR state to use for a current process. In at least one embodiment, a value passed to an operating system is similar to an application setting an AMR. If accelerator integration circuit **1236** and graphics acceleration module **1246** implementations do not support a User Authority Mask Override Register (UAMOR), an operating system may apply a current UAMOR value to an AMR value before passing an AMR in a hypervisor call. Hypervisor **1296** may optionally apply a current Authority Mask Override Register (AMOR) value before placing an AMR into process element **1283**. In at least one embodiment, CSRP is one of registers **1245** containing an effective address of an area in an application's effective address space **1282** for graphics acceleration module **1246** to save and restore context state. This pointer is optional if no state is required to be saved between jobs or when a job is preempted. In at least one embodiment, context save/restore area may be pinned system memory.

Upon receiving a system call, operating system **1295** may verify that application **1280** has registered and been given authority to use graphics acceleration module **1246**. Operating system **1295** then calls hypervisor **1296** with information shown in Table 3.

TABLE 3

OS to Hypervisor Call Parameters	
1	A work descriptor (WD)
2	An Authority Mask Register (AMR) value (potentially masked)
3	An effective address (EA) Context Save/Restore Area Pointer (CSRP)
4	A process ID (PID) and optional thread ID (TID)
5	A virtual address (VA) accelerator utilization record pointer (AURP)
6	Virtual address of storage segment table pointer (SSTP)
7	A logical interrupt service number (LISN)

Upon receiving a hypervisor call, hypervisor **1296** verifies that operating system **1295** has registered and been given authority to use graphics acceleration module **1246**. Hypervisor **1296** then puts process element **1283** into a process element linked list for a corresponding graphics acceleration module **1246** type. A process element may include information shown in Table 4.

TABLE 4

Process Element Information	
1	A work descriptor (WD)
2	An Authority Mask Register (AMR) value (potentially masked).
3	An effective address (EA) Context Save/Restore Area Pointer (CSRP)
4	A process ID (PID) and optional thread ID (TID)
5	A virtual address (VA) accelerator utilization record pointer (AURP)
6	Virtual address of storage segment table pointer (SSTP)
7	A logical interrupt service number (LISN)
8	Interrupt vector table, derived from hypervisor call parameters
9	A state register (SR) value
10	A logical partition ID (LPID)
11	A real address (RA) hypervisor accelerator utilization record pointer
12	Storage Descriptor Register (SDR)

In at least one embodiment, hypervisor initializes a plurality of accelerator integration slice **1290** registers **1245**.

As illustrated in FIG. **12F**, in at least one embodiment, a unified memory is used, addressable via a common virtual memory address space used to access physical processor memories **1201-1202** and GPU memories **1220-1223**. In this implementation, operations executed on GPUs **1210-1213** utilize a same virtual/effective memory address space to access processor memories **1201-1202** and vice versa, thereby simplifying programmability. In one embodiment, a first portion of a virtual/effective address space is allocated to processor memory **1201**, a second portion to second processor memory **1202**, a third portion to GPU memory **1220**, and so on. In at least one embodiment, an entire virtual/effective memory space (sometimes referred to as an effective address space) is thereby distributed across each of processor memories **1201-1202** and GPU memories **1220-1223**, allowing any processor or GPU to access any physical memory with a virtual address mapped to that memory.

In one embodiment, bias/coherence management circuitry **1294A-1294E** within one or more of MMUs **1239A-1239E** ensures cache coherence between caches of one or more host processors (e.g., **1205**) and GPUs **1210-1213** and implements biasing techniques indicating physical memories in which certain types of data should be stored. While multiple instances of bias/coherence management circuitry **1294A-1294E** are illustrated in FIG. **12F**, bias/coherence circuitry may be implemented within an MMU of one or more host processors **1205** and/or within accelerator integration circuit **1236**.

One embodiment allows GPU-attached memory **1220-1223** to be mapped as part of system memory, and accessed using shared virtual memory (SVM) technology, but without suffering performance drawbacks associated with full system cache coherence. In at least one embodiment, an ability for GPU-attached memory **1220-1223** to be accessed as system memory without onerous cache coherence overhead provides a beneficial operating environment for GPU offload. This arrangement allows host processor **1205** software to setup operands and access computation results, without overhead of tradition I/O DMA data copies. Such traditional copies involve driver calls, interrupts and memory mapped I/O (MMIO) accesses that are all inefficient relative to simple memory accesses. In at least one embodiment, an ability to access GPU attached memory **1220-1223** without cache coherence overheads can be critical to execution time of an offloaded computation. In cases with substantial streaming write memory traffic, for example, cache coherence overhead can significantly reduce an effective write bandwidth seen by a GPU **1210-1213**. In at least one embodiment, efficiency of operand setup, efficiency of results access, and efficiency of GPU computation may play a role in determining effectiveness of a GPU offload.

In at least one embodiment, selection of GPU bias and host processor bias is driven by a bias tracker data structure. A bias table may be used, for example, which may be a page-granular structure (i.e., controlled at a granularity of a memory page) that includes 1 or 2 bits per GPU-attached memory page. In at least one embodiment, a bias table may be implemented in a stolen memory range of one or more GPU-attached memories **1220-1223**, with or without a bias cache in GPU **1210-1213** (e.g., to cache frequently/recently used entries of a bias table). Alternatively, an entire bias table may be maintained within a GPU.

In at least one embodiment, a bias table entry associated with each access to GPU-attached memory **1220-1223** is accessed prior to actual access to a GPU memory, causing the following operations. First, local requests from GPU **1210-1213** that find their page in GPU bias are forwarded

directly to a corresponding GPU memory **1220-1223**. Local requests from a GPU that find their page in host bias are forwarded to processor **1205** (e.g., over a high-speed link as discussed above). In one embodiment, requests from processor **1205** that find a requested page in host processor bias complete a request like a normal memory read. Alternatively, requests directed to a GPU-biased page may be forwarded to GPU **1210-1213**. In at least one embodiment, a GPU may then transition a page to a host processor bias if it is not currently using a page. In at least one embodiment, bias state of a page can be changed either by a software-based mechanism, a hardware-assisted software-based mechanism, or, for a limited set of cases, a purely hardware-based mechanism.

One mechanism for changing bias state employs an API call (e.g., OpenCL), which, in turn, calls a GPU's device driver which, in turn, sends a message (or enqueues a command descriptor) to a GPU directing it to change a bias state and, for some transitions, perform a cache flushing operation in a host. In at least one embodiment, cache flushing operation is used for a transition from host processor **1205** bias to GPU bias, but is not for an opposite transition.

In one embodiment, cache coherency is maintained by temporarily rendering GPU-biased pages uncacheable by host processor **1205**. To access these pages, processor **1205** may request access from GPU **1210** which may or may not grant access right away. Thus, to reduce communication between processor **1205** and GPU **1210** it is beneficial to ensure that GPU-biased pages are those which are required by a GPU but not host processor **1205** and vice versa.

Inference and/or training logic **615** are used to perform one or more embodiments. Details regarding the inference and/or training logic **615** are provided below in conjunction with FIGS. **6A** and/or **6B**.

Inference and/or training logic **615** are used to perform inferencing and/or training operations associated with one or more embodiments. In at least one embodiment, this logic can be used with components of these figures to generate one or more images with visual effects approximated from similar effects applied to one or more lower resolution versions of those images.

FIG. **13** illustrates exemplary integrated circuits and associated graphics processors that may be fabricated using one or more IP cores, according to various embodiments described herein. In addition to what is illustrated, other logic and circuits may be included in at least one embodiment, including additional graphics processors/cores, peripheral interface controllers, or general-purpose processor cores.

FIG. **13** is a block diagram illustrating an exemplary system on a chip integrated circuit **1300** that may be fabricated using one or more IP cores, according to at least one embodiment. In at least one embodiment, integrated circuit **1300** includes one or more application processor(s) **1305** (e.g., CPUs), at least one graphics processor **1310**, and may additionally include an image processor **1315** and/or a video processor **1320**, any of which may be a modular IP core. In at least one embodiment, integrated circuit **1300** includes peripheral or bus logic including a USB controller **1325**, UART controller **1330**, an SPI/SDIO controller **1335**, and an I²S/I²C controller **1340**. In at least one embodiment, integrated circuit **1300** can include a display device **1345** coupled to one or more of a high-definition multimedia interface (HDMI) controller **1350** and a mobile industry processor interface (MIPI) display interface **1355**. In at least one embodiment, storage may be provided by a flash

memory subsystem **1360** including flash memory and a flash memory controller. In at least one embodiment, memory interface may be provided via a memory controller **1365** for access to SDRAM or SRAM memory devices. In at least one embodiment, some integrated circuits additionally include an embedded security engine **1370**.

Inference and/or training logic **615** are used to perform inferencing and/or training operations associated with one or more embodiments. Details regarding inference and/or training logic **615** are provided below in conjunction with FIGS. **6A** and/or **6B**. In at least one embodiment, inference and/or training logic **615** may be used in integrated circuit **1300** for inferencing or predicting operations based, at least in part, on weight parameters calculated using neural network training operations, neural network functions and/or architectures, or neural network use cases described herein.

Inference and/or training logic **615** are used to perform inferencing and/or training operations associated with one or more embodiments. In at least one embodiment, this logic can be used with components of these figures to generate one or more images with visual effects approximated from similar effects applied to one or more lower resolution versions of those images.

FIGS. **14A-14B** illustrate exemplary integrated circuits and associated graphics processors that may be fabricated using one or more IP cores, according to various embodiments described herein. In addition to what is illustrated, other logic and circuits may be included in at least one embodiment, including additional graphics processors/cores, peripheral interface controllers, or general-purpose processor cores.

FIGS. **14A-14B** are block diagrams illustrating exemplary graphics processors for use within an SoC, according to embodiments described herein. FIG. **14A** illustrates an exemplary graphics processor **1410** of a system on a chip integrated circuit that may be fabricated using one or more IP cores, according to at least one embodiment. FIG. **14B** illustrates an additional exemplary graphics processor **1440** of a system on a chip integrated circuit that may be fabricated using one or more IP cores, according to at least one embodiment. In at least one embodiment, graphics processor **1410** of FIG. **14A** is a low power graphics processor core. In at least one embodiment, graphics processor **1440** of FIG. **14B** is a higher performance graphics processor core. In at least one embodiment, each of graphics processors **1410**, **1440** can be variants of graphics processor **1310** of FIG. **13**.

In at least one embodiment, graphics processor **1410** includes a vertex processor **1405** and one or more fragment processor(s) **1415A-1415N** (e.g., **1415A**, **1415B**, **1415C**, **1415D**, through **1415N-1**, and **1415N**). In at least one embodiment, graphics processor **1410** can execute different shader programs via separate logic, such that vertex processor **1405** is optimized to execute operations for vertex shader programs, while one or more fragment processor(s) **1415A-1415N** execute fragment (e.g., pixel) shading operations for fragment or pixel shader programs. In at least one embodiment, vertex processor **1405** performs a vertex processing stage of a 3D graphics pipeline and generates primitives and vertex data. In at least one embodiment, fragment processor(s) **1415A-1415N** use primitive and vertex data generated by vertex processor **1405** to produce a framebuffer that is displayed on a display device. In at least one embodiment, fragment processor(s) **1415A-1415N** are optimized to execute fragment shader programs as provided for in an OpenGL API, which may be used to perform similar operations as a pixel shader program as provided for in a Direct 3D API.

In at least one embodiment, graphics processor **1410** additionally includes one or more memory management units (MMUs) **1420A-1420B**, cache(s) **1425A-1425B**, and circuit interconnect(s) **1430A-1430B**. In at least one embodiment, one or more MMU(s) **1420A-1420B** provide for virtual to physical address mapping for graphics processor **1410**, including for vertex processor **1405** and/or fragment processor(s) **1415A-1415N**, which may reference vertex or image/texture data stored in memory, in addition to vertex or image/texture data stored in one or more cache(s) **1425A-1425B**. In at least one embodiment, one or more MMU(s) **1420A-1420B** may be synchronized with other MMUs within system, including one or more MMUs associated with one or more application processor(s) **1305**, image processors **1315**, and/or video processors **1320** of FIG. **13**, such that each processor **1305-1320** can participate in a shared or unified virtual memory system. In at least one embodiment, one or more circuit interconnect(s) **1430A-1430B** enable graphics processor **1410** to interface with other IP cores within SoC, either via an internal bus of SoC or via a direct connection.

In at least one embodiment, graphics processor **1440** includes one or more MMU(s) **1420A-1420B**, cache(s) **1425A-1425B**, and circuit interconnect(s) **1430A-1430B** of graphics processor **1410** of FIG. **14A**. In at least one embodiment, graphics processor **1440** includes one or more shader core(s) **1455A-1455N** (e.g., **1455A**, **1455B**, **1455C**, **1455D**, **1455E**, **1455F**, through **1455N-1**, and **1455N**), which provides for a unified shader core architecture in which a single core or type or core can execute all types of programmable shader code, including shader program code to implement vertex shaders, fragment shaders, and/or compute shaders. In at least one embodiment, a number of shader cores can vary. In at least one embodiment, graphics processor **1440** includes an inter-core task manager **1445**, which acts as a thread dispatcher to dispatch execution threads to one or more shader cores **1455A-1455N** and a tiling unit **1458** to accelerate tiling operations for tile-based rendering, in which rendering operations for a scene are subdivided in image space, for example to exploit local spatial coherence within a scene or to optimize use of internal caches.

Inference and/or training logic **615** are used to perform inferencing and/or training operations associated with one or more embodiments. Details regarding inference and/or training logic **615** are provided below in conjunction with FIGS. **6A** and/or **6B**. In at least one embodiment, inference and/or training logic **615** may be used in integrated circuit **14A** and/or **14B** for inferencing or predicting operations based, at least in part, on weight parameters calculated using neural network training operations, neural network functions and/or architectures, or neural network use cases described herein. Inference and/or training logic **615** are used to perform inferencing and/or training operations associated with one or more embodiments. In at least one embodiment, this logic can be used with components of these figures to generate one or more images with visual effects approximated from similar effects applied to one or more lower resolution versions of those images.

FIGS. **15A-15B** illustrate additional exemplary graphics processor logic according to embodiments described herein. FIG. **15A** illustrates a graphics core **1500** that may be included within graphics processor **1310** of FIG. **13**, in at least one embodiment, and may be a unified shader core **1455A-1455N** as in FIG. **14B** in at least one embodiment. FIG. **15B** illustrates a highly-parallel general-purpose graphics processing unit **1530** suitable for deployment on a multi-chip module in at least one embodiment.

In at least one embodiment, graphics core **1500** includes a shared instruction cache **1502**, a texture unit **1518**, and a cache/shared memory **1520** that are common to execution resources within graphics core **1500**. In at least one embodiment, graphics core **1500** can include multiple slices **1501A-1501N** or partition for each core, and a graphics processor can include multiple instances of graphics core **1500**. Slices **1501A-1501N** can include support logic including a local instruction cache **1504A-1504N**, a thread scheduler **1506A-1506N**, a thread dispatcher **1508A-1508N**, and a set of registers **1510A-1510N**. In at least one embodiment, slices **1501A-1501N** can include a set of additional function units (AFUs **1512A-1512N**), floating-point units (FPU **1514A-1514N**), integer arithmetic logic units (ALUs **1516-1516N**), address computational units (ACU **1513A-1513N**), double-precision floating-point units (DPFPU **1515A-1515N**), and matrix processing units (MPU **1517A-1517N**).

In at least one embodiment, FPUs **1514A-1514N** can perform single-precision (32-bit) and half-precision (16-bit) floating point operations, while DPFPU **1515A-1515N** perform double precision (64-bit) floating point operations. In at least one embodiment, ALUs **1516A-1516N** can perform variable precision integer operations at 8-bit, 16-bit, and 32-bit precision, and can be configured for mixed precision operations. In at least one embodiment, MPUs **1517A-1517N** can also be configured for mixed precision matrix operations, including half-precision floating point and 8-bit integer operations. In at least one embodiment, MPUs **1517A-1517N** can perform a variety of matrix operations to accelerate machine learning application frameworks, including enabling support for accelerated general matrix to matrix multiplication (GEMM). In at least one embodiment, AFUs **1512A-1512N** can perform additional logic operations not supported by floating-point or integer units, including trigonometric operations (e.g., Sine, Cosine, etc.).

Inference and/or training logic **615** are used to perform inferencing and/or training operations associated with one or more embodiments. Details regarding inference and/or training logic **615** are provided below in conjunction with FIGS. **6A** and/or **6B**. In at least one embodiment, inference and/or training logic **615** may be used in graphics core **1500** for inferencing or predicting operations based, at least in part, on weight parameters calculated using neural network training operations, neural network functions and/or architectures, or neural network use cases described herein.

Inference and/or training logic **615** are used to perform inferencing and/or training operations associated with one or more embodiments. In at least one embodiment, this logic can be used with components of these figures to generate one or more images with visual effects approximated from similar effects applied to one or more lower resolution versions of those images.

FIG. **15B** illustrates a general-purpose processing unit (GPGPU) **1530** that can be configured to enable highly-parallel compute operations to be performed by an array of graphics processing units, in at least one embodiment. In at least one embodiment, GPGPU **1530** can be linked directly to other instances of GPGPU **1530** to create a multi-GPU cluster to improve training speed for deep neural networks. In at least one embodiment, GPGPU **1530** includes a host interface **1532** to enable a connection with a host processor. In at least one embodiment, host interface **1532** is a PCI Express interface. In at least one embodiment, host interface **1532** can be a vendor specific communications interface or communications fabric. In at least one embodiment, GPGPU **1530** receives commands from a host processor and uses a

global scheduler **1534** to distribute execution threads associated with those commands to a set of compute clusters **1536A-1536H**. In at least one embodiment, compute clusters **1536A-1536H** share a cache memory **1538**. In at least one embodiment, cache memory **1538** can serve as a higher-level cache for cache memories within compute clusters **1536A-1536H**.

In at least one embodiment, GPGPU **1530** includes memory **1544A-1544B** coupled with compute clusters **1536A-1536H** via a set of memory controllers **1542A-1542B**. In at least one embodiment, memory **1544A-1544B** can include various types of memory devices including dynamic random access memory (DRAM) or graphics random access memory, such as synchronous graphics random access memory (SGRAM), including graphics double data rate (GDDR) memory.

In at least one embodiment, compute clusters **1536A-1536H** each include a set of graphics cores, such as graphics core **1500** of FIG. **15A**, which can include multiple types of integer and floating point logic units that can perform computational operations at a range of precisions including suited for machine learning computations. For example, in at least one embodiment, at least a subset of floating point units in each of compute clusters **1536A-1536H** can be configured to perform 16-bit or 32-bit floating point operations, while a different subset of floating point units can be configured to perform 64-bit floating point operations.

In at least one embodiment, multiple instances of GPGPU **1530** can be configured to operate as a compute cluster. In at least one embodiment, communication used by compute clusters **1536A-1536H** for synchronization and data exchange varies across embodiments. In at least one embodiment, multiple instances of GPGPU **1530** communicate over host interface **1532**. In at least one embodiment, GPGPU **1530** includes an I/O hub **1539** that couples GPGPU **1530** with a GPU link **1540** that enables a direct connection to other instances of GPGPU **1530**. In at least one embodiment, GPU link **1540** is coupled to a dedicated GPU-to-GPU bridge that enables communication and synchronization between multiple instances of GPGPU **1530**. In at least one embodiment, GPU link **1540** couples with a high speed interconnect to transmit and receive data to other GPGPUs or parallel processors. In at least one embodiment, multiple instances of GPGPU **1530** are located in separate data processing systems and communicate via a network device that is accessible via host interface **1532**. In at least one embodiment GPU, link **1540** can be configured to enable a connection to a host processor in addition to or as an alternative to host interface **1532**.

In at least one embodiment, GPGPU **1530** can be configured to train neural networks. In at least one embodiment, GPGPU **1530** can be used within a inferencing platform. In at least one embodiment, in which GPGPU **1530** is used for inferencing, GPGPU may include fewer compute clusters **1536A-1536H** relative to when GPGPU is used for training a neural network. In at least one embodiment, memory technology associated with memory **1544A-1544B** may differ between inferencing and training configurations, with higher bandwidth memory technologies devoted to training configurations. In at least one embodiment, inferencing configuration of GPGPU **1530** can support inferencing specific instructions. For example, in at least one embodiment, an inferencing configuration can provide support for one or more 8-bit integer dot product instructions, which may be used during inferencing operations for deployed neural networks.

Inference and/or training logic **615** are used to perform inferencing and/or training operations associated with one or more embodiments. Details regarding inference and/or training logic **615** are provided below in conjunction with FIGS. **6A** and/or **6B**. In at least one embodiment, inference and/or training logic **615** may be used in GPGPU **1530** for inferencing or predicting operations based, at least in part, on weight parameters calculated using neural network training operations, neural network functions and/or architectures, or neural network use cases described herein.

Inference and/or training logic **615** are used to perform inferencing and/or training operations associated with one or more embodiments. In at least one embodiment, this logic can be used with components of these figures to generate one or more images with visual effects approximated from similar effects applied to one or more lower resolution versions of those images.

FIG. **16** is a block diagram illustrating a computing system **1600** according to at least one embodiment. In at least one embodiment, computing system **1600** includes a processing subsystem **1601** having one or more processor(s) **1602** and a system memory **1604** communicating via an interconnection path that may include a memory hub **1605**. In at least one embodiment, memory hub **1605** may be a separate component within a chipset component or may be integrated within one or more processor(s) **1602**. In at least one embodiment, memory hub **1605** couples with an I/O subsystem **1611** via a communication link **1606**. In at least one embodiment, I/O subsystem **1611** includes an I/O hub **1607** that can enable computing system **1600** to receive input from one or more input device(s) **1608**. In at least one embodiment, I/O hub **1607** can enable a display controller, which may be included in one or more processor(s) **1602**, to provide outputs to one or more display device(s) **1610A**. In at least one embodiment, one or more display device(s) **1610A** coupled with I/O hub **1607** can include a local, internal, or embedded display device.

In at least one embodiment, processing subsystem **1601** includes one or more parallel processor(s) **1612** coupled to memory hub **1605** via a bus or other communication link **1613**. In at least one embodiment, communication link **1613** may be one of any number of standards based communication link technologies or protocols, such as, but not limited to PCI Express, or may be a vendor specific communications interface or communications fabric. In at least one embodiment, one or more parallel processor(s) **1612** form a computationally focused parallel or vector processing system that can include a large number of processing cores and/or processing clusters, such as a many integrated core (MIC) processor. In at least one embodiment, one or more parallel processor(s) **1612** form a graphics processing subsystem that can output pixels to one of one or more display device(s) **1610A** coupled via I/O Hub **1607**. In at least one embodiment, one or more parallel processor(s) **1612** can also include a display controller and display interface (not shown) to enable a direct connection to one or more display device(s) **1610B**.

In at least one embodiment, a system storage unit **1614** can connect to I/O hub **1607** to provide a storage mechanism for computing system **1600**. In at least one embodiment, an I/O switch **1616** can be used to provide an interface mechanism to enable connections between I/O hub **1607** and other components, such as a network adapter **1618** and/or wireless network adapter **1619** that may be integrated into a platform(s), and various other devices that can be added via one or more add-in device(s) **1620**. In at least one embodiment, network adapter **1618** can be an Ethernet adapter or

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another wired network adapter. In at least one embodiment, wireless network adapter **1619** can include one or more of a Wi-Fi, Bluetooth, near field communication (NFC), or other network device that includes one or more wireless radios.

In at least one embodiment, computing system **1600** can include other components not explicitly shown, including USB or other port connections, optical storage drives, video capture devices, and like, may also be connected to I/O hub **1607**. In at least one embodiment, communication paths interconnecting various components in FIG. **16** may be implemented using any suitable protocols, such as PCI (Peripheral Component Interconnect) based protocols (e.g., PCI-Express), or other bus or point-to-point communication interfaces and/or protocol(s), such as NV-Link high-speed interconnect, or interconnect protocols.

In at least one embodiment, one or more parallel processor(s) **1612** incorporate circuitry optimized for graphics and video processing, including, for example, video output circuitry, and constitutes a graphics processing unit (GPU). In at least one embodiment, one or more parallel processor(s) **1612** incorporate circuitry optimized for general purpose processing. In at least one embodiment, components of computing system **1600** may be integrated with one or more other system elements on a single integrated circuit. For example, in at least one embodiment, one or more parallel processor(s) **1612**, memory hub **1605**, processor(s) **1602**, and I/O hub **1607** can be integrated into a system on chip (SoC) integrated circuit. In at least one embodiment, components of computing system **1600** can be integrated into a single package to form a system in package (SIP) configuration. In at least one embodiment, at least a portion of components of computing system **1600** can be integrated into a multi-chip module (MCM), which can be interconnected with other multi-chip modules into a modular computing system.

Inference and/or training logic **615** are used to perform inferencing and/or training operations associated with one or more embodiments. Details regarding inference and/or training logic **615** are provided below in conjunction with FIGS. **6A** and/or **6B**. In at least one embodiment, inference and/or training logic **615** may be used in system FIG. **1600** for inferencing or predicting operations based, at least in part, on weight parameters calculated using neural network training operations, neural network functions and/or architectures, or neural network use cases described herein.

Inference and/or training logic **615** are used to perform inferencing and/or training operations associated with one or more embodiments. In at least one embodiment, this logic can be used with components of these figures to generate one or more images with visual effects approximated from similar effects applied to one or more lower resolution versions of those images.

Processors

FIG. **17A** illustrates a parallel processor **1700** according to at least one embodiment. In at least one embodiment, various components of parallel processor **1700** may be implemented using one or more integrated circuit devices, such as programmable processors, application specific integrated circuits (ASICs), or field programmable gate arrays (FPGA). In at least one embodiment, illustrated parallel processor **1700** is a variant of one or more parallel processor(s) **1612** shown in FIG. **16** according to an exemplary embodiment.

In at least one embodiment, parallel processor **1700** includes a parallel processing unit **1702**. In at least one embodiment, parallel processing unit **1702** includes an I/O unit **1704** that enables communication with other devices,

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including other instances of parallel processing unit **1702**. In at least one embodiment, I/O unit **1704** may be directly connected to other devices. In at least one embodiment, I/O unit **1704** connects with other devices via use of a hub or switch interface, such as memory hub **1605**. In at least one embodiment, connections between memory hub **1605** and I/O unit **1704** form a communication link **1613**. In at least one embodiment, I/O unit **1704** connects with a host interface **1706** and a memory crossbar **1716**, where host interface **1706** receives commands directed to performing processing operations and memory crossbar **1716** receives commands directed to performing memory operations.

In at least one embodiment, when host interface **1706** receives a command buffer via I/O unit **1704**, host interface **1706** can direct work operations to perform those commands to a front end **1708**. In at least one embodiment, front end **1708** couples with a scheduler **1710**, which is configured to distribute commands or other work items to a processing cluster array **1712**. In at least one embodiment, scheduler **1710** ensures that processing cluster array **1712** is properly configured and in a valid state before tasks are distributed to processing cluster array **1712**. In at least one embodiment, scheduler **1710** is implemented via firmware logic executing on a microcontroller. In at least one embodiment, microcontroller implemented scheduler **1710** is configurable to perform complex scheduling and work distribution operations at coarse and fine granularity, enabling rapid preemption and context switching of threads executing on processing array **1712**. In at least one embodiment, host software can prove workloads for scheduling on processing array **1712** via one of multiple graphics processing doorbells. In at least one embodiment, workloads can then be automatically distributed across processing array **1712** by scheduler **1710** logic within a microcontroller including scheduler **1710**.

In at least one embodiment, processing cluster array **1712** can include up to “N” processing clusters (e.g., cluster **1714A**, cluster **1714B**, through cluster **1714N**). In at least one embodiment, each cluster **1714A-1714N** of processing cluster array **1712** can execute a large number of concurrent threads. In at least one embodiment, scheduler **1710** can allocate work to clusters **1714A-1714N** of processing cluster array **1712** using various scheduling and/or work distribution algorithms, which may vary depending on workload arising for each type of program or computation. In at least one embodiment, scheduling can be handled dynamically by scheduler **1710**, or can be assisted in part by compiler logic during compilation of program logic configured for execution by processing cluster array **1712**. In at least one embodiment, different clusters **1714A-1714N** of processing cluster array **1712** can be allocated for processing different types of programs or for performing different types of computations.

In at least one embodiment, processing cluster array **1712** can be configured to perform various types of parallel processing operations. In at least one embodiment, processing cluster array **1712** is configured to perform general-purpose parallel compute operations. For example, in at least one embodiment, processing cluster array **1712** can include logic to execute processing tasks including filtering of video and/or audio data, performing modeling operations, including physics operations, and performing data transformations.

In at least one embodiment, processing cluster array **1712** is configured to perform parallel graphics processing operations. In at least one embodiment, processing cluster array **1712** can include additional logic to support execution of such graphics processing operations, including, but not

limited to texture sampling logic to perform texture operations, as well as tessellation logic and other vertex processing logic. In at least one embodiment, processing cluster array 1712 can be configured to execute graphics processing related shader programs such as, but not limited to vertex shaders, tessellation shaders, geometry shaders, and pixel shaders. In at least one embodiment, parallel processing unit 1702 can transfer data from system memory via I/O unit 1704 for processing. In at least one embodiment, during processing, transferred data can be stored to on-chip memory (e.g., parallel processor memory 1722) during processing, then written back to system memory.

In at least one embodiment, when parallel processing unit 1702 is used to perform graphics processing, scheduler 1710 can be configured to divide a processing workload into approximately equal sized tasks, to better enable distribution of graphics processing operations to multiple clusters 1714A-1714N of processing cluster array 1712. In at least one embodiment, portions of processing cluster array 1712 can be configured to perform different types of processing. For example, in at least one embodiment, a first portion may be configured to perform vertex shading and topology generation, a second portion may be configured to perform tessellation and geometry shading, and a third portion may be configured to perform pixel shading or other screen space operations, to produce a rendered image for display. In at least one embodiment, intermediate data produced by one or more of clusters 1714A-1714N may be stored in buffers to allow intermediate data to be transmitted between clusters 1714A-1714N for further processing.

In at least one embodiment, processing cluster array 1712 can receive processing tasks to be executed via scheduler 1710, which receives commands defining processing tasks from front end 1708. In at least one embodiment, processing tasks can include indices of data to be processed, e.g., surface (patch) data, primitive data, vertex data, and/or pixel data, as well as state parameters and commands defining how data is to be processed (e.g., what program is to be executed). In at least one embodiment, scheduler 1710 may be configured to fetch indices corresponding to tasks or may receive indices from front end 1708. In at least one embodiment, front end 1708 can be configured to ensure processing cluster array 1712 is configured to a valid state before a workload specified by incoming command buffers (e.g., batch-buffers, push buffers, etc.) is initiated.

In at least one embodiment, each of one or more instances of parallel processing unit 1702 can couple with parallel processor memory 1722. In at least one embodiment, parallel processor memory 1722 can be accessed via memory crossbar 1716, which can receive memory requests from processing cluster array 1712 as well as I/O unit 1704. In at least one embodiment, memory crossbar 1716 can access parallel processor memory 1722 via a memory interface 1718. In at least one embodiment, memory interface 1718 can include multiple partition units (e.g., partition unit 1720A, partition unit 1720B, through partition unit 1720N) that can each couple to a portion (e.g., memory unit) of parallel processor memory 1722. In at least one embodiment, a number of partition units 1720A-1720N is configured to be equal to a number of memory units, such that a first partition unit 1720A has a corresponding first memory unit 1724A, a second partition unit 1720B has a corresponding memory unit 1724B, and a Nth partition unit 1720N has a corresponding Nth memory unit 1724N. In at least one embodiment, a number of partition units 1720A-1720N may not be equal to a number of memory devices.

In at least one embodiment, memory units 1724A-1724N can include various types of memory devices, including dynamic random access memory (DRAM) or graphics random access memory, such as synchronous graphics random access memory (SGRAM), including graphics double data rate (GDDR) memory. In at least one embodiment, memory units 1724A-1724N may also include 3D stacked memory, including but not limited to high bandwidth memory (HBM). In at least one embodiment, render targets, such as frame buffers or texture maps may be stored across memory units 1724A-1724N, allowing partition units 1720A-1720N to write portions of each render target in parallel to efficiently use available bandwidth of parallel processor memory 1722. In at least one embodiment, a local instance of parallel processor memory 1722 may be excluded in favor of a unified memory design that utilizes system memory in conjunction with local cache memory.

In at least one embodiment, any one of clusters 1714A-1714N of processing cluster array 1712 can process data that will be written to any of memory units 1724A-1724N within parallel processor memory 1722. In at least one embodiment, memory crossbar 1716 can be configured to transfer an output of each cluster 1714A-1714N to any partition unit 1720A-1720N or to another cluster 1714A-1714N, which can perform additional processing operations on an output. In at least one embodiment, each cluster 1714A-1714N can communicate with memory interface 1718 through memory crossbar 1716 to read from or write to various external memory devices. In at least one embodiment, memory crossbar 1716 has a connection to memory interface 1718 to communicate with I/O unit 1704, as well as a connection to a local instance of parallel processor memory 1722, enabling processing units within different processing clusters 1714A-1714N to communicate with system memory or other memory that is not local to parallel processing unit 1702. In at least one embodiment, memory crossbar 1716 can use virtual channels to separate traffic streams between clusters 1714A-1714N and partition units 1720A-1720N.

In at least one embodiment, multiple instances of parallel processing unit 1702 can be provided on a single add-in card, or multiple add-in cards can be interconnected. In at least one embodiment, different instances of parallel processing unit 1702 can be configured to inter-operate even if different instances have different numbers of processing cores, different amounts of local parallel processor memory, and/or other configuration differences. For example, in at least one embodiment, some instances of parallel processing unit 1702 can include higher precision floating point units relative to other instances. In at least one embodiment, systems incorporating one or more instances of parallel processing unit 1702 or parallel processor 1700 can be implemented in a variety of configurations and form factors, including but not limited to desktop, laptop, or handheld personal computers, servers, workstations, game consoles, and/or embedded systems.

FIG. 17B is a block diagram of a partition unit 1720 according to at least one embodiment. In at least one embodiment, partition unit 1720 is an instance of one of partition units 1720A-1720N of FIG. 17A. In at least one embodiment, partition unit 1720 includes an L2 cache 1721, a frame buffer interface 1725, and a raster operations unit ("ROP") 1726. L2 cache 1721 is a read/write cache that is configured to perform load and store operations received from memory crossbar 1716 and ROP 1726. In at least one embodiment, read misses and urgent write-back requests are output by L2 cache 1721 to frame buffer interface 1725 for processing. In at least one embodiment, updates can also be

sent to a frame buffer via frame buffer interface **1725** for processing. In at least one embodiment, frame buffer interface **1725** interfaces with one of memory units in parallel processor memory, such as memory units **1724A-1724N** of FIG. **17** (e.g., within parallel processor memory **1722**).

In at least one embodiment, ROP **1726** is a processing unit that performs raster operations such as stencil, z test, blending, and so forth. In at least one embodiment, ROP **1726** then outputs processed graphics data that is stored in graphics memory. In at least one embodiment, ROP **1726** includes compression logic to compress depth or color data that is written to memory and decompress depth or color data that is read from memory. In at least one embodiment, compression logic can be lossless compression logic that makes use of one or more of multiple compression algorithms. Compression logic that is performed by ROP **1726** can vary based on statistical characteristics of data to be compressed. For example, in at least one embodiment, delta color compression is performed on depth and color data on a per-tile basis.

In at least one embodiment, ROP **1726** is included within each processing cluster (e.g., cluster **1714A-1714N** of FIG. **17A**) instead of within partition unit **1720**. In at least one embodiment, read and write requests for pixel data are transmitted over memory crossbar **1716** instead of pixel fragment data. In at least one embodiment, processed graphics data may be displayed on a display device, such as one of one or more display device(s) **1610** of FIG. **16**, routed for further processing by processor(s) **1602**, or routed for further processing by one of processing entities within parallel processor **1700** of FIG. **17A**.

FIG. **17C** is a block diagram of a processing cluster **1714** within a parallel processing unit according to at least one embodiment. In at least one embodiment, a processing cluster is an instance of one of processing clusters **1714A-1714N** of FIG. **17A**. In at least one embodiment, one of more of processing cluster(s) **1714** can be configured to execute many threads in parallel, where “thread” refers to an instance of a particular program executing on a particular set of input data. In at least one embodiment, single-instruction, multiple-data (SIMD) instruction issue techniques are used to support parallel execution of a large number of threads without providing multiple independent instruction units. In at least one embodiment, single-instruction, multiple-thread (SMT) techniques are used to support parallel execution of a large number of generally synchronized threads, using a common instruction unit configured to issue instructions to a set of processing engines within each one of processing clusters.

In at least one embodiment, operation of processing cluster **1714** can be controlled via a pipeline manager **1732** that distributes processing tasks to SMT parallel processors. In at least one embodiment, pipeline manager **1732** receives instructions from scheduler **1710** of FIG. **17A** and manages execution of those instructions via a graphics multiprocessor **1734** and/or a texture unit **1736**. In at least one embodiment, graphics multiprocessor **1734** is an exemplary instance of a SMT parallel processor. However, in at least one embodiment, various types of SMT parallel processors of differing architectures may be included within processing cluster **1714**. In at least one embodiment, one or more instances of graphics multiprocessor **1734** can be included within a processing cluster **1714**. In at least one embodiment, graphics multiprocessor **1734** can process data and a data crossbar **1740** can be used to distribute processed data to one of multiple possible destinations, including other shader units. In at least one embodiment, pipeline manager **1732** can

facilitate distribution of processed data by specifying destinations for processed data to be distributed via data crossbar **1740**.

In at least one embodiment, each graphics multiprocessor **1734** within processing cluster **1714** can include an identical set of functional execution logic (e.g., arithmetic logic units, load-store units, etc.). In at least one embodiment, functional execution logic can be configured in a pipelined manner in which new instructions can be issued before previous instructions are complete. In at least one embodiment, functional execution logic supports a variety of operations including integer and floating point arithmetic, comparison operations, Boolean operations, bit-shifting, and computation of various algebraic functions. In at least one embodiment, same functional-unit hardware can be leveraged to perform different operations and any combination of functional units may be present.

In at least one embodiment, instructions transmitted to processing cluster **1714** constitute a thread. In at least one embodiment, a set of threads executing across a set of parallel processing engines is a thread group. In at least one embodiment, thread group executes a program on different input data. In at least one embodiment, each thread within a thread group can be assigned to a different processing engine within a graphics multiprocessor **1734**. In at least one embodiment, a thread group may include fewer threads than a number of processing engines within graphics multiprocessor **1734**. In at least one embodiment, when a thread group includes fewer threads than a number of processing engines, one or more processing engines may be idle during cycles in which that thread group is being processed. In at least one embodiment, a thread group may also include more threads than a number of processing engines within graphics multiprocessor **1734**. In at least one embodiment, when a thread group includes more threads than processing engines within graphics multiprocessor **1734**, processing can be performed over consecutive clock cycles. In at least one embodiment, multiple thread groups can be executed concurrently on a graphics multiprocessor **1734**.

In at least one embodiment, graphics multiprocessor **1734** includes an internal cache memory to perform load and store operations. In at least one embodiment, graphics multiprocessor **1734** can forego an internal cache and use a cache memory (e.g., L1 cache **1748**) within processing cluster **1714**. In at least one embodiment, each graphics multiprocessor **1734** also has access to L2 caches within partition units (e.g., partition units **1720A-1720N** of FIG. **17A**) that are shared among all processing clusters **1714** and may be used to transfer data between threads. In at least one embodiment, graphics multiprocessor **1734** may also access off-chip global memory, which can include one or more of local parallel processor memory and/or system memory. In at least one embodiment, any memory external to parallel processing unit **1702** may be used as global memory. In at least one embodiment, processing cluster **1714** includes multiple instances of graphics multiprocessor **1734** can share common instructions and data, which may be stored in L1 cache **1748**.

In at least one embodiment, each processing cluster **1714** may include a memory management unit (“MMU”) **1745** that is configured to map virtual addresses into physical addresses. In at least one embodiment, one or more instances of MMU **1745** may reside within memory interface **1718** of FIG. **17A**. In at least one embodiment, MMU **1745** includes a set of page table entries (PTEs) used to map a virtual address to a physical address of a tile and optionally a cache line index. In at least one embodiment, MMU **1745** may

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include address translation lookaside buffers (TLB) or caches that may reside within graphics multiprocessor **1734** or L1 cache or processing cluster **1714**. In at least one embodiment, physical address is processed to distribute surface data access locality to allow efficient request inter-leaving among partition units. In at least one embodiment, cache line index may be used to determine whether a request for a cache line is a hit or miss.

In at least one embodiment, a processing cluster **1714** may be configured such that each graphics multiprocessor **1734** is coupled to a texture unit **1736** for performing texture mapping operations, e.g., determining texture sample positions, reading texture data, and filtering texture data. In at least one embodiment, texture data is read from an internal texture L1 cache (not shown) or from an L1 cache within graphics multiprocessor **1734** and is fetched from an L2 cache, local parallel processor memory, or system memory, as needed. In at least one embodiment, each graphics multiprocessor **1734** outputs processed tasks to data crossbar **1740** to provide processed task(s) to another processing cluster **1714** for further processing or to store processed task(s) in an L2 cache, local parallel processor memory, or system memory via memory crossbar **1716**. In at least one embodiment, preROP **1742** (pre-raster operations unit) is configured to receive data from graphics multiprocessor **1734**, direct data to ROP units, which may be located with partition units as described herein (e.g., partition units **1720A-1720N** of FIG. **17A**). In at least one embodiment, PreROP **1742** unit can perform optimizations for color blending, organize pixel color data, and perform address translations.

Inference and/or training logic **615** are used to perform inferencing and/or training operations associated with one or more embodiments. Details regarding inference and/or training logic **615** are provided below in conjunction with FIGS. **6A** and/or **6B**. In at least one embodiment, inference and/or training logic **615** may be used in graphics processing cluster **1714** for inferencing or predicting operations based, at least in part, on weight parameters calculated using neural network training operations, neural network functions and/or architectures, or neural network use cases described herein.

Inference and/or training logic **615** are used to perform inferencing and/or training operations associated with one or more embodiments. In at least one embodiment, this logic can be used with components of these figures to generate one or more images with visual effects approximated from similar effects applied to one or more lower resolution versions of those images.

FIG. **17D** shows a graphics multiprocessor **1734** according to at least one embodiment. In at least one embodiment, graphics multiprocessor **1734** couples with pipeline manager **1732** of processing cluster **1714**. In at least one embodiment, graphics multiprocessor **1734** has an execution pipeline including but not limited to an instruction cache **1752**, an instruction unit **1754**, an address mapping unit **1756**, a register file **1758**, one or more general purpose graphics processing unit (GPGPU) cores **1762**, and one or more load/store units **1766**. GPGPU core(s) **1762** and load/store unit(s) **1766** are coupled with cache memory **1772** and shared memory **1770** via a memory and cache interconnect **1768**.

In at least one embodiment, instruction cache **1752** receives a stream of instructions to execute from pipeline manager **1732**. In at least one embodiment, instructions are cached in instruction cache **1752** and dispatched for execution by instruction unit **1754**. In at least one embodiment,

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instruction unit **1754** can dispatch instructions as thread groups (e.g., warps), with each thread group assigned to a different execution unit within GPGPU core(s) **1762**. In at least one embodiment, an instruction can access any of a local, shared, or global address space by specifying an address within a unified address space. In at least one embodiment, address mapping unit **1756** can be used to translate addresses in a unified address space into a distinct memory address that can be accessed by load/store unit(s) **1766**.

In at least one embodiment, register file **1758** provides a set of registers for functional units of graphics multiprocessor **1734**. In at least one embodiment, register file **1758** provides temporary storage for operands connected to data paths of functional units (e.g., GPGPU cores **1762**, load/store units **1766**) of graphics multiprocessor **1734**. In at least one embodiment, register file **1758** is divided between each of functional units such that each functional unit is allocated a dedicated portion of register file **1758**. In at least one embodiment, register file **1758** is divided between different warps being executed by graphics multiprocessor **1734**.

In at least one embodiment, GPGPU cores **1762** can each include floating point units (FPUs) and/or integer arithmetic logic units (ALUs) that are used to execute instructions of graphics multiprocessor **1734**. GPGPU cores **1762** can be similar in architecture or can differ in architecture. In at least one embodiment, a first portion of GPGPU cores **1762** include a single precision FPU and an integer ALU while a second portion of GPGPU cores include a double precision FPU. In at least one embodiment, FPUs can implement IEEE 754-2008 standard for floating point arithmetic or enable variable precision floating point arithmetic. In at least one embodiment, graphics multiprocessor **1734** can additionally include one or more fixed function or special function units to perform specific functions such as copy rectangle or pixel blending operations. In at least one embodiment one or more of GPGPU cores can also include fixed or special function logic.

In at least one embodiment, GPGPU cores **1762** include SIMD logic capable of performing a single instruction on multiple sets of data. In at least one embodiment GPGPU cores **1762** can physically execute SIMD4, SIMD8, and SIMD16 instructions and logically execute SIMD1, SIMD2, and SIMD32 instructions. In at least one embodiment, SIMD instructions for GPGPU cores can be generated at compile time by a shader compiler or automatically generated when executing programs written and compiled for single program multiple data (SPMD) or SIMT architectures. In at least one embodiment, multiple threads of a program configured for an SIMT execution model can be executed via a single SIMD instruction. For example, in at least one embodiment, eight SIMT threads that perform same or similar operations can be executed in parallel via a single SIMD8 logic unit.

In at least one embodiment, memory and cache interconnect **1768** is an interconnect network that connects each functional unit of graphics multiprocessor **1734** to register file **1758** and to shared memory **1770**. In at least one embodiment, memory and cache interconnect **1768** is a crossbar interconnect that allows load/store unit **1766** to implement load and store operations between shared memory **1770** and register file **1758**. In at least one embodiment, register file **1758** can operate at a same frequency as GPGPU cores **1762**, thus data transfer between GPGPU cores **1762** and register file **1758** is very low latency. In at least one embodiment, shared memory **1770** can be used to enable communication between threads that execute on

functional units within graphics multiprocessor **1734**. In at least one embodiment, cache memory **1772** can be used as a data cache for example, to cache texture data communicated between functional units and texture unit **1736**. In at least one embodiment, shared memory **1770** can also be used as a program managed cache. In at least one embodiment, threads executing on GPGPU cores **1762** can programmatically store data within shared memory in addition to automatically cached data that is stored within cache memory **1772**.

In at least one embodiment, a parallel processor or GPGPU as described herein is communicatively coupled to host/processor cores to accelerate graphics operations, machine-learning operations, pattern analysis operations, and various general purpose GPU (GPGPU) functions. In at least one embodiment, GPU may be communicatively coupled to host processor/cores over a bus or other interconnect (e.g., a high speed interconnect such as PCIe or NVLink). In at least one embodiment, GPU may be integrated on same package or chip as cores and communicatively coupled to cores over an internal processor bus/interconnect (i.e., internal to package or chip). In at least one embodiment, regardless of manner in which GPU is connected, processor cores may allocate work to GPU in form of sequences of commands/instructions contained in a work descriptor. In at least one embodiment, GPU then uses dedicated circuitry/logic for efficiently processing these commands/instructions.

Inference and/or training logic **615** are used to perform inferencing and/or training operations associated with one or more embodiments. Details regarding inference and/or training logic **615** are provided below in conjunction with FIGS. **6A** and/or **6B**. In at least one embodiment, inference and/or training logic **615** may be used in graphics multiprocessor **1734** for inferencing or predicting operations based, at least in part, on weight parameters calculated using neural network training operations, neural network functions and/or architectures, or neural network use cases described herein.

Inference and/or training logic **615** are used to perform inferencing and/or training operations associated with one or more embodiments. In at least one embodiment, this logic can be used with components of these figures to generate one or more images with visual effects approximated from similar effects applied to one or more lower resolution versions of those images.

FIG. **18** illustrates a multi-GPU computing system **1800**, according to at least one embodiment. In at least one embodiment, multi-GPU computing system **1800** can include a processor **1802** coupled to multiple general purpose graphics processing units (GPGPUs) **1806A-D** via a host interface switch **1804**. In at least one embodiment, host interface switch **1804** is a PCI express switch device that couples processor **1802** to a PCI express bus over which processor **1802** can communicate with GPGPUs **1806A-D**. GPGPUs **1806A-D** can interconnect via a set of high-speed point to point GPU to GPU links **1816**. In at least one embodiment, GPU to GPU links **1816** connect to each of GPGPUs **1806A-D** via a dedicated GPU link. In at least one embodiment, P2P GPU links **1816** enable direct communication between each of GPGPUs **1806A-D** without requiring communication over host interface bus **1804** to which processor **1802** is connected. In at least one embodiment, with GPU-to-GPU traffic directed to P2P GPU links **1816**, host interface bus **1804** remains available for system memory access or to communicate with other instances of multi-GPU computing system **1800**, for example, via one or more network devices. While in at least one embodiment GPGPUs

1806A-D connect to processor **1802** via host interface switch **1804**, in at least one embodiment processor **1802** includes direct support for P2P GPU links **1816** and can connect directly to GPGPUs **1806A-D**.

Inference and/or training logic **615** are used to perform inferencing and/or training operations associated with one or more embodiments. Details regarding inference and/or training logic **615** are provided below in conjunction with FIGS. **6A** and/or **6B**. In at least one embodiment, inference and/or training logic **615** may be used in multi-GPU computing system **1800** for inferencing or predicting operations based, at least in part, on weight parameters calculated using neural network training operations, neural network functions and/or architectures, or neural network use cases described herein.

Inference and/or training logic **615** are used to perform inferencing and/or training operations associated with one or more embodiments. In at least one embodiment, this logic can be used with components of these figures to generate one or more images with visual effects approximated from similar effects applied to one or more lower resolution versions of those images.

FIG. **19** is a block diagram of a graphics processor **1900**, according to at least one embodiment. In at least one embodiment, graphics processor **1900** includes a ring interconnect **1902**, a pipeline front-end **1904**, a media engine **1937**, and graphics cores **1980A-1980N**. In at least one embodiment, ring interconnect **1902** couples graphics processor **1900** to other processing units, including other graphics processors or one or more general-purpose processor cores. In at least one embodiment, graphics processor **1900** is one of many processors integrated within a multi-core processing system.

In at least one embodiment, graphics processor **1900** receives batches of commands via ring interconnect **1902**. In at least one embodiment, incoming commands are interpreted by a command streamer **1903** in pipeline front-end **1904**. In at least one embodiment, graphics processor **1900** includes scalable execution logic to perform 3D geometry processing and media processing via graphics core(s) **1980A-1980N**. In at least one embodiment, for 3D geometry processing commands, command streamer **1903** supplies commands to geometry pipeline **1936**. In at least one embodiment, for at least some media processing commands, command streamer **1903** supplies commands to a video front end **1934**, which couples with a media engine **1937**. In at least one embodiment, media engine **1937** includes a Video Quality Engine (VQE) **1930** for video and image post-processing and a multi-format encode/decode (MFX) **1933** engine to provide hardware-accelerated media data encode and decode. In at least one embodiment, geometry pipeline **1936** and media engine **1937** each generate execution threads for thread execution resources provided by at least one graphics core **1980A**.

In at least one embodiment, graphics processor **1900** includes scalable thread execution resources featuring modular cores **1980A-1980N** (sometimes referred to as core slices), each having multiple sub-cores **1950A-1950N**, **1960A-1960N** (sometimes referred to as core sub-slices). In at least one embodiment, graphics processor **1900** can have any number of graphics cores **1980A** through **1980N**. In at least one embodiment, graphics processor **1900** includes a graphics core **1980A** having at least a first sub-core **1950A** and a second sub-core **1960A**. In at least one embodiment, graphics processor **1900** is a low power processor with a single sub-core (e.g., **1950A**). In at least one embodiment, graphics processor **1900** includes multiple graphics cores

1980A-1980N, each including a set of first sub-cores **1950A-1950N** and a set of second sub-cores **1960A-1960N**. In at least one embodiment, each sub-core in first sub-cores **1950A-1950N** includes at least a first set of execution units **1952A-1952N** and media/texture samplers **1954A-1954N**. In at least one embodiment, each sub-core in second sub-cores **1960A-1960N** includes at least a second set of execution units **1962A-1962N** and samplers **1964A-1964N**. In at least one embodiment, each sub-core **1950A-1950N**, **1960A-1960N** shares a set of shared resources **1970A-1970N**. In at least one embodiment, shared resources include shared cache memory and pixel operation logic.

Inference and/or training logic **615** are used to perform inferencing and/or training operations associated with one or more embodiments. Details regarding inference and/or training logic **615** are provided below in conjunction with FIGS. **6A** and/or **6B**. In at least one embodiment, inference and/or training logic **615** may be used in graphics processor **1900** for inferencing or predicting operations based, at least in part, on weight parameters calculated using neural network training operations, neural network functions and/or architectures, or neural network use cases described herein.

Inference and/or training logic **615** are used to perform inferencing and/or training operations associated with one or more embodiments. In at least one embodiment, this logic can be used with components of these figures to generate one or more images with visual effects approximated from similar effects applied to one or more lower resolution versions of those images.

FIG. **20** is a block diagram illustrating micro-architecture for a processor **2000** that may include logic circuits to perform instructions, according to at least one embodiment. In at least one embodiment, processor **2000** may perform instructions, including x86 instructions, ARM instructions, specialized instructions for application-specific integrated circuits (ASICs), etc. In at least one embodiment, processor **2000** may include registers to store packed data, such as 64-bit wide MMX™ registers in microprocessors enabled with MMX technology from Intel Corporation of Santa Clara, Calif. In at least one embodiment, MMX registers, available in both integer and floating point forms, may operate with packed data elements that accompany single instruction, multiple data (“SIMD”) and streaming SIMD extensions (“SSE”) instructions. In at least one embodiment, 128-bit wide XMM registers relating to SSE2, SSE3, SSE4, AVX, or beyond (referred to generically as “SSEx”) technology may hold such packed data operands. In at least one embodiment, processor **2000** may perform instructions to accelerate machine learning or deep learning algorithms, training, or inferencing.

In at least one embodiment, processor **2000** includes an in-order front end (“front end”) **2001** to fetch instructions to be executed and prepare instructions to be used later in processor pipeline. In at least one embodiment, front end **2001** may include several units. In at least one embodiment, an instruction prefetcher **2026** fetches instructions from memory and feeds instructions to an instruction decoder **2028** which in turn decodes or interprets instructions. For example, in at least one embodiment, instruction decoder **2028** decodes a received instruction into one or more operations called “micro-instructions” or “micro-operations” (also called “micro ops” or “uops”) that machine may execute. In at least one embodiment, instruction decoder **2028** parses instruction into an opcode and corresponding data and control fields that may be used by micro-architecture to perform operations in accordance with at least one embodiment. In at least one embodiment, a trace cache **2030**

may assemble decoded uops into program ordered sequences or traces in a uop queue **2034** for execution. In at least one embodiment, when trace cache **2030** encounters a complex instruction, a microcode ROM **2032** provides uops needed to complete operation.

In at least one embodiment, some instructions may be converted into a single micro-op, whereas others need several micro-ops to complete full operation. In at least one embodiment, if more than four micro-ops are needed to complete an instruction, instruction decoder **2028** may access microcode ROM **2032** to perform instruction. In at least one embodiment, an instruction may be decoded into a small number of micro-ops for processing at instruction decoder **2028**. In at least one embodiment, an instruction may be stored within microcode ROM **2032** should a number of micro-ops be needed to accomplish operation. In at least one embodiment, trace cache **2030** refers to an entry point programmable logic array (“PLA”) to determine a correct micro-instruction pointer for reading microcode sequences to complete one or more instructions from microcode ROM **2032** in accordance with at least one embodiment. In at least one embodiment, after microcode ROM **2032** finishes sequencing micro-ops for an instruction, front end **2001** of machine may resume fetching micro-ops from trace cache **2030**.

In at least one embodiment, out-of-order execution engine (“out of order engine”) **2003** may prepare instructions for execution. In at least one embodiment, out-of-order execution logic has a number of buffers to smooth out and re-order flow of instructions to optimize performance as they go down pipeline and get scheduled for execution. In at least one embodiment, out-of-order execution engine **2003** includes, without limitation, an allocator/register renamer **2040**, a memory uop queue **2042**, an integer/floating point uop queue **2044**, a memory scheduler **2046**, a fast scheduler **2002**, a slow/general floating point scheduler (“slow/general FP scheduler”) **2004**, and a simple floating point scheduler (“simple FP scheduler”) **2006**. In at least one embodiment, fast scheduler **2002**, slow/general floating point scheduler **2004**, and simple floating point scheduler **2006** are also collectively referred to herein as “uop schedulers **2002**, **2004**, **2006**.” In at least one embodiment, allocator/register renamer **2040** allocates machine buffers and resources that each uop needs in order to execute. In at least one embodiment, allocator/register renamer **2040** renames logic registers onto entries in a register file. In at least one embodiment, allocator/register renamer **2040** also allocates an entry for each uop in one of two uop queues, memory uop queue **2042** for memory operations and integer/floating point uop queue **2044** for non-memory operations, in front of memory scheduler **2046** and uop schedulers **2002**, **2004**, **2006**. In at least one embodiment, uop schedulers **2002**, **2004**, **2006** determine when a uop is ready to execute based on readiness of their dependent input register operand sources and availability of execution resources uops need to complete their operation. In at least one embodiment, fast scheduler **2002** of at least one embodiment may schedule on each half of main clock cycle while slow/general floating point scheduler **2004** and simple floating point scheduler **2006** may schedule once per main processor clock cycle. In at least one embodiment, uop schedulers **2002**, **2004**, **2006** arbitrate for dispatch ports to schedule uops for execution.

In at least one embodiment, execution block **2011** includes, without limitation, an integer register file/bypass network **2008**, a floating point register file/bypass network (“FP register file/bypass network”) **2010**, address generation units (“AGUs”) **2012** and **2014**, fast Arithmetic Logic Units

(ALUs) (“fast ALUs”) **2016** and **2018**, a slow Arithmetic Logic Unit (“slow ALU”) **2020**, a floating point ALU (“FP”) **2022**, and a floating point move unit (“FP move”) **2024**. In at least one embodiment, integer register file/bypass network **2008** and floating point register file/bypass network **2010** are also referred to herein as “register files **2008**, **2010**.” In at least one embodiment, AGUs **2012** and **2014**, fast ALUs **2016** and **2018**, slow ALU **2020**, floating point ALU **2022**, and floating point move unit **2024** are also referred to herein as “execution units **2012**, **2014**, **2016**, **2018**, **2020**, **2022**, and **2024**.” In at least one embodiment, execution block **b 11** may include, without limitation, any number (including zero) and type of register files, bypass networks, address generation units, and execution units, in any combination.

In at least one embodiment, register files **2008**, **2010** may be arranged between uop schedulers **2002**, **2004**, **2006**, and execution units **2012**, **2014**, **2016**, **2018**, **2020**, **2022**, and **2024**. In at least one embodiment, integer register file/bypass network **2008** performs integer operations. In at least one embodiment, floating point register file/bypass network **2010** performs floating point operations. In at least one embodiment, each of register files **2008**, **2010** may include, without limitation, a bypass network that may bypass or forward just completed results that have not yet been written into register file to new dependent uops. In at least one embodiment, register files **2008**, **2010** may communicate data with each other. In at least one embodiment, integer register file/bypass network **2008** may include, without limitation, two separate register files, one register file for low-order thirty-two bits of data and a second register file for high order thirty-two bits of data. In at least one embodiment, floating point register file/bypass network **2010** may include, without limitation, 128-bit wide entries because floating point instructions typically have operands from 64 to 128 bits in width.

In at least one embodiment, execution units **2012**, **2014**, **2016**, **2018**, **2020**, **2022**, **2024** may execute instructions. In at least one embodiment, register files **2008**, **2010** store integer and floating point data operand values that micro-instructions need to execute. In at least one embodiment, processor **2000** may include, without limitation, any number and combination of execution units **2012**, **2014**, **2016**, **2018**, **2020**, **2022**, **2024**. In at least one embodiment, floating point ALU **2022** and floating point move unit **2024**, may execute floating point, MMX, SIMD, AVX and SSE, or other operations, including specialized machine learning instructions. In at least one embodiment, floating point ALU **2022** may include, without limitation, a 64-bit by 64-bit floating point divider to execute divide, square root, and remainder micro ops. In at least one embodiment, instructions involving a floating point value may be handled with floating point hardware. In at least one embodiment, ALU operations may be passed to fast ALUs **2016**, **2018**. In at least one embodiment, fast ALUS **2016**, **2018** may execute fast operations with an effective latency of half a clock cycle. In at least one embodiment, most complex integer operations go to slow ALU **2020** as slow ALU **2020** may include, without limitation, integer execution hardware for long-latency type of operations, such as a multiplier, shifts, flag logic, and branch processing. In at least one embodiment, memory load/store operations may be executed by AGUS **2012**, **2014**. In at least one embodiment, fast ALU **2016**, fast ALU **2018**, and slow ALU **2020** may perform integer operations on 64-bit data operands. In at least one embodiment, fast ALU **2016**, fast ALU **2018**, and slow ALU **2020** may be implemented to support a variety of data bit sizes including sixteen, thirty-two, 128, 256, etc. In at least one embodiment, floating point

ALU **2022** and floating point move unit **2024** may be implemented to support a range of operands having bits of various widths. In at least one embodiment, floating point ALU **2022** and floating point move unit **2024** may operate on 128-bit wide packed data operands in conjunction with SIMD and multimedia instructions.

In at least one embodiment, uop schedulers **2002**, **2004**, **2006**, dispatch dependent operations before parent load has finished executing. In at least one embodiment, as uops may be speculatively scheduled and executed in processor **2000**, processor **2000** may also include logic to handle memory misses. In at least one embodiment, if a data load misses in data cache, there may be dependent operations in flight in pipeline that have left scheduler with temporarily incorrect data. In at least one embodiment, a replay mechanism tracks and re-executes instructions that use incorrect data. In at least one embodiment, dependent operations might need to be replayed and independent ones may be allowed to complete. In at least one embodiment, schedulers and replay mechanism of at least one embodiment of a processor may also be designed to catch instruction sequences for text string comparison operations.

In at least one embodiment, term “registers” may refer to on-board processor storage locations that may be used as part of instructions to identify operands. In at least one embodiment, registers may be those that may be usable from outside of processor (from a programmer’s perspective). In at least one embodiment, registers might not be limited to a particular type of circuit. Rather, in at least one embodiment, a register may store data, provide data, and perform functions described herein. In at least one embodiment, registers described herein may be implemented by circuitry within a processor using any number of different techniques, such as dedicated physical registers, dynamically allocated physical registers using register renaming, combinations of dedicated and dynamically allocated physical registers, etc. In at least one embodiment, integer registers store 32-bit integer data. A register file of at least one embodiment also contains eight multimedia SIMD registers for packed data.

Inference and/or training logic **615** are used to perform inferencing and/or training operations associated with one or more embodiments. Details regarding inference and/or training logic **615** are provided below in conjunction with FIGS. **6A** and/or **6B**. In at least one embodiment portions or all of inference and/or training logic **615** may be incorporated into execution block **2011** and other memory or registers shown or not shown. For example, in at least one embodiment, training and/or inferencing techniques described herein may use one or more of ALUs illustrated in execution block **2011**. Moreover, weight parameters may be stored in on-chip or off-chip memory and/or registers (shown or not shown) that configure ALUs of execution block **2011** to perform one or more machine learning algorithms, neural network architectures, use cases, or training techniques described herein.

Inference and/or training logic **615** are used to perform inferencing and/or training operations associated with one or more embodiments. In at least one embodiment, this logic can be used with components of these figures to generate one or more images with visual effects approximated from similar effects applied to one or more lower resolution versions of those images.

FIG. **21** illustrates a deep learning application processor **2100**, according to at least one embodiment. In at least one embodiment, deep learning application processor **2100** uses instructions that, if executed by deep learning application processor **2100**, cause deep learning application processor **2100** to perform some or all of processes and techniques

described throughout this disclosure. In at least one embodiment, deep learning application processor **2100** is an application-specific integrated circuit (ASIC). In at least one embodiment, application processor **2100** performs matrix multiply operations either “hard-wired” into hardware as a result of performing one or more instructions or both. In at least one embodiment, deep learning application processor **2100** includes, without limitation, processing clusters **2110** (1)-**2110**(12), Inter-Chip Links (“ICLs”) **2120**(1)-**2120**(12), Inter-Chip Controllers (“ICCs”) **2130**(1)-**2130**(2), memory controllers (“Mem Ctrls”) **2142**(1)-**2142**(4), high bandwidth memory physical layer (“HBM PHY”) **2144**(1)-**2144**(4), a management-controller central processing unit (“management-controller CPU”) **2150**, a peripheral component interconnect express controller and direct memory access block (“PCIe Controller and DMA”) **2170**, and a sixteen-lane peripheral component interconnect express port (“PCI Express×16”) **2180**.

In at least one embodiment, processing clusters **2110** may perform deep learning operations, including inference or prediction operations based on weight parameters calculated one or more training techniques, including those described herein. In at least one embodiment, each processing cluster **2110** may include, without limitation, any number and type of processors. In at least one embodiment, deep learning application processor **2100** may include any number and type of processing clusters **2100**. In at least one embodiment, Inter-Chip Links **2120** are bi-directional. In at least one embodiment, Inter-Chip Links **2120** and Inter-Chip Controllers **2130** enable multiple deep learning application processors **2100** to exchange information, including activation information resulting from performing one or more machine learning algorithms embodied in one or more neural networks. In at least one embodiment, deep learning application processor **2100** may include any number (including zero) and type of ICLs **2120** and ICCs **2130**.

In at least one embodiment, HBM2s **2140** provide a total of 32 Gigabytes (GB) of memory. HBM2 **2140**(*i*) is associated with both memory controller **2142**(*i*) and HBM PHY **2144**(*i*). In at least one embodiment, any number of HBM2s **2140** may provide any type and total amount of high bandwidth memory and may be associated with any number (including zero) and type of memory controllers **2142** and HBM PHYs **2144**. In at least one embodiment, SPI, I2C, GPIO **2160**, PCIe Controller and DMA **2170**, and/or PCIe **2180** may be replaced with any number and type of blocks that enable any number and type of communication standards in any technically feasible fashion.

Inference and/or training logic **615** are used to perform inferencing and/or training operations associated with one or more embodiments. Details regarding inference and/or training logic **615** are provided below in conjunction with FIGS. 6A and/or 6B. In at least one embodiment, deep learning application processor **2100** is used to train a machine learning model, such as a neural network, to predict or infer information provided to deep learning application processor **2100**. In at least one embodiment, deep learning application processor **2100** is used to infer or predict information based on a trained machine learning model (e.g., neural network) that has been trained by another processor or system or by deep learning application processor **2100**. In at least one embodiment, processor **2100** may be used to perform one or more neural network use cases described herein.

Inference and/or training logic **615** are used to perform inferencing and/or training operations associated with one or more embodiments. In at least one embodiment, this logic can be used with components of these figures to generate one

or more images with visual effects approximated from similar effects applied to one or more lower resolution versions of those images.

FIG. 22 is a block diagram of a neuromorphic processor **2200**, according to at least one embodiment. In at least one embodiment, neuromorphic processor **2200** may receive one or more inputs from sources external to neuromorphic processor **2200**. In at least one embodiment, these inputs may be transmitted to one or more neurons **2202** within neuromorphic processor **2200**. In at least one embodiment, neurons **2202** and components thereof may be implemented using circuitry or logic, including one or more arithmetic logic units (ALUs). In at least one embodiment, neuromorphic processor **2200** may include, without limitation, thousands or millions of instances of neurons **2202**, but any suitable number of neurons **2202** may be used. In at least one embodiment, each instance of neuron **2202** may include a neuron input **2204** and a neuron output **2206**. In at least one embodiment, neurons **2202** may generate outputs that may be transmitted to inputs of other instances of neurons **2202**. For example, in at least one embodiment, neuron inputs **2204** and neuron outputs **2206** may be interconnected via synapses **2208**.

In at least one embodiment, neurons **2202** and synapses **2208** may be interconnected such that neuromorphic processor **2200** operates to process or analyze information received by neuromorphic processor **2200**. In at least one embodiment, neurons **2202** may transmit an output pulse (or “fire” or “spike”) when inputs received through neuron input **2204** exceed a threshold. In at least one embodiment, neurons **2202** may sum or integrate signals received at neuron inputs **2204**. For example, in at least one embodiment, neurons **2202** may be implemented as leaky integrate-and-fire neurons, wherein if a sum (referred to as a “membrane potential”) exceeds a threshold value, neuron **2202** may generate an output (or “fire”) using a transfer function such as a sigmoid or threshold function. In at least one embodiment, a leaky integrate-and-fire neuron may sum signals received at neuron inputs **2204** into a membrane potential and may also apply a decay factor (or leak) to reduce a membrane potential. In at least one embodiment, a leaky integrate-and-fire neuron may fire if multiple input signals are received at neuron inputs **2204** rapidly enough to exceed a threshold value (i.e., before a membrane potential decays too low to fire). In at least one embodiment, neurons **2202** may be implemented using circuits or logic that receive inputs, integrate inputs into a membrane potential, and decay a membrane potential. In at least one embodiment, inputs may be averaged, or any other suitable transfer function may be used. Furthermore, in at least one embodiment, neurons **2202** may include, without limitation, comparator circuits or logic that generate an output spike at neuron output **2206** when result of applying a transfer function to neuron input **2204** exceeds a threshold. In at least one embodiment, once neuron **2202** fires, it may disregard previously received input information by, for example, resetting a membrane potential to 0 or another suitable default value. In at least one embodiment, once membrane potential is reset to 0, neuron **2202** may resume normal operation after a suitable period of time (or refractory period).

In at least one embodiment, neurons **2202** may be interconnected through synapses **2208**. In at least one embodiment, synapses **2208** may operate to transmit signals from an output of a first neuron **2202** to an input of a second neuron **2202**. In at least one embodiment, neurons **2202** may transmit information over more than one instance of synapse **2208**. In at least one embodiment, one or more instances of

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neuron output **2206** may be connected, via an instance of synapse **2208**, to an instance of neuron input **2204** in same neuron **2202**. In at least one embodiment, an instance of neuron **2202** generating an output to be transmitted over an instance of synapse **2208** may be referred to as a “pre-synaptic neuron” with respect to that instance of synapse **2208**. In at least one embodiment, an instance of neuron **2202** receiving an input transmitted over an instance of synapse **2208** may be referred to as a “post-synaptic neuron” with respect to that instance of synapse **2208**. Because an instance of neuron **2202** may receive inputs from one or more instances of synapse **2208**, and may also transmit outputs over one or more instances of synapse **2208**, a single instance of neuron **2202** may therefore be both a “pre-synaptic neuron” and “post-synaptic neuron,” with respect to various instances of synapses **2208**, in at least one embodiment.

In at least one embodiment, neurons **2202** may be organized into one or more layers. Each instance of neuron **2202** may have one neuron output **2206** that may fan out through one or more synapses **2208** to one or more neuron inputs **2204**. In at least one embodiment, neuron outputs **2206** of neurons **2202** in a first layer **2210** may be connected to neuron inputs **2204** of neurons **2202** in a second layer **2212**. In at least one embodiment, layer **2210** may be referred to as a “feed-forward layer.” In at least one embodiment, each instance of neuron **2202** in an instance of first layer **2210** may fan out to each instance of neuron **2202** in second layer **2212**. In at least one embodiment, first layer **2210** may be referred to as a “fully connected feed-forward layer.” In at least one embodiment, each instance of neuron **2202** in an instance of second layer **2212** may fan out to fewer than all instances of neuron **2202** in a third layer **2214**. In at least one embodiment, second layer **2212** may be referred to as a “sparsely connected feed-forward layer.” In at least one embodiment, neurons **2202** in second layer **2212** may fan out to neurons **2202** in multiple other layers, including to neurons **2202** in (same) second layer **2212**. In at least one embodiment, second layer **2212** may be referred to as a “recurrent layer.” In at least one embodiment, neuromorphic processor **2200** may include, without limitation, any suitable combination of recurrent layers and feed-forward layers, including, without limitation, both sparsely connected feed-forward layers and fully connected feed-forward layers.

In at least one embodiment, neuromorphic processor **2200** may include, without limitation, a reconfigurable interconnect architecture or dedicated hard wired interconnects to connect synapse **2208** to neurons **2202**. In at least one embodiment, neuromorphic processor **2200** may include, without limitation, circuitry or logic that allows synapses to be allocated to different neurons **2202** as needed based on neural network topology and neuron fan-in/out. For example, in at least one embodiment, synapses **2208** may be connected to neurons **2202** using an interconnect fabric, such as network-on-chip, or with dedicated connections. In at least one embodiment, synapse interconnections and components thereof may be implemented using circuitry or logic.

Inference and/or training logic **615** are used to perform inferencing and/or training operations associated with one or more embodiments. In at least one embodiment, this logic can be used with components of these figures to generate one or more images with visual effects approximated from similar effects applied to one or more lower resolution versions of those images.

FIG. **23** is a block diagram of a processing system, according to at least one embodiment. In at least one

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embodiment, system **2300** includes one or more processors **2302** and one or more graphics processors **2308**, and may be a single processor desktop system, a multiprocessor workstation system, or a server system having a large number of processors **2302** or processor cores **2307**. In at least one embodiment, system **2300** is a processing platform incorporated within a system-on-a-chip (SoC) integrated circuit for use in mobile, handheld, or embedded devices.

In at least one embodiment, system **2300** can include, or be incorporated within a server-based gaming platform, a game console, including a game and media console, a mobile gaming console, a handheld game console, or an online game console. In at least one embodiment, system **2300** is a mobile phone, smart phone, tablet computing device or mobile Internet device. In at least one embodiment, processing system **2300** can also include, couple with, or be integrated within a wearable device, such as a smart watch wearable device, smart eyewear device, augmented reality device, or virtual reality device. In at least one embodiment, processing system **2300** is a television or set top box device having one or more processors **2302** and a graphical interface generated by one or more graphics processors **2308**.

In at least one embodiment, one or more processors **2302** each include one or more processor cores **2307** to process instructions which, when executed, perform operations for system and user software. In at least one embodiment, each of one or more processor cores **2307** is configured to process a specific instruction set **2309**. In at least one embodiment, instruction set **2309** may facilitate Complex Instruction Set Computing (CISC), Reduced Instruction Set Computing (RISC), or computing via a Very Long Instruction Word (VLIW). In at least one embodiment, processor cores **2307** may each process a different instruction set **2309**, which may include instructions to facilitate emulation of other instruction sets. In at least one embodiment, processor core **2307** may also include other processing devices, such as a Digital Signal Processor (DSP).

In at least one embodiment, processor **2302** includes cache memory **2304**. In at least one embodiment, processor **2302** can have a single internal cache or multiple levels of internal cache. In at least one embodiment, cache memory is shared among various components of processor **2302**. In at least one embodiment, processor **2302** also uses an external cache (e.g., a Level-3 (L3) cache or Last Level Cache (LLC)) (not shown), which may be shared among processor cores **2307** using known cache coherency techniques. In at least one embodiment, register file **2306** is additionally included in processor **2302** which may include different types of registers for storing different types of data (e.g., integer registers, floating point registers, status registers, and an instruction pointer register). In at least one embodiment, register file **2306** may include general-purpose registers or other registers.

In at least one embodiment, one or more processor(s) **2302** are coupled with one or more interface bus(es) **2310** to transmit communication signals such as address, data, or control signals between processor **2302** and other components in system **2300**. In at least one embodiment, interface bus **2310**, in one embodiment, can be a processor bus, such as a version of a Direct Media Interface (DMI) bus. In at least one embodiment, interface **2310** is not limited to a DMI bus, and may include one or more Peripheral Component Interconnect buses (e.g., PCI, PCI Express), memory busses, or other types of interface busses. In at least one embodiment processor(s) **2302** include an integrated memory controller **2316** and a platform controller hub **2330**. In at least

one embodiment, memory controller **2316** facilitates communication between a memory device and other components of system **2300**, while platform controller hub (PCH) **2330** provides connections to I/O devices via a local I/O bus.

In at least one embodiment, memory device **2320** can be a dynamic random access memory (DRAM) device, a static random access memory (SRAM) device, flash memory device, phase-change memory device, or some other memory device having suitable performance to serve as process memory. In at least one embodiment memory device **2320** can operate as system memory for system **2300**, to store data **2322** and instructions **2321** for use when one or more processors **2302** executes an application or process. In at least one embodiment, memory controller **2316** also couples with an optional external graphics processor **2312**, which may communicate with one or more graphics processors **2308** in processors **2302** to perform graphics and media operations. In at least one embodiment, a display device **2311** can connect to processor(s) **2302**. In at least one embodiment display device **2311** can include one or more of an internal display device, as in a mobile electronic device or a laptop device or an external display device attached via a display interface (e.g., DisplayPort, etc.). In at least one embodiment, display device **2311** can include a head mounted display (HMD) such as a stereoscopic display device for use in virtual reality (VR) applications or augmented reality (AR) applications.

In at least one embodiment, platform controller hub **2330** enables peripherals to connect to memory device **2320** and processor **2302** via a high-speed I/O bus. In at least one embodiment, I/O peripherals include, but are not limited to, an audio controller **2346**, a network controller **2334**, a firmware interface **2328**, a wireless transceiver **2326**, touch sensors **2325**, a data storage device **2324** (e.g., hard disk drive, flash memory, etc.). In at least one embodiment, data storage device **2324** can connect via a storage interface (e.g., SATA) or via a peripheral bus, such as a Peripheral Component Interconnect bus (e.g., PCI, PCI Express). In at least one embodiment, touch sensors **2325** can include touch screen sensors, pressure sensors, or fingerprint sensors. In at least one embodiment, wireless transceiver **2326** can be a Wi-Fi transceiver, a Bluetooth transceiver, or a mobile network transceiver such as a 3G, 4G, or Long Term Evolution (LTE) transceiver. In at least one embodiment, firmware interface **2328** enables communication with system firmware, and can be, for example, a unified extensible firmware interface (UEFI). In at least one embodiment, network controller **2334** can enable a network connection to a wired network. In at least one embodiment, a high-performance network controller (not shown) couples with interface bus **2310**. In at least one embodiment, audio controller **2346** is a multi-channel high definition audio controller. In at least one embodiment, system **2300** includes an optional legacy I/O controller **2340** for coupling legacy (e.g., Personal System 2 (PS/2)) devices to system. In at least one embodiment, platform controller hub **2330** can also connect to one or more Universal Serial Bus (USB) controllers **2342** connect input devices, such as keyboard and mouse **2343** combinations, a camera **2344**, or other USB input devices.

In at least one embodiment, an instance of memory controller **2316** and platform controller hub **2330** may be integrated into a discreet external graphics processor, such as external graphics processor **2312**. In at least one embodiment, platform controller hub **2330** and/or memory controller **2316** may be external to one or more processor(s) **2302**. For example, in at least one embodiment, system **2300** can

include an external memory controller **2316** and platform controller hub **2330**, which may be configured as a memory controller hub and peripheral controller hub within a system chipset that is in communication with processor(s) **2302**.

Inference and/or training logic **615** are used to perform inferencing and/or training operations associated with one or more embodiments. Details regarding inference and/or training logic **615** are provided below in conjunction with FIGS. **6A** and/or **6B**. In at least one embodiment portions or all of inference and/or training logic **615** may be incorporated into graphics processor **2300**. For example, in at least one embodiment, training and/or inferencing techniques described herein may use one or more of ALUs embodied in graphics processor **2312**. Moreover, in at least one embodiment, inferencing and/or training operations described herein may be done using logic other than logic illustrated in FIG. **6A** or **6B**. In at least one embodiment, weight parameters may be stored in on-chip or off-chip memory and/or registers (shown or not shown) that configure ALUs of graphics processor **2300** to perform one or more machine learning algorithms, neural network architectures, use cases, or training techniques described herein.

Inference and/or training logic **615** are used to perform inferencing and/or training operations associated with one or more embodiments. In at least one embodiment, this logic can be used with components of these figures to generate one or more images with visual effects approximated from similar effects applied to one or more lower resolution versions of those images.

FIG. **24** is a block diagram of a processor **2400** having one or more processor cores **2402A-2402N**, an integrated memory controller **2414**, and an integrated graphics processor **2408**, according to at least one embodiment. In at least one embodiment, processor **2400** can include additional cores up to and including additional core **2402N** represented by dashed lined boxes. In at least one embodiment, each of processor cores **2402A-2402N** includes one or more internal cache units **2404A-2404N**. In at least one embodiment, each processor core also has access to one or more shared cached units **2406**.

In at least one embodiment, internal cache units **2404A-2404N** and shared cache units **2406** represent a cache memory hierarchy within processor **2400**. In at least one embodiment, cache memory units **2404A-2404N** may include at least one level of instruction and data cache within each processor core and one or more levels of shared mid-level cache, such as a Level 2 (L2), Level 3 (L3), Level 4 (L4), or other levels of cache, where a highest level of cache before external memory is classified as an LLC. In at least one embodiment, cache coherency logic maintains coherency between various cache units **2406** and **2404A-2404N**.

In at least one embodiment, processor **2400** may also include a set of one or more bus controller units **2416** and a system agent core **2410**. In at least one embodiment, one or more bus controller units **2416** manage a set of peripheral buses, such as one or more PCI or PCI express busses. In at least one embodiment, system agent core **2410** provides management functionality for various processor components. In at least one embodiment, system agent core **2410** includes one or more integrated memory controllers **2414** to manage access to various external memory devices (not shown).

In at least one embodiment, one or more of processor cores **2402A-2402N** include support for simultaneous multi-threading. In at least one embodiment, system agent core **2410** includes components for coordinating and operating

cores **2402A-2402N** during multi-threaded processing. In at least one embodiment, system agent core **2410** may additionally include a power control unit (PCU), which includes logic and components to regulate one or more power states of processor cores **2402A-2402N** and graphics processor **2408**.

In at least one embodiment, processor **2400** additionally includes graphics processor **2408** to execute graphics processing operations. In at least one embodiment, graphics processor **2408** couples with shared cache units **2406**, and system agent core **2410**, including one or more integrated memory controllers **2414**. In at least one embodiment, system agent core **2410** also includes a display controller **2411** to drive graphics processor output to one or more coupled displays. In at least one embodiment, display controller **2411** may also be a separate module coupled with graphics processor **2408** via at least one interconnect, or may be integrated within graphics processor **2408**.

In at least one embodiment, a ring based interconnect unit **2412** is used to couple internal components of processor **2400**. In at least one embodiment, an alternative interconnect unit may be used, such as a point-to-point interconnect, a switched interconnect, or other techniques. In at least one embodiment, graphics processor **2408** couples with ring interconnect **2412** via an I/O link **2413**.

In at least one embodiment, I/O link **2413** represents at least one of multiple varieties of I/O interconnects, including an on package I/O interconnect which facilitates communication between various processor components and a high-performance embedded memory module **2418**, such as an eDRAM module. In at least one embodiment, each of processor cores **2402A-2402N** and graphics processor **2408** use embedded memory modules **2418** as a shared Last Level Cache.

In at least one embodiment, processor cores **2402A-2402N** are homogenous cores executing a common instruction set architecture. In at least one embodiment, processor cores **2402A-2402N** are heterogeneous in terms of instruction set architecture (ISA), where one or more of processor cores **2402A-2402N** execute a common instruction set, while one or more other cores of processor cores **2402A-2402N** executes a subset of a common instruction set or a different instruction set. In at least one embodiment, processor cores **2402A-2402N** are heterogeneous in terms of microarchitecture, where one or more cores having a relatively higher power consumption couple with one or more power cores having a lower power consumption. In at least one embodiment, processor **2400** can be implemented on one or more chips or as an SoC integrated circuit.

Inference and/or training logic **615** are used to perform inferencing and/or training operations associated with one or more embodiments. Details regarding inference and/or training logic **615** are provided below in conjunction with FIGS. **6A** and/or **6B**. In at least one embodiment portions or all of inference and/or training logic **615** may be incorporated into processor **2400**. For example, in at least one embodiment, training and/or inferencing techniques described herein may use one or more of ALUs embodied in graphics processor **2312**, graphics core(s) **2402A-2402N**, or other components in FIG. **24**. Moreover, in at least one embodiment, inferencing and/or training operations described herein may be done using logic other than logic illustrated in FIG. **6A** or **6B**. In at least one embodiment, weight parameters may be stored in on-chip or off-chip memory and/or registers (shown or not shown) that configure ALUs of graphics processor **2400** to

perform one or more machine learning algorithms, neural network architectures, use cases, or training techniques described herein.

Inference and/or training logic **615** are used to perform inferencing and/or training operations associated with one or more embodiments. In at least one embodiment, this logic can be used with components of these figures to generate one or more images with visual effects approximated from similar effects applied to one or more lower resolution versions of those images.

FIG. **25** is a block diagram of hardware logic of a graphics processor core **2500**, according to at least one embodiment described herein. In at least one embodiment, graphics processor core **2500** is included within a graphics core array. In at least one embodiment, graphics processor core **2500**, sometimes referred to as a core slice, can be one or multiple graphics cores within a modular graphics processor. In at least one embodiment, graphics processor core **2500** is exemplary of one graphics core slice, and a graphics processor as described herein may include multiple graphics core slices based on target power and performance envelopes. In at least one embodiment, each graphics core **2500** can include a fixed function block **2530** coupled with multiple sub-cores **2501A-2501F**, also referred to as sub-slices, that include modular blocks of general-purpose and fixed function logic.

In at least one embodiment, fixed function block **2530** includes a geometry/fixed function pipeline **2536** that can be shared by all sub-cores in graphics processor **2500**, for example, in lower performance and/or lower power graphics processor implementations. In at least one embodiment, geometry/fixed function pipeline **2536** includes a 3D fixed function pipeline, a video front-end unit, a thread spawner and thread dispatcher, and a unified return buffer manager, which manages unified return buffers.

In at least one embodiment fixed, function block **2530** also includes a graphics SoC interface **2537**, a graphics microcontroller **2538**, and a media pipeline **2539**. In at least one embodiment fixed, graphics SoC interface **2537** provides an interface between graphics core **2500** and other processor cores within a system on a chip integrated circuit. In at least one embodiment, graphics microcontroller **2538** is a programmable sub-processor that is configurable to manage various functions of graphics processor **2500**, including thread dispatch, scheduling, and pre-emption. In at least one embodiment, media pipeline **2539** includes logic to facilitate decoding, encoding, pre-processing, and/or post-processing of multimedia data, including image and video data. In at least one embodiment, media pipeline **2539** implements media operations via requests to compute or sampling logic within sub-cores **2501-2501F**.

In at least one embodiment, SoC interface **2537** enables graphics core **2500** to communicate with general-purpose application processor cores (e.g., CPUs) and/or other components within an SoC, including memory hierarchy elements such as a shared last level cache memory, system RAM, and/or embedded on-chip or on-package DRAM. In at least one embodiment, SoC interface **2537** can also enable communication with fixed function devices within an SoC, such as camera imaging pipelines, and enables use of and/or implements global memory atomics that may be shared between graphics core **2500** and CPUs within an SoC. In at least one embodiment, SoC interface **2537** can also implement power management controls for graphics core **2500** and enable an interface between a clock domain of graphic core **2500** and other clock domains within an SoC. In at least one embodiment, SoC interface **2537** enables receipt of

command buffers from a command streamer and global thread dispatcher that are configured to provide commands and instructions to each of one or more graphics cores within a graphics processor. In at least one embodiment, commands and instructions can be dispatched to media pipeline **2539**, when media operations are to be performed, or a geometry and fixed function pipeline (e.g., geometry and fixed function pipeline **2536**, geometry and fixed function pipeline **2514**) when graphics processing operations are to be performed.

In at least one embodiment, graphics microcontroller **2538** can be configured to perform various scheduling and management tasks for graphics core **2500**. In at least one embodiment, graphics microcontroller **2538** can perform graphics and/or compute workload scheduling on various graphics parallel engines within execution unit (EU) arrays **2502A-2502F**, **2504A-2504F** within sub-cores **2501A-2501F**. In at least one embodiment, host software executing on a CPU core of an SoC including graphics core **2500** can submit workloads one of multiple graphic processor doorbells, which invokes a scheduling operation on an appropriate graphics engine. In at least one embodiment, scheduling operations include determining which workload to run next, submitting a workload to a command streamer, preempting existing workloads running on an engine, monitoring progress of a workload, and notifying host software when a workload is complete. In at least one embodiment, graphics microcontroller **2538** can also facilitate low-power or idle states for graphics core **2500**, providing graphics core **2500** with an ability to save and restore registers within graphics core **2500** across low-power state transitions independently from an operating system and/or graphics driver software on a system.

In at least one embodiment, graphics core **2500** may have greater than or fewer than illustrated sub-cores **2501A-2501F**, up to N modular sub-cores. For each set of N sub-cores, in at least one embodiment, graphics core **2500** can also include shared function logic **2510**, shared and/or cache memory **2512**, a geometry/fixed function pipeline **2514**, as well as additional/fixed function logic **2516** to accelerate various graphics and compute processing operations. In at least one embodiment, shared function logic **2510** can include logic units (e.g., sampler, math, and/or inter-thread communication logic) that can be shared by each N sub-cores within graphics core **2500**. In at least one embodiment fixed, shared and/or cache memory **2512** can be a last-level cache for N sub-cores **2501A-2501F** within graphics core **2500** and can also serve as shared memory that is accessible by multiple sub-cores. In at least one embodiment, geometry/fixed function pipeline **2514** can be included instead of geometry/fixed function pipeline **2536** within fixed function block **2530** and can include same or similar logic units.

In at least one embodiment, graphics core **2500** includes additional fixed function logic **2516** that can include various fixed function acceleration logic for use by graphics core **2500**. In at least one embodiment, additional fixed function logic **2516** includes an additional geometry pipeline for use in position only shading. In position-only shading, at least two geometry pipelines exist, whereas in a full geometry pipeline within geometry/fixed function pipeline **2516**, **2536**, and a cull pipeline, which is an additional geometry pipeline which may be included within additional fixed function logic **2516**. In at least one embodiment, cull pipeline is a trimmed down version of a full geometry pipeline. In at least one embodiment, a full pipeline and a cull pipeline can execute different instances of an application, each

instance having a separate context. In at least one embodiment, position only shading can hide long cull runs of discarded triangles, enabling shading to be completed earlier in some instances. For example, in at least one embodiment, cull pipeline logic within additional fixed function logic **2516** can execute position shaders in parallel with a main application and generally generates critical results faster than a full pipeline, as cull pipeline fetches and shades position attribute of vertices, without performing rasterization and rendering of pixels to a frame buffer. In at least one embodiment, cull pipeline can use generated critical results to compute visibility information for all triangles without regard to whether those triangles are culled. In at least one embodiment, full pipeline (which in this instance may be referred to as a replay pipeline) can consume visibility information to skip culled triangles to shade only visible triangles that are finally passed to a rasterization phase.

In at least one embodiment, additional fixed function logic **2516** can also include machine-learning acceleration logic, such as fixed function matrix multiplication logic, for implementations including optimizations for machine learning training or inferencing.

In at least one embodiment, within each graphics sub-core **2501A-2501F** includes a set of execution resources that may be used to perform graphics, media, and compute operations in response to requests by graphics pipeline, media pipeline, or shader programs. In at least one embodiment, graphics sub-cores **2501A-2501F** include multiple EU arrays **2502A-2502F**, **2504A-2504F**, thread dispatch and inter-thread communication (TD/IC) logic **2503A-2503F**, a 3D (e.g., texture) sampler **2505A-2505F**, a media sampler **2506A-2506F**, a shader processor **2507A-2507F**, and shared local memory (SLM) **2508A-2508F**. EU arrays **2502A-2502F**, **2504A-2504F** each include multiple execution units, which are general-purpose graphics processing units capable of performing floating-point and integer/fixed-point logic operations in service of a graphics, media, or compute operation, including graphics, media, or compute shader programs. In at least one embodiment, TD/IC logic **2503A-2503F** performs local thread dispatch and thread control operations for execution units within a sub-core and facilitate communication between threads executing on execution units of a sub-core. In at least one embodiment, 3D sampler **2505A-2505F** can read texture or other 3D graphics related data into memory. In at least one embodiment, 3D sampler can read texture data differently based on a configured sample state and texture format associated with a given texture. In at least one embodiment, media sampler **2506A-2506F** can perform similar read operations based on a type and format associated with media data. In at least one embodiment, each graphics sub-core **2501A-2501F** can alternately include a unified 3D and media sampler. In at least one embodiment, threads executing on execution units within each of sub-cores **2501A-2501F** can make use of shared local memory **2508A-2508F** within each sub-core, to enable threads executing within a thread group to execute using a common pool of on-chip memory.

Inference and/or training logic **615** are used to perform inferencing and/or training operations associated with one or more embodiments. Details regarding inference and/or training logic **615** are provided below in conjunction with FIGS. **6A** and/or **6B**. In at least one embodiment, portions or all of inference and/or training logic **615** may be incorporated into graphics processor **2510**. For example, in at least one embodiment, training and/or inferencing techniques described herein may use one or more of ALUs embodied in graphics processor **2312**, graphics microcontroller **2538**,

geometry & fixed function pipeline **2514** and **2536**, or other logic in FIG. **24**. Moreover, in at least one embodiment, inferencing and/or training operations described herein may be done using logic other than logic illustrated in FIG. **6A** or **6B**. In at least one embodiment, weight parameters may be stored in on-chip or off-chip memory and/or registers (shown or not shown) that configure ALUs of graphics processor **2500** to perform one or more machine learning algorithms, neural network architectures, use cases, or training techniques described herein.

Inference and/or training logic **615** are used to perform inferencing and/or training operations associated with one or more embodiments. In at least one embodiment, this logic can be used with components of these figures to generate one or more images with visual effects approximated from similar effects applied to one or more lower resolution versions of those images.

FIGS. **26A-26B** illustrate thread execution logic **2600** including an array of processing elements of a graphics processor core according to at least one embodiment. FIG. **26A** illustrates at least one embodiment, in which thread execution logic **2600** is used. FIG. **26B** illustrates exemplary internal details of an execution unit, according to at least one embodiment.

As illustrated in FIG. **26A**, in at least one embodiment, thread execution logic **2600** includes a shader processor **2602**, a thread dispatcher **2604**, instruction cache **2606**, a scalable execution unit array including a plurality of execution units **2608A-2608N**, sampler(s) **2610**, a data cache **2612**, and a data port **2614**. In at least one embodiment a scalable execution unit array can dynamically scale by enabling or disabling one or more execution units (e.g., any of execution unit **2608A**, **2608B**, **2608C**, **2608D**, through **2608N-1** and **2608N**) based on computational requirements of a workload, for example. In at least one embodiment, scalable execution units are interconnected via an interconnect fabric that links to each of execution unit. In at least one embodiment, thread execution logic **2600** includes one or more connections to memory, such as system memory or cache memory, through one or more of instruction cache **2606**, data port **2614**, sampler **2610**, and execution units **2608A-2608N**. In at least one embodiment, each execution unit (e.g., **2608A**) is a stand-alone programmable general-purpose computational unit that is capable of executing multiple simultaneous hardware threads while processing multiple data elements in parallel for each thread. In at least one embodiment, array of execution units **2608A-2608N** is scalable to include any number individual execution units.

In at least one embodiment, execution units **2608A-2608N** are primarily used to execute shader programs. In at least one embodiment, shader processor **2602** can process various shader programs and dispatch execution threads associated with shader programs via a thread dispatcher **2604**. In at least one embodiment, thread dispatcher **2604** includes logic to arbitrate thread initiation requests from graphics and media pipelines and instantiate requested threads on one or more execution units in execution units **2608A-2608N**. For example, in at least one embodiment, a geometry pipeline can dispatch vertex, tessellation, or geometry shaders to thread execution logic for processing. In at least one embodiment, thread dispatcher **2604** can also process runtime thread spawning requests from executing shader programs.

In at least one embodiment, execution units **2608A-2608N** support an instruction set that includes native support for many standard 3D graphics shader instructions, such that shader programs from graphics libraries (e.g., Direct 3D and

OpenGL) are executed with a minimal translation. In at least one embodiment, execution units support vertex and geometry processing (e.g., vertex programs, geometry programs, vertex shaders), pixel processing (e.g., pixel shaders, fragment shaders) and general-purpose processing (e.g., compute and media shaders). In at least one embodiment, each of execution units **2608A-2608N**, which include one or more arithmetic logic units (ALUs), is capable of multi-issue single instruction multiple data (SIMD) execution and multi-threaded operation enables an efficient execution environment despite higher latency memory accesses. In at least one embodiment, each hardware thread within each execution unit has a dedicated high-bandwidth register file and associated independent thread-state. In at least one embodiment, execution is multi-issue per clock to pipelines capable of integer, single and double precision floating point operations, SIMD branch capability, logical operations, transcendental operations, and other miscellaneous operations. In at least one embodiment, while waiting for data from memory or one of shared functions, dependency logic within execution units **2608A-2608N** causes a waiting thread to sleep until requested data has been returned. In at least one embodiment, while a waiting thread is sleeping, hardware resources may be devoted to processing other threads. For example, in at least one embodiment, during a delay associated with a vertex shader operation, an execution unit can perform operations for a pixel shader, fragment shader, or another type of shader program, including a different vertex shader.

In at least one embodiment, each execution unit in execution units **2608A-2608N** operates on arrays of data elements. In at least one embodiment, a number of data elements is "execution size," or number of channels for an instruction. In at least one embodiment, an execution channel is a logical unit of execution for data element access, masking, and flow control within instructions. In at least one embodiment, a number of channels may be independent of a number of physical Arithmetic Logic Units (ALUs) or Floating Point Units (FPUs) for a particular graphics processor. In at least one embodiment, execution units **2608A-2608N** support integer and floating-point data types.

In at least one embodiment, an execution unit instruction set includes SIMD instructions. In at least one embodiment, various data elements can be stored as a packed data type in a register and execution unit will process various elements based on data size of elements. For example, in at least one embodiment, when operating on a 256-bit wide vector, 256 bits of a vector are stored in a register and an execution unit operates on a vector as four separate 64-bit packed data elements (Quad-Word (QW) size data elements), eight separate 32-bit packed data elements (Double Word (DW) size data elements), sixteen separate 16-bit packed data elements (Word (W) size data elements), or thirty-two separate 8-bit data elements (byte (B) size data elements). However, in at least one embodiment, different vector widths and register sizes are possible.

In at least one embodiment, one or more execution units can be combined into a fused execution unit **2609A-2609N** having thread control logic (**2607A-2607N**) that is common to fused EUs. In at least one embodiment, multiple EUs can be fused into an EU group. In at least one embodiment, each EU in fused EU group can be configured to execute a separate SIMD hardware thread. Number of EUs in a fused EU group can vary according to various embodiments. In at least one embodiment, various SIMD widths can be performed per-EU, including but not limited to SIMD8, SIMD16, and SIMD32. In at least one embodiment, each

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fused graphics execution unit **2609A-2609N** includes at least two execution units. For example, in at least one embodiment, fused execution unit **2609A** includes a first EU **2608A**, second EU **2608B**, and thread control logic **2607A** that is common to first EU **2608A** and second EU **2608B**. In at least one embodiment, thread control logic **2607A** controls threads executed on fused graphics execution unit **2609A**, allowing each EU within fused execution units **2609A-2609N** to execute using a common instruction pointer register.

In at least one embodiment, one or more internal instruction caches (e.g., **2606**) are included in thread execution logic **2600** to cache thread instructions for execution units. In at least one embodiment, one or more data caches (e.g., **2612**) are included to cache thread data during thread execution. In at least one embodiment, a sampler **2610** is included to provide texture sampling for 3D operations and media sampling for media operations. In at least one embodiment, sampler **2610** includes specialized texture or media sampling functionality to process texture or media data during a sampling process before providing sampled data to an execution unit.

During execution, in at least one embodiment, graphics and media pipelines send thread initiation requests to thread execution logic **2600** via thread spawning and dispatch logic. In at least one embodiment, once a group of geometric objects has been processed and rasterized into pixel data, pixel processor logic (e.g., pixel shader logic, fragment shader logic, etc.) within shader processor **2602** is invoked to further compute output information and cause results to be written to output surfaces (e.g., color buffers, depth buffers, stencil buffers, etc.). In at least one embodiment, a pixel shader or fragment shader calculates values of various vertex attributes that are to be interpolated across a rasterized object. In at least one embodiment, pixel processor logic within shader processor **2602** then executes an application programming interface (API)-supplied pixel or fragment shader program. In at least one embodiment, to execute a shader program, shader processor **2602** dispatches threads to an execution unit (e.g., **2608A**) via thread dispatcher **2604**. In at least one embodiment, shader processor **2602** uses texture sampling logic in sampler **2610** to access texture data in texture maps stored in memory. In at least one embodiment, arithmetic operations on texture data and input geometry data compute pixel color data for each geometric fragment, or discards one or more pixels from further processing.

In at least one embodiment, data port **2614** provides a memory access mechanism for thread execution logic **2600** to output processed data to memory for further processing on a graphics processor output pipeline. In at least one embodiment, data port **2614** includes or couples to one or more cache memories (e.g., data cache **2612**) to cache data for memory access via a data port.

As illustrated in FIG. 26B, in at least one embodiment, a graphics execution unit **2608** can include an instruction fetch unit **2637**, a general register file array (GRF) **2624**, an architectural register file array (ARF) **2626**, a thread arbiter **2622**, a send unit **2630**, a branch unit **2632**, a set of SIMD floating point units (FPUs) **2634**, and, in at least one embodiment, a set of dedicated integer SIMD ALUs **2635**. In at least one embodiment, GRF **2624** and ARF **2626** includes a set of general register files and architecture register files associated with each simultaneous hardware thread that may be active in graphics execution unit **2608**. In at least one embodiment, per thread architectural state is maintained in ARF **2626**, while data used during thread

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execution is stored in GRF **2624**. In at least one embodiment, execution state of each thread, including instruction pointers for each thread, can be held in thread-specific registers in ARF **2626**.

In at least one embodiment, graphics execution unit **2608** has an architecture that is a combination of Simultaneous Multi-Threading (SMT) and fine-grained Interleaved Multi-Threading (IMT). In at least one embodiment, architecture has a modular configuration that can be fine-tuned at design time based on a target number of simultaneous threads and number of registers per execution unit, where execution unit resources are divided across logic used to execute multiple simultaneous threads.

In at least one embodiment, graphics execution unit **2608** can co-issue multiple instructions, which may each be different instructions. In at least one embodiment, thread arbiter **2622** of graphics execution unit thread **2608** can dispatch instructions to one of send unit **2630**, branch unit **2642**, or SIMD FPU(s) **2634** for execution. In at least one embodiment, each execution thread can access **128** general-purpose registers within GRF **2624**, where each register can store 32 bytes, accessible as a SIMD 8-element vector of 32-bit data elements. In at least one embodiment, each execution unit thread has access to 4 Kbytes within GRF **2624**, although embodiments are not so limited, and greater or fewer register resources may be provided in other embodiments. In at least one embodiment, up to seven threads can execute simultaneously, although a number of threads per execution unit can also vary according to embodiments. In at least one embodiment, in which seven threads may access 4 Kbytes, GRF **2624** can store a total of 28 Kbytes. In at least one embodiment, flexible addressing modes can permit registers to be addressed together to build effectively wider registers or to represent strided rectangular block data structures.

In at least one embodiment, memory operations, sampler operations, and other longer-latency system communications are dispatched via "send" instructions that are executed by message passing send unit **2630**. In at least one embodiment, branch instructions are dispatched to a dedicated branch unit **2632** to facilitate SIMD divergence and eventual convergence.

In at least one embodiment graphics execution unit **2608** includes one or more SIMD floating point units (FPU(s)) **2634** to perform floating-point operations. In at least one embodiment, FPU(s) **2634** also support integer computation. In at least one embodiment FPU(s) **2634** can SIMD execute up to M number of 32-bit floating-point (or integer) operations, or SIMD execute up to 2M 16-bit integer or 16-bit floating-point operations. In at least one embodiment, at least one of FPU(s) provides extended math capability to support high-throughput transcendental math functions and double precision 64-bit floating-point. In at least one embodiment, a set of 8-bit integer SIMD ALUs **2635** are also present, and may be specifically optimized to perform operations associated with machine learning computations.

In at least one embodiment, arrays of multiple instances of graphics execution unit **2608** can be instantiated in a graphics sub-core grouping (e.g., a sub-slice). In at least one embodiment, execution unit **2608** can execute instructions across a plurality of execution channels. In at least one embodiment, each thread executed on graphics execution unit **2608** is executed on a different channel.

Inference and/or training logic **615** are used to perform inferencing and/or training operations associated with one or more embodiments. Details regarding inference and/or training logic **615** are provided below in conjunction with FIGS.

6A and/or 6B. In at least one embodiment, portions or all of inference and/or training logic 615 may be incorporated into execution logic 2600. Moreover, in at least one embodiment, inferencing and/or training operations described herein may be done using logic other than logic illustrated in FIG. 6A or 6B. In at least one embodiment, weight parameters may be stored in on-chip or off-chip memory and/or registers (shown or not shown) that configure ALUs of execution logic 2600 to perform one or more machine learning algorithms, neural network architectures, use cases, or training techniques described herein.

Inference and/or training logic 615 are used to perform inferencing and/or training operations associated with one or more embodiments. In at least one embodiment, this logic can be used with components of these figures to generate one or more images with visual effects approximated from similar effects applied to one or more lower resolution versions of those images.

FIG. 27 illustrates a parallel processing unit (“PPU”) 2700, according to at least one embodiment. In at least one embodiment, PPU 2700 is configured with machine-readable code that, if executed by PPU 2700, causes PPU 2700 to perform some or all of processes and techniques described throughout this disclosure. In at least one embodiment, PPU 2700 is a multi-threaded processor that is implemented on one or more integrated circuit devices and that utilizes multithreading as a latency-hiding technique designed to process computer-readable instructions (also referred to as machine-readable instructions or simply instructions) on multiple threads in parallel. In at least one embodiment, a thread refers to a thread of execution and is an instantiation of a set of instructions configured to be executed by PPU 2700. In at least one embodiment, PPU 2700 is a graphics processing unit (“GPU”) configured to implement a graphics rendering pipeline for processing three-dimensional (“3D”) graphics data in order to generate two-dimensional (“2D”) image data for display on a display device such as a liquid crystal display (“LCD”) device. In at least one embodiment, PPU 2700 is utilized to perform computations such as linear algebra operations and machine-learning operations. FIG. 27 illustrates an example parallel processor for illustrative purposes only and should be construed as a non-limiting example of processor architectures contemplated within scope of this disclosure and that any suitable processor may be employed to supplement and/or substitute for same.

In at least one embodiment, one or more PPUs 2700 are configured to accelerate High Performance Computing (“HPC”), data center, and machine learning applications. In at least one embodiment, PPU 2700 is configured to accelerate deep learning systems and applications including following non-limiting examples: autonomous vehicle platforms, deep learning, high-accuracy speech, image, text recognition systems, intelligent video analytics, molecular simulations, drug discovery, disease diagnosis, weather forecasting, big data analytics, astronomy, molecular dynamics simulation, financial modeling, robotics, factory automation, real-time language translation, online search optimizations, and personalized user recommendations, and more.

In at least one embodiment, PPU 2700 includes, without limitation, an Input/Output (“I/O”) unit 2706, a front-end unit 2710, a scheduler unit 2712, a work distribution unit 2714, a hub 2716, a crossbar (“Xbar”) 2720, one or more general processing clusters (“GPCs”) 2718, and one or more partition units (“memory partition units”) 2722. In at least one embodiment, PPU 2700 is connected to a host processor or other PPUs 2700 via one or more high-speed GPU interconnects (“GPU interconnects”) 2708. In at least one

embodiment, PPU 2700 is connected to a host processor or other peripheral devices via an interconnect 2702. In at least one embodiment, PPU 2700 is connected to a local memory comprising one or more memory devices (“memory”) 2704. In at least one embodiment, memory devices 2704 include, without limitation, one or more dynamic random access memory (“DRAM”) devices. In at least one embodiment, one or more DRAM devices are configured and/or configurable as high-bandwidth memory (“HBM”) subsystems, with multiple DRAM dies stacked within each device.

In at least one embodiment, high-speed GPU interconnect 2708 may refer to a wire-based multi-lane communications link that is used by systems to scale and include one or more PPUs 2700 combined with one or more central processing units (“CPUs”), supports cache coherence between PPUs 2700 and CPUs, and CPU mastering. In at least one embodiment, data and/or commands are transmitted by high-speed GPU interconnect 2708 through hub 2716 to/from other units of PPU 2700 such as one or more copy engines, video encoders, video decoders, power management units, and other components which may not be explicitly illustrated in FIG. 27.

In at least one embodiment, I/O unit 2706 is configured to transmit and receive communications (e.g., commands, data) from a host processor (not illustrated in FIG. 27) over system bus 2702. In at least one embodiment, I/O unit 2706 communicates with host processor directly via system bus 2702 or through one or more intermediate devices such as a memory bridge. In at least one embodiment, I/O unit 2706 may communicate with one or more other processors, such as one or more of PPUs 2700 via system bus 2702. In at least one embodiment, I/O unit 2706 implements a Peripheral Component Interconnect Express (“PCIe”) interface for communications over a PCIe bus. In at least one embodiment, I/O unit 2706 implements interfaces for communicating with external devices.

In at least one embodiment, I/O unit 2706 decodes packets received via system bus 2702. In at least one embodiment, at least some packets represent commands configured to cause PPU 2700 to perform various operations. In at least one embodiment, I/O unit 2706 transmits decoded commands to various other units of PPU 2700 as specified by commands. In at least one embodiment, commands are transmitted to front-end unit 2710 and/or transmitted to hub 2716 or other units of PPU 2700 such as one or more copy engines, a video encoder, a video decoder, a power management unit, etc. (not explicitly illustrated in FIG. 27). In at least one embodiment, I/O unit 2706 is configured to route communications between and among various logical units of PPU 2700.

In at least one embodiment, a program executed by host processor encodes a command stream in a buffer that provides workloads to PPU 2700 for processing. In at least one embodiment, a workload comprises instructions and data to be processed by those instructions. In at least one embodiment, buffer is a region in a memory that is accessible (e.g., read/write) by both host processor and PPU 2700—a host interface unit may be configured to access buffer in a system memory connected to system bus 2702 via memory requests transmitted over system bus 2702 by I/O unit 2706. In at least one embodiment, host processor writes command stream to buffer and then transmits a pointer to start of command stream to PPU 2700 such that front-end unit 2710 receives pointers to one or more command streams and manages one or more command streams, reading commands from command streams and forwarding commands to various units of PPU 2700.

In at least one embodiment, front-end unit **2710** is coupled to scheduler unit **2712** that configures various GPCs **2718** to process tasks defined by one or more command streams. In at least one embodiment, scheduler unit **2712** is configured to track state information related to various tasks managed by scheduler unit **2712** where state information may indicate which of GPCs **2718** a task is assigned to, whether task is active or inactive, a priority level associated with task, and so forth. In at least one embodiment, scheduler unit **2712** manages execution of a plurality of tasks on one or more of GPCs **2718**.

In at least one embodiment, scheduler unit **2712** is coupled to work distribution unit **2714** that is configured to dispatch tasks for execution on GPCs **2718**. In at least one embodiment, work distribution unit **2714** tracks a number of scheduled tasks received from scheduler unit **2712** and work distribution unit **2714** manages a pending task pool and an active task pool for each of GPCs **2718**. In at least one embodiment, pending task pool comprises a number of slots (e.g., 32 slots) that contain tasks assigned to be processed by a particular GPC **2718**; active task pool may comprise a number of slots (e.g., 4 slots) for tasks that are actively being processed by GPCs **2718** such that as one of GPCs **2718** completes execution of a task, that task is evicted from active task pool for GPC **2718** and one of other tasks from pending task pool is selected and scheduled for execution on GPC **2718**. In at least one embodiment, if an active task is idle on GPC **2718**, such as while waiting for a data dependency to be resolved, then active task is evicted from GPC **2718** and returned to pending task pool while another task in pending task pool is selected and scheduled for execution on GPC **2718**.

In at least one embodiment, work distribution unit **2714** communicates with one or more GPCs **2718** via XBar **2720**. In at least one embodiment, XBar **2720** is an interconnect network that couples many of units of PPU **2700** to other units of PPU **2700** and can be configured to couple work distribution unit **2714** to a particular GPC **2718**. In at least one embodiment, one or more other units of PPU **2700** may also be connected to XBar **2720** via hub **2716**.

In at least one embodiment, tasks are managed by scheduler unit **2712** and dispatched to one of GPCs **2718** by work distribution unit **2714**. GPC **2718** is configured to process task and generate results. In at least one embodiment, results may be consumed by other tasks within GPC **2718**, routed to a different GPC **2718** via XBar **2720**, or stored in memory **2704**. In at least one embodiment, results can be written to memory **2704** via partition units **2722**, which implement a memory interface for reading and writing data to/from memory **2704**. In at least one embodiment, results can be transmitted to another PPU **2704** or CPU via high-speed GPU interconnect **2708**. In at least one embodiment, PPU **2700** includes, without limitation, a number U of partition units **2722** that is equal to number of separate and distinct memory devices **2704** coupled to PPU **2700**. In at least one embodiment, partition unit **2722** will be described in more detail below in conjunction with FIG. **29**.

In at least one embodiment, a host processor executes a driver kernel that implements an application programming interface (“API”) that enables one or more applications executing on host processor to schedule operations for execution on PPU **2700**. In at least one embodiment, multiple compute applications are simultaneously executed by PPU **2700** and PPU **2700** provides isolation, quality of service (“QoS”), and independent address spaces for multiple compute applications. In at least one embodiment, an application generates instructions (e.g., in form of API calls)

that cause driver kernel to generate one or more tasks for execution by PPU **2700** and driver kernel outputs tasks to one or more streams being processed by PPU **2700**. In at least one embodiment, each task comprises one or more groups of related threads, which may be referred to as a warp. In at least one embodiment, a warp comprises a plurality of related threads (e.g., 32 threads) that can be executed in parallel. In at least one embodiment, cooperating threads can refer to a plurality of threads including instructions to perform task and that exchange data through shared memory. In at least one embodiment, threads and cooperating threads are described in more detail, in accordance with at least one embodiment, in conjunction with FIG. **29**.

Inference and/or training logic **615** are used to perform inferencing and/or training operations associated with one or more embodiments. Details regarding inference and/or training logic **615** are provided below in conjunction with FIGS. **6A** and/or **6B**. In at least one embodiment, deep learning application processor is used to train a machine learning model, such as a neural network, to predict or infer information provided to PPU **2700**. In at least one embodiment, PPU **2700** is used to infer or predict information based on a trained machine learning model (e.g., neural network) that has been trained by another processor or system or by PPU **2700**. In at least one embodiment, PPU **2700** may be used to perform one or more neural network use cases described herein.

Inference and/or training logic **615** are used to perform inferencing and/or training operations associated with one or more embodiments. In at least one embodiment, this logic can be used with components of these figures to generate one or more images with visual effects approximated from similar effects applied to one or more lower resolution versions of those images.

FIG. **28** illustrates a general processing cluster (“GPC”) **2800**, according to at least one embodiment. In at least one embodiment, GPC **2800** is GPC **2718** of FIG. **27**. In at least one embodiment, each GPC **2800** includes, without limitation, a number of hardware units for processing tasks and each GPC **2800** includes, without limitation, a pipeline manager **2802**, a pre-raster operations unit (“PROP”) **2804**, a raster engine **2808**, a work distribution crossbar (“WDX”) **2816**, a memory management unit (“MMU”) **2818**, one or more Data Processing Clusters (“DPCs”) **2806**, and any suitable combination of parts.

In at least one embodiment, operation of GPC **2800** is controlled by pipeline manager **2802**. In at least one embodiment, pipeline manager **2802** manages configuration of one or more DPCs **2806** for processing tasks allocated to GPC **2800**. In at least one embodiment, pipeline manager **2802** configures at least one of one or more DPCs **2806** to implement at least a portion of a graphics rendering pipeline. In at least one embodiment, DPC **2806** is configured to execute a vertex shader program on a programmable streaming multi-processor (“SM”) **2814**. In at least one embodiment, pipeline manager **2802** is configured to route packets received from a work distribution unit to appropriate logical units within GPC **2800**, in at least one embodiment, and some packets may be routed to fixed function hardware units in PROP **2804** and/or raster engine **2808** while other packets may be routed to DPCs **2806** for processing by a primitive engine **2812** or SM **2814**. In at least one embodiment, pipeline manager **2802** configures at least one of DPCs **2806** to implement a neural network model and/or a computing pipeline.

In at least one embodiment, PROP unit **2804** is configured, in at least one embodiment, to route data generated by

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raster engine **2808** and DPCs **2806** to a Raster Operations (“ROP”) unit in partition unit **2722**, described in more detail above in conjunction with FIG. 27. In at least one embodiment, PROP unit **2804** is configured to perform optimizations for color blending, organize pixel data, perform address translations, and more. In at least one embodiment, raster engine **2808** includes, without limitation, a number of fixed function hardware units configured to perform various raster operations, in at least one embodiment, and raster engine **2808** includes, without limitation, a setup engine, a coarse raster engine, a culling engine, a clipping engine, a fine raster engine, a tile coalescing engine, and any suitable combination thereof. In at least one embodiment, setup engine receives transformed vertices and generates plane equations associated with geometric primitive defined by vertices; plane equations are transmitted to coarse raster engine to generate coverage information (e.g., an x, y coverage mask for a tile) for primitive; output of coarse raster engine is transmitted to culling engine where fragments associated with primitive that fail a z-test are culled, and transmitted to a clipping engine where fragments lying outside a viewing frustum are clipped. In at least one embodiment, fragments that survive clipping and culling are passed to fine raster engine to generate attributes for pixel fragments based on plane equations generated by setup engine. In at least one embodiment, output of raster engine **2808** comprises fragments to be processed by any suitable entity such as by a fragment shader implemented within DPC **2806**.

In at least one embodiment, each DPC **2806** included in GPC **2800** comprise, without limitation, an M-Pipe Controller (“MPC”) **2810**; primitive engine **2812**; one or more SMs **2814**; and any suitable combination thereof. In at least one embodiment, MPC **2810** controls operation of DPC **2806**, routing packets received from pipeline manager **2802** to appropriate units in DPC **2806**. In at least one embodiment, packets associated with a vertex are routed to primitive engine **2812**, which is configured to fetch vertex attributes associated with vertex from memory; in contrast, packets associated with a shader program may be transmitted to SM **2814**.

In at least one embodiment, SM **2814** comprises, without limitation, a programmable streaming processor that is configured to process tasks represented by a number of threads. In at least one embodiment, SM **2814** is multi-threaded and configured to execute a plurality of threads (e.g., 32 threads) from a particular group of threads concurrently and implements a Single-Instruction, Multiple-Data (“SIMD”) architecture where each thread in a group of threads (e.g., a warp) is configured to process a different set of data based on same set of instructions. In at least one embodiment, all threads in group of threads execute same instructions. In at least one embodiment, SM **2814** implements a Single-Instruction, Multiple Thread (“SIMT”) architecture wherein each thread in a group of threads is configured to process a different set of data based on same set of instructions, but where individual threads in group of threads are allowed to diverge during execution. In at least one embodiment, a program counter, call stack, and execution state is maintained for each warp, enabling concurrency between warps and serial execution within warps when threads within warp diverge. In another embodiment, a program counter, call stack, and execution state is maintained for each individual thread, enabling equal concurrency between all threads, within and between warps. In at least one embodiment, execution state is maintained for each individual thread and threads executing same instructions may be converged and executed in

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parallel for better efficiency. At least one embodiment of SM **2814** are described in more detail below.

In at least one embodiment, MMU **2818** provides an interface between GPC **2800** and memory partition unit (e.g., partition unit **2722** of FIG. 27) and MMU **2818** provides translation of virtual addresses into physical addresses, memory protection, and arbitration of memory requests. In at least one embodiment, MMU **2818** provides one or more translation lookaside buffers (“TLBs”) for performing translation of virtual addresses into physical addresses in memory.

Inference and/or training logic **615** are used to perform inferencing and/or training operations associated with one or more embodiments. Details regarding inference and/or training logic **615** are provided below in conjunction with FIGS. 6A and/or 6B. In at least one embodiment, deep learning application processor is used to train a machine learning model, such as a neural network, to predict or infer information provided to GPC **2800**. In at least one embodiment, GPC **2800** is used to infer or predict information based on a trained machine learning model (e.g., neural network) that has been trained by another processor or system or by GPC **2800**. In at least one embodiment, GPC **2800** may be used to perform one or more neural network use cases described herein.

Inference and/or training logic **615** are used to perform inferencing and/or training operations associated with one or more embodiments. In at least one embodiment, this logic can be used with components of these figures to generate one or more images with visual effects approximated from similar effects applied to one or more lower resolution versions of those images.

FIG. 29 illustrates a memory partition unit **2900** of a parallel processing unit (“PPU”), in accordance with at least one embodiment. In at least one embodiment, memory partition unit **2900** includes, without limitation, a Raster Operations (“ROP”) unit **2902**; a level two (“L2”) cache **2904**; a memory interface **2906**; and any suitable combination thereof. In at least one embodiment, memory interface **2906** is coupled to memory. In at least one embodiment, memory interface **2906** may implement 32, 64, 128, 1024-bit data buses, or similar implementations, for high-speed data transfer. In at least one embodiment, PPU incorporates U memory interfaces **2906**, one memory interface **2906** per pair of partition units **2900**, where each pair of partition units **2900** is connected to a corresponding memory device. For example, in at least one embodiment, PPU may be connected to up to Y memory devices, such as high bandwidth memory stacks or graphics double-data-rate, version 5, synchronous dynamic random access memory (“GDDR5 SDRAM”).

In at least one embodiment, memory interface **2906** implements a high bandwidth memory second generation (“HBM2”) memory interface and Y equals half U. In at least one embodiment, HBM2 memory stacks are located on same physical package as PPU, providing substantial power and area savings compared with conventional GDDR5 SDRAM systems. In at least one embodiment, each HBM2 stack includes, without limitation, four memory dies and Y equals 4, with each HBM2 stack including two 128-bit channels per die for a total of 8 channels and a data bus width of 1024 bits. In at least one embodiment, memory supports Single-Error Correcting Double-Error Detecting (“SECCED”) Error Correction Code (“ECC”) to protect data. In at least one embodiment, ECC provides higher reliability for compute applications that are sensitive to data corruption.

In at least one embodiment, PPU implements a multi-level memory hierarchy. In at least one embodiment, memory partition unit **2900** supports a unified memory to provide a single unified virtual address space for central processing unit (“CPU”) and PPU memory, enabling data sharing between virtual memory systems. In at least one embodiment, frequency of accesses by a PPU to memory located on other processors is traced to ensure that memory pages are moved to physical memory of PPU that is accessing pages more frequently. In at least one embodiment, high-speed GPU interconnect **2708** supports address translation services allowing PPU to directly access a CPU’s page tables and providing full access to CPU memory by PPU.

In at least one embodiment, copy engines transfer data between multiple PPUs or between PPUs and CPUs. In at least one embodiment, copy engines can generate page faults for addresses that are not mapped into page tables and memory partition unit **2900** then services page faults, mapping addresses into page table, after which copy engine performs transfer. In at least one embodiment, memory is pinned (i.e., non-pageable) for multiple copy engine operations between multiple processors, substantially reducing available memory. In at least one embodiment, with hardware page faulting, addresses can be passed to copy engines without regard as to whether memory pages are resident, and copy process is transparent.

Data from memory **2704** of FIG. **27** or other system memory is fetched by memory partition unit **2900** and stored in L2 cache **2904**, which is located on-chip and is shared between various GPCs, in accordance with at least one embodiment. Each memory partition unit **2900**, in at least one embodiment, includes, without limitation, at least a portion of L2 cache associated with a corresponding memory device. In at least one embodiment, lower level caches are implemented in various units within GPCs. In at least one embodiment, each of SMs **2814** may implement a level one (“L1”) cache wherein L1 cache is private memory that is dedicated to a particular SM **2814** and data from L2 cache **2904** is fetched and stored in each of L1 caches for processing in functional units of SMs **2814**. In at least one embodiment, L2 cache **2904** is coupled to memory interface **2906** and XBar **2720**.

ROP unit **2902** performs graphics raster operations related to pixel color, such as color compression, pixel blending, and more, in at least one embodiment. ROP unit **2902**, in at least one embodiment, implements depth testing in conjunction with raster engine **2808**, receiving a depth for a sample location associated with a pixel fragment from culling engine of raster engine **2808**. In at least one embodiment, depth is tested against a corresponding depth in a depth buffer for a sample location associated with fragment. In at least one embodiment, if fragment passes depth test for sample location, then ROP unit **2902** updates depth buffer and transmits a result of depth test to raster engine **2808**. It will be appreciated that number of partition units **2900** may be different than number of GPCs and, therefore, each ROP unit **2902** can, in at least one embodiment, be coupled to each of GPCs. In at least one embodiment, ROP unit **2902** tracks packets received from different GPCs and determines which that a result generated by ROP unit **2902** is routed to through XBar **2720**.

FIG. **30** illustrates a streaming multi-processor (“SM”) **3000**, according to at least one embodiment. In at least one embodiment, SM **3000** is SM **2814** of FIG. **28**. In at least one embodiment, SM **3000** includes, without limitation, an instruction cache **3002**; one or more scheduler units **3004**; a

register file **3008**; one or more processing cores (“cores”) **3010**; one or more special function units (“SFUs”) **3012**; one or more load/store units (“LSUs”) **3014**; an interconnect network **3016**; a shared memory/level one (“L1”) cache **3018**; and any suitable combination thereof. In at least one embodiment, a work distribution unit dispatches tasks for execution on general processing clusters (“GPCs”) of parallel processing units (“PPUs”) and each task is allocated to a particular Data Processing Cluster (“DPC”) within a GPC and, if task is associated with a shader program, task is allocated to one of SMs **3000**. In at least one embodiment, scheduler unit **3004** receives tasks from work distribution unit and manages instruction scheduling for one or more thread blocks assigned to SM **3000**. In at least one embodiment, scheduler unit **3004** schedules thread blocks for execution as warps of parallel threads, wherein each thread block is allocated at least one warp. In at least one embodiment, each warp executes threads. In at least one embodiment, scheduler unit **3004** manages a plurality of different thread blocks, allocating warps to different thread blocks and then dispatching instructions from plurality of different cooperative groups to various functional units (e.g., processing cores **3010**, SFUs **3012**, and LSUs **3014**) during each clock cycle.

In at least one embodiment, Cooperative Groups may refer to a programming model for organizing groups of communicating threads that allows developers to express granularity at which threads are communicating, enabling expression of richer, more efficient parallel decompositions. In at least one embodiment, cooperative launch APIs support synchronization amongst thread blocks for execution of parallel algorithms. In at least one embodiment, applications of conventional programming models provide a single, simple construct for synchronizing cooperating threads: a barrier across all threads of a thread block (e.g., syncthreads() function). However, In at least one embodiment, programmers may define groups of threads at smaller than thread block granularities and synchronize within defined groups to enable greater performance, design flexibility, and software reuse in form of collective group-wide function interfaces. In at least one embodiment, Cooperative Groups enables programmers to define groups of threads explicitly at sub-block (i.e., as small as a single thread) and multi-block granularities, and to perform collective operations such as synchronization on threads in a cooperative group. In at least one embodiment, programming model supports clean composition across software boundaries, so that libraries and utility functions can synchronize safely within their local context without having to make assumptions about convergence. In at least one embodiment, Cooperative Groups primitives enable new patterns of cooperative parallelism, including, without limitation, producer-consumer parallelism, opportunistic parallelism, and global synchronization across an entire grid of thread blocks.

In at least one embodiment, a dispatch unit **3006** is configured to transmit instructions to one or more of functional units and scheduler unit **3004** includes, without limitation, two dispatch units **3006** that enable two different instructions from same warp to be dispatched during each clock cycle. In at least one embodiment, each scheduler unit **3004** includes a single dispatch unit **3006** or additional dispatch units **3006**.

In at least one embodiment, each SM **3000**, in at least one embodiment, includes, without limitation, register file **3008** that provides a set of registers for functional units of SM **3000**. In at least one embodiment, register file **3008** is divided between each of functional units such that each

functional unit is allocated a dedicated portion of register file **3008**. In at least one embodiment, register file **3008** is divided between different warps being executed by SM **3000** and register file **3008** provides temporary storage for operands connected to data paths of functional units. In at least one embodiment, each SM **3000** comprises, without limitation, a plurality of L processing cores **3010**. In at least one embodiment, SM **3000** includes, without limitation, a large number (e.g., 128 or more) of distinct processing cores **3010**. In at least one embodiment, each processing core **3010**, in at least one embodiment, includes, without limitation, a fully-pipelined, single-precision, double-precision, and/or mixed precision processing unit that includes, without limitation, a floating point arithmetic logic unit and an integer arithmetic logic unit. In at least one embodiment, floating point arithmetic logic units implement IEEE 754-2008 standard for floating point arithmetic. In at least one embodiment, processing cores **3010** include, without limitation, 64 single-precision (32-bit) floating point cores, 64 integer cores, 32 double-precision (64-bit) floating point cores, and 8 tensor cores.

Tensor cores are configured to perform matrix operations in accordance with at least one embodiment. In at least one embodiment, one or more tensor cores are included in processing cores **3010**. In at least one embodiment, tensor cores are configured to perform deep learning matrix arithmetic, such as convolution operations for neural network training and inferencing. In at least one embodiment, each tensor core operates on a 4×4 matrix and performs a matrix multiply and accumulate operation $D=A \times B + C$, where A, B, C, and D are 4×4 matrices.

In at least one embodiment, matrix multiply inputs A and B are 16-bit floating point matrices and accumulation matrices C and D are 16-bit floating point or 32-bit floating point matrices. In at least one embodiment, tensor cores operate on 16-bit floating point input data with 32-bit floating point accumulation. In at least one embodiment, 16-bit floating point multiply uses 64 operations and results in a full precision product that is then accumulated using 32-bit floating point addition with other intermediate products for a 4×4×4 matrix multiply. Tensor cores are used to perform much larger two-dimensional or higher dimensional matrix operations, built up from these smaller elements, in at least one embodiment. In at least one embodiment, an API, such as CUDA 9 C++ API, exposes specialized matrix load, matrix multiply and accumulate, and matrix store operations to efficiently use tensor cores from a CUDA-C++ program. In at least one embodiment, at CUDA level, warp-level interface assumes 16×16 size matrices spanning all 32 threads of warp.

In at least one embodiment, each SM **3000** comprises, without limitation, M SFUs **3012** that perform special functions (e.g., attribute evaluation, reciprocal square root, etc.). In at least one embodiment, SFUs **3012** include, without limitation, a tree traversal unit configured to traverse a hierarchical tree data structure. In at least one embodiment, SFUs **3012** include, without limitation, a texture unit configured to perform texture map filtering operations. In at least one embodiment, texture units are configured to load texture maps (e.g., a 2D array of texels) from memory and sample texture maps to produce sampled texture values for use in shader programs executed by SM **3000**. In at least one embodiment, texture maps are stored in shared memory/L1 cache **3018**. In at least one embodiment, texture units implement texture operations such as filtering operations using mip-maps (e.g., texture maps of varying levels of

detail), in accordance with at least one embodiment. In at least one embodiment, each SM **3000** includes, without limitation, two texture units.

Each SM **3000** comprises, without limitation, N LSUs **3014** that implement load and store operations between shared memory/L1 cache **3018** and register file **3008**, in at least one embodiment. Each SM **3000** includes, without limitation, interconnect network **3016** that connects each of functional units to register file **3008** and LSU **3014** to register file **3008** and shared memory/L1 cache **3018** in at least one embodiment. In at least one embodiment, interconnect network **3016** is a crossbar that can be configured to connect any of functional units to any of registers in register file **3008** and connect LSUs **3014** to register file **3008** and memory locations in shared memory/L1 cache **3018**.

In at least one embodiment, shared memory/L1 cache **3018** is an array of on-chip memory that allows for data storage and communication between SM **3000** and primitive engine and between threads in SM **3000**, in at least one embodiment. In at least one embodiment, shared memory/L1 cache **3018** comprises, without limitation, 128 KB of storage capacity and is in path from SM **3000** to partition unit. In at least one embodiment, shared memory/L1 cache **3018**, in at least one embodiment, is used to cache reads and writes. In at least one embodiment, one or more of shared memory/L1 cache **3018**, L2 cache, and memory are backing stores.

Combining data cache and shared memory functionality into a single memory block provides improved performance for both types of memory accesses, in at least one embodiment. In at least one embodiment, capacity is used or is usable as a cache by programs that do not use shared memory, such as if shared memory is configured to use half of capacity, texture and load/store operations can use remaining capacity. Integration within shared memory/L1 cache **3018** enables shared memory/L1 cache **3018** to function as a high-throughput conduit for streaming data while simultaneously providing high-bandwidth and low-latency access to frequently reused data, in accordance with at least one embodiment. In at least one embodiment, when configured for general purpose parallel computation, a simpler configuration can be used compared with graphics processing. In at least one embodiment, fixed function graphics processing units are bypassed, creating a much simpler programming model. In general purpose parallel computation configuration, work distribution unit assigns and distributes blocks of threads directly to DPCs, in at least one embodiment. In at least one embodiment, threads in a block execute same program, using a unique thread ID in calculation to ensure each thread generates unique results, using SM **3000** to execute program and perform calculations, shared memory/L1 cache **3018** to communicate between threads, and LSU **3014** to read and write global memory through shared memory/L1 cache **3018** and memory partition unit. In at least one embodiment, when configured for general purpose parallel computation, SM **3000** writes commands that scheduler unit **3004** can use to launch new work on DPCs.

In at least one embodiment, PPU is included in or coupled to a desktop computer, a laptop computer, a tablet computer, servers, supercomputers, a smart-phone (e.g., a wireless, hand-held device), personal digital assistant ("PDA"), a digital camera, a vehicle, a head mounted display, a hand-held electronic device, and more. In at least one embodiment, PPU is embodied on a single semiconductor substrate. In at least one embodiment, PPU is included in a system-on-a-chip ("SoC") along with one or more other devices

such as additional PPUs, memory, a reduced instruction set computer (“RISC”) CPU, a memory management unit (“MMU”), a digital-to-analog converter (“DAC”), and like.

In at least one embodiment, PPU may be included on a graphics card that includes one or more memory devices. A graphics card may be configured to interface with a PCIe slot on a motherboard of a desktop computer. In at least one embodiment, PPU may be an integrated graphics processing unit (“iGPU”) included in chipset of motherboard.

Inference and/or training logic **615** are used to perform inferencing and/or training operations associated with one or more embodiments. Details regarding inference and/or training logic **615** are provided below in conjunction with FIGS. **6A** and/or **6B**. In at least one embodiment, deep learning application processor is used to train a machine learning model, such as a neural network, to predict or infer information provided to SM **3000**. In at least one embodiment, SM **3000** is used to infer or predict information based on a trained machine learning model (e.g., neural network) that has been trained by another processor or system or by SM **3000**. In at least one embodiment, SM **3000** may be used to perform one or more neural network use cases described herein.

Inference and/or training logic **615** are used to perform inferencing and/or training operations associated with one or more embodiments. In at least one embodiment, this logic can be used with components of these figures to generate one or more images with visual effects approximated from similar effects applied to one or more lower resolution versions of those images.

In at least one embodiment, a single semiconductor platform may refer to a sole unitary semiconductor-based integrated circuit or chip. In at least one embodiment, multi-chip modules may be used with increased connectivity which simulate on-chip operation, and make substantial improvements over utilizing a conventional central processing unit (“CPU”) and bus implementation. In at least one embodiment, various modules may also be situated separately or in various combinations of semiconductor platforms per desires of user.

In at least one embodiment, computer programs in form of machine-readable executable code or computer control logic algorithms are stored in main memory **1004** and/or secondary storage. Computer programs, if executed by one or more processors, enable system **1000** to perform various functions in accordance with at least one embodiment. In at least one embodiment, memory **1004**, storage, and/or any other storage are possible examples of computer-readable media. In at least one embodiment, secondary storage may refer to any suitable storage device or system such as a hard disk drive and/or a removable storage drive, representing a floppy disk drive, a magnetic tape drive, a compact disk drive, digital versatile disk (“DVD”) drive, recording device, universal serial bus (“USB”) flash memory, etc. In at least one embodiment, architecture and/or functionality of various previous figures are implemented in context of CPU **1002**; parallel processing system **1012**; an integrated circuit capable of at least a portion of capabilities of both CPU **1002**; parallel processing system **1012**; a chipset (e.g., a group of integrated circuits designed to work and sold as a unit for performing related functions, etc.); and any suitable combination of integrated circuit(s).

In at least one embodiment, architecture and/or functionality of various previous figures are implemented in context of a general computer system, a circuit board system, a game console system dedicated for entertainment purposes, an application-specific system, and more. In at least one

embodiment, computer system **1000** may take form of a desktop computer, a laptop computer, a tablet computer, servers, supercomputers, a smart-phone (e.g., a wireless, hand-held device), personal digital assistant (“PDA”), a digital camera, a vehicle, a head mounted display, a hand-held electronic device, a mobile phone device, a television, workstation, game consoles, embedded system, and/or any other type of logic.

In at least one embodiment, parallel processing system **1012** includes, without limitation, a plurality of parallel processing units (“PPUs”) **1014** and associated memories **1016**. In at least one embodiment, PPUs **1014** are connected to a host processor or other peripheral devices via an interconnect **1018** and a switch **1020** or multiplexer. In at least one embodiment, parallel processing system **1012** distributes computational tasks across PPUs **1014** which can be parallelizable—for example, as part of distribution of computational tasks across multiple graphics processing unit (“GPU”) thread blocks. In at least one embodiment, memory is shared and accessible (e.g., for read and/or write access) across some or all of PPUs **1014**, although such shared memory may incur performance penalties relative to use of local memory and registers resident to a PPU **1014**. In at least one embodiment, operation of PPUs **1014** is synchronized through use of a command such as `_syncthreads()` wherein all threads in a block (e.g., executed across multiple PPUs **1014**) to reach a certain point of execution of code before proceeding.

Virtualized Computing Platform

Embodiments are disclosed related a virtualized computing platform for advanced computing, such as image inferencing and image processing. With reference to FIG. **31** is an example data flow diagram for a process **3100** of generating and deploying an image processing and inferencing pipeline, in accordance with at least one embodiment. In at least one embodiment, process **3100** may be deployed for use with imaging devices, processing devices, genomics devices, gene sequencing devices, radiology devices, and/or other device types at one or more facilities **3102**, such as medical facilities, hospitals, healthcare institutes, clinics, research or diagnostic labs, etc. In at least one embodiment, process **3100** may be deployed to perform genomics analysis and inferencing on sequencing data. Examples of genomic analyses that may be performed using systems and processes described herein include, without limitation, variant calling, mutation detection, and gene expression quantification. Process **3100** may be executed within a training system **3104** and/or a deployment system **3106**. In at least one embodiment, training system **3104** may be used to perform training, deployment, and implementation of machine learning models (e.g., neural networks, object detection algorithms, computer vision algorithms, etc.) for use in deployment system **3106**. In at least one embodiment, deployment system **3106** may be configured to offload processing and compute resources among a distributed computing environment to reduce infrastructure requirements at facility **3102**. In at least one embodiment, deployment system **3106** may provide a streamlined platform for selecting, customizing, and implementing virtual instruments for use with imaging devices (e.g., MRI, CT Scan, X-Ray, Ultrasound, etc.) or sequencing devices at facility **3102**. In at least one embodiment, virtual instruments may include software-defined applications for performing one or more processing operations with respect to imaging data generated by imaging devices, sequencing devices, radiology devices, and/or other device types. In at least one embodiment, one or more applications in a pipeline may use or call upon services (e.g.,

inference, visualization, compute, AI, etc.) of deployment system **3106** during execution of applications.

In at least one embodiment, some of applications used in advanced processing and inferring pipelines may use machine learning models or other AI to perform one or more processing steps. In at least one embodiment, machine learning models may be trained at facility **3102** using data **3108** (such as imaging data) generated at facility **3102** (and stored on one or more picture archiving and communication system (PACS) servers at facility **3102**), may be trained using imaging or sequencing data **3108** from another facility(ies) (e.g., a different hospital, lab, clinic, etc.), or a combination thereof. In at least one embodiment, training system **3104** may be used to provide applications, services, and/or other resources for generating working, deployable machine learning models for deployment system **3106**.

In at least one embodiment, model registry **3124** may be backed by object storage that may support versioning and object metadata. In at least one embodiment, object storage may be accessible through, for example, a cloud storage (e.g., cloud **3226** of FIG. **32**) compatible application programming interface (API) from within a cloud platform. In at least one embodiment, machine learning models within model registry **3124** may be uploaded, listed, modified, or deleted by developers or partners of a system interacting with an API. In at least one embodiment, an API may provide access to methods that allow users with appropriate credentials to associate models with applications, such that models may be executed as part of execution of containerized instantiations of applications.

In at least one embodiment, training pipeline **3204** (FIG. **32**) may include a scenario where facility **3102** is training their own machine learning model, or has an existing machine learning model that needs to be optimized or updated. In at least one embodiment, imaging data **3108** generated by imaging device(s), sequencing devices, and/or other device types may be received. In at least one embodiment, once imaging data **3108** is received, AI-assisted annotation **3110** may be used to aid in generating annotations corresponding to imaging data **3108** to be used as ground truth data for a machine learning model. In at least one embodiment, AI-assisted annotation **3110** may include one or more machine learning models (e.g., convolutional neural networks (CNNs)) that may be trained to generate annotations corresponding to certain types of imaging data **3108** (e.g., from certain devices) and/or certain types of anomalies in imaging data **3108**. In at least one embodiment, AI-assisted annotations **3110** may then be used directly, or may be adjusted or fine-tuned using an annotation tool (e.g., by a researcher, a clinician, a doctor, a scientist, etc.), to generate ground truth data. In at least one embodiment, in some examples, labeled clinic data **3112** (e.g., annotations provided by a clinician, doctor, scientist, technician, etc.) may be used as ground truth data for training a machine learning model. In at least one embodiment, AI-assisted annotations **3110**, labeled clinic data **3112**, or a combination thereof may be used as ground truth data for training a machine learning model. In at least one embodiment, a trained machine learning model may be referred to as output model **3116**, and may be used by deployment system **3106**, as described herein.

In at least one embodiment, training pipeline **3204** (FIG. **32**) may include a scenario where facility **3102** needs a machine learning model for use in performing one or more processing tasks for one or more applications in deployment system **3106**, but facility **3102** may not currently have such a machine learning model (or may not have a model that is

optimized, efficient, or effective for such purposes). In at least one embodiment, an existing machine learning model may be selected from a model registry **3124**. In at least one embodiment, model registry **3124** may include machine learning models trained to perform a variety of different inference tasks on imaging data. In at least one embodiment, machine learning models in model registry **3124** may have been trained on imaging data from different facilities than facility **3102** (e.g., facilities remotely located). In at least one embodiment, machine learning models may have been trained on imaging data from one location, two locations, or any number of locations. In at least one embodiment, when being trained on imaging data from a specific location, training may take place at that location, or at least in a manner that protects confidentiality of imaging data or restricts imaging data from being transferred off-premises (e.g., to comply with HIPAA regulations, privacy regulations, etc.). In at least one embodiment, once a model is trained—or partially trained—at one location, a machine learning model may be added to model registry **3124**. In at least one embodiment, a machine learning model may then be retrained, or updated, at any number of other facilities, and a retrained or updated model may be made available in model registry **3124**. In at least one embodiment, a machine learning model may then be selected from model registry **3124**—and referred to as output model **3116**—and may be used in deployment system **3106** to perform one or more processing tasks for one or more applications of a deployment system.

In at least one embodiment, training pipeline **3204** (FIG. **32**), a scenario may include facility **3102** requiring a machine learning model for use in performing one or more processing tasks for one or more applications in deployment system **3106**, but facility **3102** may not currently have such a machine learning model (or may not have a model that is optimized, efficient, or effective for such purposes). In at least one embodiment, a machine learning model selected from model registry **3124** may not be fine-tuned or optimized for imaging data **3108** generated at facility **3102** because of differences in populations, genetic variations, robustness of training data used to train a machine learning model, diversity in anomalies of training data, and/or other issues with training data. In at least one embodiment, AI-assisted annotation **3110** may be used to aid in generating annotations corresponding to imaging data **3108** to be used as ground truth data for retraining or updating a machine learning model. In at least one embodiment, labeled clinic data **3112** (e.g., annotations provided by a clinician, doctor, scientist, etc.) may be used as ground truth data for training a machine learning model. In at least one embodiment, retraining or updating a machine learning model may be referred to as model training **3114**. In at least one embodiment, model training **3114**—e.g., AI-assisted annotations **3110**, labeled clinic data **3112**, or a combination thereof—may be used as ground truth data for retraining or updating a machine learning model. In at least one embodiment, a trained machine learning model may be referred to as output model **3116**, and may be used by deployment system **3106**, as described herein.

In at least one embodiment, deployment system **3106** may include software **3118**, services **3120**, hardware **3122**, and/or other components, features, and functionality. In at least one embodiment, deployment system **3106** may include a software “stack,” such that software **3118** may be built on top of services **3120** and may use services **3120** to perform some or all of processing tasks, and services **3120** and software **3118** may be built on top of hardware **3122** and use hardware

3122 to execute processing, storage, and/or other compute tasks of deployment system **3106**. In at least one embodiment, software **3118** may include any number of different containers, where each container may execute an instantiation of an application. In at least one embodiment, each application may perform one or more processing tasks in an advanced processing and inferencing pipeline (e.g., inferencing, object detection, feature detection, segmentation, image enhancement, calibration, etc.). In at least one embodiment, for each type of imaging device (e.g., CT, MM, X-Ray, ultrasound, sonography, echocardiography, etc.), sequencing device, radiology device, genomics device, etc., there may be any number of containers that may perform a data processing task with respect to imaging data **3108** (or other data types, such as those described herein) generated by a device. In at least one embodiment, an advanced processing and inferencing pipeline may be defined based on selections of different containers that are desired or required for processing imaging data **3108**, in addition to containers that receive and configure imaging data for use by each container and/or for use by facility **3102** after processing through a pipeline (e.g., to convert outputs back to a usable data type, such as digital imaging and communications in medicine (DICOM) data, radiology information system (RIS) data, clinical information system (CIS) data, remote procedure call (RPC) data, data substantially compliant with a representation state transfer (REST) interface, data substantially compliant with a file-based interface, and/or raw data, for storage and display at facility **3102**). In at least one embodiment, a combination of containers within software **3118** (e.g., that make up a pipeline) may be referred to as a virtual instrument (as described in more detail herein), and a virtual instrument may leverage services **3120** and hardware **3122** to execute some or all processing tasks of applications instantiated in containers.

In at least one embodiment, a data processing pipeline may receive input data (e.g., imaging data **3108**) in a DICOM, RIS, CIS, REST compliant, RPC, raw, and/or other format in response to an inference request (e.g., a request from a user of deployment system **3106**, such as a clinician, a doctor, a radiologist, etc.). In at least one embodiment, input data may be representative of one or more images, video, and/or other data representations generated by one or more imaging devices, sequencing devices, radiology devices, genomics devices, and/or other device types. In at least one embodiment, data may undergo pre-processing as part of data processing pipeline to prepare data for processing by one or more applications. In at least one embodiment, post-processing may be performed on an output of one or more inferencing tasks or other processing tasks of a pipeline to prepare an output data for a next application and/or to prepare output data for transmission and/or use by a user (e.g., as a response to an inference request). In at least one embodiment, inferencing tasks may be performed by one or more machine learning models, such as trained or deployed neural networks, which may include output models **3116** of training system **3104**.

In at least one embodiment, tasks of data processing pipeline may be encapsulated in a container(s) that each represent a discrete, fully functional instantiation of an application and virtualized computing environment that is able to reference machine learning models. In at least one embodiment, containers or applications may be published into a private (e.g., limited access) area of a container registry (described in more detail herein), and trained or deployed models may be stored in model registry **3124** and associated with one or more applications. In at least one

embodiment, images of applications (e.g., container images) may be available in a container registry, and once selected by a user from a container registry for deployment in a pipeline, an image may be used to generate a container for an instantiation of an application for use by a user's system.

In at least one embodiment, developers (e.g., software developers, clinicians, doctors, etc.) may develop, publish, and store applications (e.g., as containers) for performing image processing and/or inferencing on supplied data. In at least one embodiment, development, publishing, and/or storing may be performed using a software development kit (SDK) associated with a system (e.g., to ensure that an application and/or container developed is compliant with or compatible with a system). In at least one embodiment, an application that is developed may be tested locally (e.g., at a first facility, on data from a first facility) with an SDK which may support at least some of services **3120** as a system (e.g., system **3200** of FIG. 32). In at least one embodiment, because DICOM objects may contain anywhere from one to hundreds of images or other data types, and due to a variation in data, a developer may be responsible for managing (e.g., setting constructs for, building pre-processing into an application, etc.) extraction and preparation of incoming DICOM data. In at least one embodiment, once validated by system **3200** (e.g., for accuracy, safety, patient privacy, etc.), an application may be available in a container registry for selection and/or implementation by a user (e.g., a hospital, clinic, lab, healthcare provider, etc.) to perform one or more processing tasks with respect to data at a facility (e.g., a second facility) of a user.

In at least one embodiment, developers may then share applications or containers through a network for access and use by users of a system (e.g., system **3200** of FIG. 32). In at least one embodiment, completed and validated applications or containers may be stored in a container registry and associated machine learning models may be stored in model registry **3124**. In at least one embodiment, a requesting entity (e.g., a user at a medical facility)—who provides an inference or image processing request—may browse a container registry and/or model registry **3124** for an application, container, dataset, machine learning model, etc., select a desired combination of elements for inclusion in data processing pipeline, and submit an imaging processing request. In at least one embodiment, a request may include input data (and associated patient data, in some examples) that is necessary to perform a request, and/or may include a selection of application(s) and/or machine learning models to be executed in processing a request. In at least one embodiment, a request may then be passed to one or more components of deployment system **3106** (e.g., a cloud) to perform processing of data processing pipeline. In at least one embodiment, processing by deployment system **3106** may include referencing selected elements (e.g., applications, containers, models, etc.) from a container registry and/or model registry **3124**. In at least one embodiment, once results are generated by a pipeline, results may be returned to a user for reference (e.g., for viewing in a viewing application suite executing on a local, on-premises workstation or terminal). In at least one embodiment, a radiologist may receive results from an data processing pipeline including any number of application and/or containers, where results may include anomaly detection in X-rays, CT scans, MRIs, etc.

In at least one embodiment, to aid in processing or execution of applications or containers in pipelines, services **3120** may be leveraged. In at least one embodiment, services **3120** may include compute services, artificial intelligence

(AI) services, visualization services, and/or other service types. In at least one embodiment, services **3120** may provide functionality that is common to one or more applications in software **3118**, so functionality may be abstracted to a service that may be called upon or leveraged by applications. In at least one embodiment, functionality provided by services **3120** may run dynamically and more efficiently, while also scaling well by allowing applications to process data in parallel (e.g., using a parallel computing platform **3230** (FIG. **32**)). In at least one embodiment, rather than each application that shares a same functionality offered by a service **3120** being required to have a respective instance of service **3120**, service **3120** may be shared between and among various applications. In at least one embodiment, services may include an inference server or engine that may be used for executing detection or segmentation tasks, as non-limiting examples. In at least one embodiment, a model training service may be included that may provide machine learning model training and/or retraining capabilities. In at least one embodiment, a data augmentation service may further be included that may provide GPU accelerated data (e.g., DICOM, RIS, CIS, REST compliant, RPC, raw, etc.) extraction, resizing, scaling, and/or other augmentation. In at least one embodiment, a visualization service may be used that may add image rendering effects—such as ray-tracing, rasterization, denoising, sharpening, etc.—to add realism to two-dimensional (2D) and/or three-dimensional (3D) models. In at least one embodiment, virtual instrument services may be included that provide for beam-forming, segmentation, inferencing, imaging, and/or support for other applications within pipelines of virtual instruments.

In at least one embodiment, where a service **3120** includes an AI service (e.g., an inference service), one or more machine learning models associated with an application for anomaly detection (e.g., tumors, growth abnormalities, scarring, etc.) may be executed by calling upon (e.g., as an API call) an inference service (e.g., an inference server) to execute machine learning model(s), or processing thereof, as part of application execution. In at least one embodiment, where another application includes one or more machine learning models for segmentation tasks, an application may call upon an inference service to execute machine learning models for performing one or more of processing operations associated with segmentation tasks. In at least one embodiment, software **3118** implementing advanced processing and inferencing pipeline that includes segmentation application and anomaly detection application may be streamlined because each application may call upon a same inference service to perform one or more inferencing tasks.

In at least one embodiment, hardware **3122** may include GPUs, CPUs, graphics cards, an AI/deep learning system (e.g., an AI supercomputer, such as NVIDIA's DGX), a cloud platform, or a combination thereof. In at least one embodiment, different types of hardware **3122** may be used to provide efficient, purpose-built support for software **3118** and services **3120** in deployment system **3106**. In at least one embodiment, use of GPU processing may be implemented for processing locally (e.g., at facility **3102**), within an AI/deep learning system, in a cloud system, and/or in other processing components of deployment system **3106** to improve efficiency, accuracy, and efficacy of image processing, image reconstruction, segmentation, MRI exams, stroke or heart attack detection (e.g., in real-time), image quality in rendering, etc. In at least one embodiment, a facility may include imaging devices, genomics devices, sequencing devices, and/or other device types on-premises that may

leverage GPUs to generate imaging data representative of a subject's anatomy. In at least one embodiment, software **3118** and/or services **3120** may be optimized for GPU processing with respect to deep learning, machine learning, and/or high-performance computing, as non-limiting examples. In at least one embodiment, at least some of computing environment of deployment system **3106** and/or training system **3104** may be executed in a datacenter one or more supercomputers or high performance computing systems, with GPU optimized software (e.g., hardware and software combination of NVIDIA's DGX System). In at least one embodiment, datacenters may be compliant with provisions of HIPAA, such that receipt, processing, and transmission of imaging data and/or other patient data is securely handled with respect to privacy of patient data. In at least one embodiment, hardware **3122** may include any number of GPUs that may be called upon to perform processing of data in parallel, as described herein. In at least one embodiment, cloud platform may further include GPU processing for GPU-optimized execution of deep learning tasks, machine learning tasks, or other computing tasks. In at least one embodiment, cloud platform (e.g., NVIDIA's NGC) may be executed using an AI/deep learning supercomputer(s) and/or GPU-optimized software (e.g., as provided on NVIDIA's DGX Systems) as a hardware abstraction and scaling platform. In at least one embodiment, cloud platform may integrate an application container clustering system or orchestration system (e.g., KUBERNETES) on multiple GPUs to enable seamless scaling and load balancing.

FIG. **32** is a system diagram for an example system **3200** for generating and deploying an imaging deployment pipeline, in accordance with at least one embodiment. In at least one embodiment, system **3200** may be used to implement process **3100** of FIG. **31** and/or other processes including advanced processing and inferencing pipelines. In at least one embodiment, system **3200** may include training system **3104** and deployment system **3106**. In at least one embodiment, training system **3104** and deployment system **3106** may be implemented using software **3118**, services **3120**, and/or hardware **3122**, as described herein.

In at least one embodiment, system **3200** (e.g., training system **3104** and/or deployment system **3106**) may be implemented in a cloud computing environment (e.g., using cloud **3226**). In at least one embodiment, system **3200** may be implemented locally with respect to a healthcare services facility, or as a combination of both cloud and local computing resources. In at least one embodiment, in embodiments where cloud computing is implemented, patient data may be separated from, or unprocessed by, by one or more components of system **3200** that would render processing non-compliant with HIPAA and/or other data handling and privacy regulations or laws. In at least one embodiment, access to APIs in cloud **3226** may be restricted to authorized users through enacted security measures or protocols. In at least one embodiment, a security protocol may include web tokens that may be signed by an authentication (e.g., AuthN, AuthZ, Gluecon, etc.) service and may carry appropriate authorization. In at least one embodiment, APIs of virtual instruments (described herein), or other instantiations of system **3200**, may be restricted to a set of public IPs that have been vetted or authorized for interaction.

In at least one embodiment, various components of system **3200** may communicate between and among one another using any of a variety of different network types, including but not limited to local area networks (LANs) and/or wide area networks (WANs) via wired and/or wire-

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less communication protocols. In at least one embodiment, communication between facilities and components of system 3200 (e.g., for transmitting inference requests, for receiving results of inference requests, etc.) may be communicated over data bus(es), wireless data protocols (Wi-Fi), wired data protocols (e.g., Ethernet), etc.

In at least one embodiment, training system 3104 may execute training pipelines 3204, similar to those described herein with respect to FIG. 31. In at least one embodiment, where one or more machine learning models are to be used in deployment pipelines 3210 by deployment system 3106, training pipelines 3204 may be used to train or retrain one or more (e.g. pre-trained) models, and/or implement one or more of pre-trained models 3206 (e.g., without a need for retraining or updating). In at least one embodiment, as a result of training pipelines 3204, output model(s) 3116 may be generated. In at least one embodiment, training pipelines 3204 may include any number of processing steps, such as but not limited to imaging data (or other input data) conversion or adaption (e.g., using DICOM adapter 3202A to convert DICOM images to another format suitable for processing by respective machine learning models, such as Neuroimaging Informatics Technology Initiative (NIfTI) format), AI-assisted annotation 3110, labeling or annotating of imaging data 3108 to generate labeled clinic data 3112, model selection from a model registry, model training 3114, training, retraining, or updating models, and/or other processing steps. In at least one embodiment, for different machine learning models used by deployment system 3106, different training pipelines 3204 may be used. In at least one embodiment, training pipeline 3204 similar to a first example described with respect to FIG. 31 may be used for a first machine learning model, training pipeline 3204 similar to a second example described with respect to FIG. 31 may be used for a second machine learning model, and training pipeline 3204 similar to a third example described with respect to FIG. 31 may be used for a third machine learning model. In at least one embodiment, any combination of tasks within training system 3104 may be used depending on what is required for each respective machine learning model. In at least one embodiment, one or more of machine learning models may already be trained and ready for deployment so machine learning models may not undergo any processing by training system 3104, and may be implemented by deployment system 3106.

In at least one embodiment, output model(s) 3116 and/or pre-trained model(s) 3206 may include any types of machine learning models depending on implementation or embodiment. In at least one embodiment, and without limitation, machine learning models used by system 3200 may include machine learning model(s) using linear regression, logistic regression, decision trees, support vector machines (SVM), Naïve Bayes, k-nearest neighbor (Knn), K means clustering, random forest, dimensionality reduction algorithms, gradient boosting algorithms, neural networks (e.g., auto-encoders, convolutional, recurrent, perceptrons, Long/Short Term Memory (LSTM), Hopfield, Boltzmann, deep belief, deconvolutional, generative adversarial, liquid state machine, etc.), and/or other types of machine learning models.

In at least one embodiment, training pipelines 3204 may include AI-assisted annotation, as described in more detail herein with respect to at least FIG. 35B. In at least one embodiment, labeled clinic data 3112 (e.g., traditional annotation) may be generated by any number of techniques. In at least one embodiment, labels or other annotations may be generated within a drawing program (e.g., an annotation program), a computer aided design (CAD) program, a

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labeling program, another type of program suitable for generating annotations or labels for ground truth, and/or may be hand drawn, in some examples. In at least one embodiment, ground truth data may be synthetically produced (e.g., generated from computer models or renderings), real produced (e.g., designed and produced from real-world data), machine-automated (e.g., using feature analysis and learning to extract features from data and then generate labels), human annotated (e.g., labeler, or annotation expert, defines location of labels), and/or a combination thereof. In at least one embodiment, for each instance of imaging data 3108 (or other data type used by machine learning models), there may be corresponding ground truth data generated by training system 3104. In at least one embodiment, AI-assisted annotation may be performed as part of deployment pipelines 3210; either in addition to, or in lieu of AI-assisted annotation included in training pipelines 3204. In at least one embodiment, system 3200 may include a multi-layer platform that may include a software layer (e.g., software 3118) of diagnostic applications (or other application types) that may perform one or more medical imaging and diagnostic functions. In at least one embodiment, system 3200 may be communicatively coupled to (e.g., via encrypted links) PACS server networks of one or more facilities. In at least one embodiment, system 3200 may be configured to access and referenced data (e.g., DICOM data, RIS data, raw data, CIS data, REST compliant data, RPC data, raw data, etc.) from PACS servers (e.g., via a DICOM adapter 3202, or another data type adapter such as RIS, CIS, REST compliant, RPC, raw, etc.) to perform operations, such as training machine learning models, deploying machine learning models, image processing, inferencing, and/or other operations.

In at least one embodiment, a software layer may be implemented as a secure, encrypted, and/or authenticated API through which applications or containers may be invoked (e.g., called) from an external environment(s) (e.g., facility 3102). In at least one embodiment, applications may then call or execute one or more services 3120 for performing compute, AI, or visualization tasks associated with respective applications, and software 3118 and/or services 3120 may leverage hardware 3122 to perform processing tasks in an effective and efficient manner.

In at least one embodiment, deployment system 3106 may execute deployment pipelines 3210. In at least one embodiment, deployment pipelines 3210 may include any number of applications that may be sequentially, non-sequentially, or otherwise applied to imaging data (and/or other data types) generated by imaging devices, sequencing devices, genomics devices, etc. —including AI-assisted annotation, as described above. In at least one embodiment, as described herein, a deployment pipeline 3210 for an individual device may be referred to as a virtual instrument for a device (e.g., a virtual ultrasound instrument, a virtual CT scan instrument, a virtual sequencing instrument, etc.). In at least one embodiment, for a single device, there may be more than one deployment pipeline 3210 depending on information desired from data generated by a device. In at least one embodiment, where detections of anomalies are desired from an MRI machine, there may be a first deployment pipeline 3210, and where image enhancement is desired from output of an MRI machine, there may be a second deployment pipeline 3210.

In at least one embodiment, applications available for deployment pipelines 3210 may include any application that may be used for performing processing tasks on imaging data or other data from devices. In at least one embodiment, different applications may be responsible for image

enhancement, segmentation, reconstruction, anomaly detection, object detection, feature detection, treatment planning, dosimetry, beam planning (or other radiation treatment procedures), and/or other analysis, image processing, or inferencing tasks. In at least one embodiment, deployment system **3106** may define constructs for each of applications, such that users of deployment system **3106** (e.g., medical facilities, labs, clinics, etc.) may understand constructs and adapt applications for implementation within their respective facility. In at least one embodiment, an application for image reconstruction may be selected for inclusion in deployment pipeline **3210**, but data type generated by an imaging device may be different from a data type used within an application. In at least one embodiment, DICOM adapter **3202B** (and/or a DICOM reader) or another data type adapter or reader (e.g., RIS, CIS, REST compliant, RPC, raw, etc.) may be used within deployment pipeline **3210** to convert data to a form useable by an application within deployment system **3106**. In at least one embodiment, access to DICOM, RIS, CIS, REST compliant, RPC, raw, and/or other data type libraries may be accumulated and pre-processed, including decoding, extracting, and/or performing any convolutions, color corrections, sharpness, gamma, and/or other augmentations to data. In at least one embodiment, DICOM, RIS, CIS, REST compliant, RPC, and/or raw data may be unordered and a pre-pass may be executed to organize or sort collected data. In at least one embodiment, because various applications may share common image operations, in some embodiments, a data augmentation library (e.g., as one of services **3120**) may be used to accelerate these operations. In at least one embodiment, to avoid bottlenecks of conventional processing approaches that rely on CPU processing, parallel computing platform **3230** may be used for GPU acceleration of these processing tasks.

In at least one embodiment, an image reconstruction application may include a processing task that includes use of a machine learning model. In at least one embodiment, a user may desire to use their own machine learning model, or to select a machine learning model from model registry **3124**. In at least one embodiment, a user may implement their own machine learning model or select a machine learning model for inclusion in an application for performing a processing task. In at least one embodiment, applications may be selectable and customizable, and by defining constructs of applications, deployment and implementation of applications for a particular user are presented as a more seamless user experience. In at least one embodiment, by leveraging other features of system **3200**—such as services **3120** and hardware **3122**—deployment pipelines **3210** may be even more user friendly, provide for easier integration, and produce more accurate, efficient, and timely results.

In at least one embodiment, deployment system **3106** may include a user interface **3214** (e.g., a graphical user interface, a web interface, etc.) that may be used to select applications for inclusion in deployment pipeline(s) **3210**, arrange applications, modify or change applications or parameters or constructs thereof, use and interact with deployment pipeline(s) **3210** during set-up and/or deployment, and/or to otherwise interact with deployment system **3106**. In at least one embodiment, although not illustrated with respect to training system **3104**, user interface **3214** (or a different user interface) may be used for selecting models for use in deployment system **3106**, for selecting models for training, or retraining, in training system **3104**, and/or for otherwise interacting with training system **3104**.

In at least one embodiment, pipeline manager **3212** may be used, in addition to an application orchestration system **3228**, to manage interaction between applications or containers of deployment pipeline(s) **3210** and services **3120** and/or hardware **3122**. In at least one embodiment, pipeline manager **3212** may be configured to facilitate interactions from application to application, from application to service **3120**, and/or from application or service to hardware **3122**. In at least one embodiment, although illustrated as included in software **3118**, this is not intended to be limiting, and in some examples pipeline manager **3212** may be included in services **3120**. In at least one embodiment, application orchestration system **3228** (e.g., Kubernetes, DOCKER, etc.) may include a container orchestration system that may group applications into containers as logical units for coordination, management, scaling, and deployment. In at least one embodiment, by associating applications from deployment pipeline(s) **3210** (e.g., a reconstruction application, a segmentation application, etc.) with individual containers, each application may execute in a self-contained environment (e.g., at a kernel level) to increase speed and efficiency.

In at least one embodiment, each application and/or container (or image thereof) may be individually developed, modified, and deployed (e.g., a first user or developer may develop, modify, and deploy a first application and a second user or developer may develop, modify, and deploy a second application separate from a first user or developer), which may allow for focus on, and attention to, a task of a single application and/or container(s) without being hindered by tasks of another application(s) or container(s). In at least one embodiment, communication, and cooperation between different containers or applications may be aided by pipeline manager **3212** and application orchestration system **3228**. In at least one embodiment, so long as an expected input and/or output of each container or application is known by a system (e.g., based on constructs of applications or containers), application orchestration system **3228** and/or pipeline manager **3212** may facilitate communication among and between, and sharing of resources among and between, each of applications or containers. In at least one embodiment, because one or more of applications or containers in deployment pipeline(s) **3210** may share same services and resources, application orchestration system **3228** may orchestrate, load balance, and determine sharing of services or resources between and among various applications or containers. In at least one embodiment, a scheduler may be used to track resource requirements of applications or containers, current usage or planned usage of these resources, and resource availability. In at least one embodiment, a scheduler may thus allocate resources to different applications and distribute resources between and among applications in view of requirements and availability of a system. In some examples, a scheduler (and/or other component of application orchestration system **3228**) may determine resource availability and distribution based on constraints imposed on a system (e.g., user constraints), such as quality of service (QoS), urgency of need for data outputs (e.g., to determine whether to execute real-time processing or delayed processing), etc.

In at least one embodiment, services **3120** leveraged by and shared by applications or containers in deployment system **3106** may include compute services **3216**, AI services **3218**, visualization services **3220**, and/or other service types. In at least one embodiment, applications may call (e.g., execute) one or more of services **3120** to perform processing operations for an application. In at least one embodiment, compute services **3216** may be leveraged by

applications to perform super-computing or other high-performance computing (HPC) tasks. In at least one embodiment, compute service(s) **3216** may be leveraged to perform parallel processing (e.g., using a parallel computing platform **3230**) for processing data through one or more of applications and/or one or more tasks of a single application, substantially simultaneously. In at least one embodiment, parallel computing platform **3230** (e.g., NVIDIA's CUDA) may enable general purpose computing on GPUs (GPGPU) (e.g., GPUs **3222**). In at least one embodiment, a software layer of parallel computing platform **3230** may provide access to virtual instruction sets and parallel computational elements of GPUs, for execution of compute kernels. In at least one embodiment, parallel computing platform **3230** may include memory and, in some embodiments, a memory may be shared between and among multiple containers, and/or between and among different processing tasks within a single container. In at least one embodiment, inter-process communication (IPC) calls may be generated for multiple containers and/or for multiple processes within a container to use same data from a shared segment of memory of parallel computing platform **3230** (e.g., where multiple different stages of an application or multiple applications are processing same information). In at least one embodiment, rather than making a copy of data and moving data to different locations in memory (e.g., a read/write operation), same data in same location of a memory may be used for any number of processing tasks (e.g., at a same time, at different times, etc.). In at least one embodiment, as data is used to generate new data as a result of processing, this information of a new location of data may be stored and shared between various applications. In at least one embodiment, location of data and a location of updated or modified data may be part of a definition of how a payload is understood within containers.

In at least one embodiment, AI services **3218** may be leveraged to perform inferencing services for executing machine learning model(s) associated with applications (e.g., tasked with performing one or more processing tasks of an application). In at least one embodiment, AI services **3218** may leverage AI system **3224** to execute machine learning model(s) (e.g., neural networks, such as CNNs) for segmentation, reconstruction, object detection, feature detection, classification, and/or other inferencing tasks. In at least one embodiment, applications of deployment pipeline (s) **3210** may use one or more of output models **3116** from training system **3104** and/or other models of applications to perform inference on imaging data (e.g., DICOM data, RIS data, CIS data, REST compliant data, RPC data, raw data, etc.). In at least one embodiment, two or more examples of inferencing using application orchestration system **3228** (e.g., a scheduler) may be available. In at least one embodiment, a first category may include a high priority/low latency path that may achieve higher service level agreements, such as for performing inference on urgent requests during an emergency, or for a radiologist during diagnosis. In at least one embodiment, a second category may include a standard priority path that may be used for requests that may be non-urgent or where analysis may be performed at a later time. In at least one embodiment, application orchestration system **3228** may distribute resources (e.g., services **3120** and/or hardware **3122**) based on priority paths for different inferencing tasks of AI services **3218**.

In at least one embodiment, shared storage may be mounted to AI services **3218** within system **3200**. In at least one embodiment, shared storage may operate as a cache (or other storage device type) and may be used to process

inference requests from applications. In at least one embodiment, when an inference request is submitted, a request may be received by a set of API instances of deployment system **3106**, and one or more instances may be selected (e.g., for best fit, for load balancing, etc.) to process a request. In at least one embodiment, to process a request, a request may be entered into a database, a machine learning model may be located from model registry **3124** if not already in a cache, a validation step may ensure appropriate machine learning model is loaded into a cache (e.g., shared storage), and/or a copy of a model may be saved to a cache. In at least one embodiment, a scheduler (e.g., of pipeline manager **3212**) may be used to launch an application that is referenced in a request if an application is not already running or if there are not enough instances of an application. In at least one embodiment, if an inference server is not already launched to execute a model, an inference server may be launched. Any number of inference servers may be launched per model. In at least one embodiment, in a pull model, in which inference servers are clustered, models may be cached whenever load balancing is advantageous. In at least one embodiment, inference servers may be statically loaded in corresponding, distributed servers.

In at least one embodiment, inferencing may be performed using an inference server that runs in a container. In at least one embodiment, an instance of an inference server may be associated with a model (and optionally a plurality of versions of a model). In at least one embodiment, if an instance of an inference server does not exist when a request to perform inference on a model is received, a new instance may be loaded. In at least one embodiment, when starting an inference server, a model may be passed to an inference server such that a same container may be used to serve different models so long as inference server is running as a different instance.

In at least one embodiment, during application execution, an inference request for a given application may be received, and a container (e.g., hosting an instance of an inference server) may be loaded (if not already), and a start procedure may be called. In at least one embodiment, pre-processing logic in a container may load, decode, and/or perform any additional pre-processing on incoming data (e.g., using a CPU(s) and/or GPU(s)). In at least one embodiment, once data is prepared for inference, a container may perform inference as necessary on data. In at least one embodiment, this may include a single inference call on one image (e.g., a hand X-ray), or may require inference on hundreds of images (e.g., a chest CT). In at least one embodiment, an application may summarize results before completing, which may include, without limitation, a single confidence score, pixel level-segmentation, voxel-level segmentation, generating a visualization, or generating text to summarize findings. In at least one embodiment, different models or applications may be assigned different priorities. For example, some models may have a real-time (TAT<1 min) priority while others may have lower priority (e.g., TAT<10 min). In at least one embodiment, model execution times may be measured from requesting institution or entity and may include partner network traversal time, as well as execution on an inference service.

In at least one embodiment, transfer of requests between services **3120** and inference applications may be hidden behind a software development kit (SDK), and robust transport may be provided through a queue. In at least one embodiment, a request will be placed in a queue via an API for an individual application/tenant ID combination and an SDK will pull a request from a queue and give a request to

an application. In at least one embodiment, a name of a queue may be provided in an environment from where an SDK will pick it up. In at least one embodiment, asynchronous communication through a queue may be useful as it may allow any instance of an application to pick up work as it becomes available. Results may be transferred back through a queue, to ensure no data is lost. In at least one embodiment, queues may also provide an ability to segment work, as highest priority work may go to a queue with most instances of an application connected to it, while lowest priority work may go to a queue with a single instance connected to it that processes tasks in an order received. In at least one embodiment, an application may run on a GPU-accelerated instance generated in cloud 3226, and an inference service may perform inferencing on a GPU.

In at least one embodiment, visualization services 3220 may be leveraged to generate visualizations for viewing outputs of applications and/or deployment pipeline(s) 3210. In at least one embodiment, GPUs 3222 may be leveraged by visualization services 3220 to generate visualizations. In at least one embodiment, rendering effects, such as ray-tracing, may be implemented by visualization services 3220 to generate higher quality visualizations. In at least one embodiment, visualizations may include, without limitation, 2D image renderings, 3D volume renderings, 3D volume reconstruction, 2D tomographic slices, virtual reality displays, augmented reality displays, etc. In at least one embodiment, virtualized environments may be used to generate a virtual interactive display or environment (e.g., a virtual environment) for interaction by users of a system (e.g., doctors, nurses, radiologists, etc.). In at least one embodiment, visualization services 3220 may include an internal visualizer, cinematics, and/or other rendering or image processing capabilities or functionality (e.g., ray tracing, rasterization, internal optics, etc.).

In at least one embodiment, hardware 3122 may include GPUs 3222, AI system 3224, cloud 3226, and/or any other hardware used for executing training system 3104 and/or deployment system 3106. In at least one embodiment, GPUs 3222 (e.g., NVIDIA's TESLA and/or QUADRO GPUs) may include any number of GPUs that may be used for executing processing tasks of compute services 3216, AI services 3218, visualization services 3220, other services, and/or any of features or functionality of software 3118. For example, with respect to AI services 3218, GPUs 3222 may be used to perform pre-processing on imaging data (or other data types used by machine learning models), post-processing on outputs of machine learning models, and/or to perform inferencing (e.g., to execute machine learning models). In at least one embodiment, cloud 3226, AI system 3224, and/or other components of system 3200 may use GPUs 3222. In at least one embodiment, cloud 3226 may include a GPU-optimized platform for deep learning tasks. In at least one embodiment, AI system 3224 may use GPUs, and cloud 3226—or at least a portion tasked with deep learning or inferencing—may be executed using one or more AI systems 3224. As such, although hardware 3122 is illustrated as discrete components, this is not intended to be limiting, and any components of hardware 3122 may be combined with, or leveraged by, any other components of hardware 3122.

In at least one embodiment, AI system 3224 may include a purpose-built computing system (e.g., a super-computer or an HPC) configured for inferencing, deep learning, machine learning, and/or other artificial intelligence tasks. In at least one embodiment, AI system 3224 (e.g., NVIDIA's DGX) may include GPU-optimized software (e.g., a software stack) that may be executed using a plurality of GPUs 3222,

in addition to CPUs, RAM, storage, and/or other components, features, or functionality. In at least one embodiment, one or more AI systems 3224 may be implemented in cloud 3226 (e.g., in a data center) for performing some or all of AI-based processing tasks of system 3200.

In at least one embodiment, cloud 3226 may include a GPU-accelerated infrastructure (e.g., NVIDIA's NGC) that may provide a GPU-optimized platform for executing processing tasks of system 3200. In at least one embodiment, cloud 3226 may include an AI system(s) 3224 for performing one or more of AI-based tasks of system 3200 (e.g., as a hardware abstraction and scaling platform). In at least one embodiment, cloud 3226 may integrate with application orchestration system 3228 leveraging multiple GPUs to enable seamless scaling and load balancing between and among applications and services 3120. In at least one embodiment, cloud 3226 may tasked with executing at least some of services 3120 of system 3200, including compute services 3216, AI services 3218, and/or visualization services 3220, as described herein. In at least one embodiment, cloud 3226 may perform small and large batch inference (e.g., executing NVIDIA's TENSOR RT), provide an accelerated parallel computing API and platform 3230 (e.g., NVIDIA's CUDA), execute application orchestration system 3228 (e.g., KUBERNETES), provide a graphics rendering API and platform (e.g., for ray-tracing, 2D graphics, 3D graphics, and/or other rendering techniques to produce higher quality cinematics), and/or may provide other functionality for system 3200.

In at least one embodiment, in an effort to preserve patient confidentiality (e.g., where patient data or records are to be used off-premises), cloud 3226 may include a registry—such as a deep learning container registry. In at least one embodiment, a registry may store containers for instantiations of applications that may perform pre-processing, post-processing, or other processing tasks on patient data. In at least one embodiment, cloud 3226 may receive data that includes patient data as well as sensor data in containers, perform requested processing for just sensor data in those containers, and then forward a resultant output and/or visualizations to appropriate parties and/or devices (e.g., on-premises medical devices used for visualization or diagnoses), all without having to extract, store, or otherwise access patient data. In at least one embodiment, confidentiality of patient data is preserved in compliance with HIPAA and/or other data regulations.

FIG. 33A illustrates a data flow diagram for a process 3300 to train, retrain, or update a machine learning model, in accordance with at least one embodiment. In at least one embodiment, process 3300 may be executed using, as a non-limiting example, system 3200 of FIG. 32. In at least one embodiment, process 3300 may leverage services 3120 and/or hardware 3122 of system 3200, as described herein. In at least one embodiment, refined models 3312 generated by process 3300 may be executed by deployment system 3106 for one or more containerized applications in deployment pipelines 3210.

In at least one embodiment, model training 3114 may include retraining or updating an initial model 3304 (e.g., a pre-trained model) using new training data (e.g., new input data, such as customer dataset 3306, and/or new ground truth data associated with input data). In at least one embodiment, to retrain, or update, initial model 3304, output or loss layer(s) of initial model 3304 may be reset, or deleted, and/or replaced with an updated or new output or loss layer(s). In at least one embodiment, initial model 3304 may have previously fine-tuned parameters (e.g., weights and/or

biases) that remain from prior training, so training or retraining **3114** may not take as long or require as much processing as training a model from scratch. In at least one embodiment, during model training **3114**, by having reset or replaced output or loss layer(s) of initial model **3304**, parameters may be updated and re-tuned for a new data set based on loss calculations associated with accuracy of output or loss layer(s) at generating predictions on new, customer dataset **3306** (e.g., image data **3108** of FIG. **31**).

In at least one embodiment, pre-trained models **3206** may be stored in a data store, or registry (e.g., model registry **3124** of FIG. **31**). In at least one embodiment, pre-trained models **3206** may have been trained, at least in part, at one or more facilities other than a facility executing process **3300**. In at least one embodiment, to protect privacy and rights of patients, subjects, or clients of different facilities, pre-trained models **3206** may have been trained, on-premise, using customer or patient data generated on-premise. In at least one embodiment, pre-trained models **3206** may be trained using cloud **3226** and/or other hardware **3122**, but confidential, privacy protected patient data may not be transferred to, used by, or accessible to any components of cloud **3226** (or other off premise hardware). In at least one embodiment, where a pre-trained model **3206** is trained at using patient data from more than one facility, pre-trained model **3206** may have been individually trained for each facility prior to being trained on patient or customer data from another facility. In at least one embodiment, such as where a customer or patient data has been released of privacy concerns (e.g., by waiver, for experimental use, etc.), or where a customer or patient data is included in a public data set, a customer or patient data from any number of facilities may be used to train pre-trained model **3206** on-premise and/or off premise, such as in a datacenter or other cloud computing infrastructure.

In at least one embodiment, when selecting applications for use in deployment pipelines **3210**, a user may also select machine learning models to be used for specific applications. In at least one embodiment, a user may not have a model for use, so a user may select a pre-trained model **3206** to use with an application. In at least one embodiment, pre-trained model **3206** may not be optimized for generating accurate results on customer dataset **3306** of a facility of a user (e.g., based on patient diversity, demographics, types of medical imaging devices used, etc.). In at least one embodiment, prior to deploying pre-trained model **3206** into deployment pipeline **3210** for use with an application(s), pre-trained model **3206** may be updated, retrained, and/or fine-tuned for use at a respective facility.

In at least one embodiment, a user may select pre-trained model **3206** that is to be updated, retrained, and/or fine-tuned, and pre-trained model **3206** may be referred to as initial model **3304** for training system **3104** within process **3300**. In at least one embodiment, customer dataset **3306** (e.g., imaging data, genomics data, sequencing data, or other data types generated by devices at a facility) may be used to perform model training **3114** (which may include, without limitation, transfer learning) on initial model **3304** to generate refined model **3312**. In at least one embodiment, ground truth data corresponding to customer dataset **3306** may be generated by training system **3104**. In at least one embodiment, ground truth data may be generated, at least in part, by clinicians, scientists, doctors, practitioners, at a facility (e.g., as labeled clinic data **3112** of FIG. **31**).

In at least one embodiment, AI-assisted annotation **3110** may be used in some examples to generate ground truth data. In at least one embodiment, AI-assisted annotation **3110**

(e.g., implemented using an AI-assisted annotation SDK) may leverage machine learning models (e.g., neural networks) to generate suggested or predicted ground truth data for a customer dataset. In at least one embodiment, user **3310** may use annotation tools within a user interface (a graphical user interface (GUI)) on computing device **3308**.

In at least one embodiment, user **3310** may interact with a GUI via computing device **3308** to edit or fine-tune (auto)annotations. In at least one embodiment, a polygon editing feature may be used to move vertices of a polygon to more accurate or fine-tuned locations.

In at least one embodiment, once customer dataset **3306** has associated ground truth data, ground truth data (e.g., from AI-assisted annotation, manual labeling, etc.) may be used by during model training **3114** to generate refined model **3312**. In at least one embodiment, customer dataset **3306** may be applied to initial model **3304** any number of times, and ground truth data may be used to update parameters of initial model **3304** until an acceptable level of accuracy is attained for refined model **3312**. In at least one embodiment, once refined model **3312** is generated, refined model **3312** may be deployed within one or more deployment pipelines **3210** at a facility for performing one or more processing tasks with respect to medical imaging data.

In at least one embodiment, refined model **3312** may be uploaded to pre-trained models **3206** in model registry **3124** to be selected by another facility. In at least one embodiment, his process may be completed at any number of facilities such that refined model **3312** may be further refined on new datasets any number of times to generate a more universal model.

FIG. **33B** is an example illustration of a client-server architecture **3332** to enhance annotation tools with pre-trained annotation models, in accordance with at least one embodiment. In at least one embodiment, AI-assisted annotation tools **3336** may be instantiated based on a client-server architecture **3332**. In at least one embodiment, annotation tools **3336** in imaging applications may aid radiologists, for example, identify organs and abnormalities. In at least one embodiment, imaging applications may include software tools that help user **3310** to identify, as a non-limiting example, a few extreme points on a particular organ of interest in raw images **3334** (e.g., in a 3D MRI or CT scan) and receive auto-annotated results for all 2D slices of a particular organ. In at least one embodiment, results may be stored in a data store as training data **3338** and used as (for example and without limitation) ground truth data for training. In at least one embodiment, when computing device **3308** sends extreme points for AI-assisted annotation **3110**, a deep learning model, for example, may receive this data as input and return inference results of a segmented organ or abnormality. In at least one embodiment, pre-instantiated annotation tools, such as AI-Assisted Annotation Tool **3336B** in FIG. **33B**, may be enhanced by making API calls (e.g., API Call **3344**) to a server, such as an Annotation Assistant Server **3340** that may include a set of pre-trained models **3342** stored in an annotation model registry, for example. In at least one embodiment, an annotation model registry may store pre-trained models **3342** (e.g., machine learning models, such as deep learning models) that are pre-trained to perform AI-assisted annotation on a particular organ or abnormality. These models may be further updated by using training pipelines **3204**. In at least one embodiment, pre-installed annotation tools may be improved over time as new labeled clinic data **3112** is added.

Inference and/or training logic **615** are used to perform inferencing and/or training operations associated with one or

more embodiments. In at least one embodiment, this logic can be used with components of these figures to generate one or more images with visual effects approximated from similar effects applied to one or more lower resolution versions of those images.

Other variations are within spirit of present disclosure. Thus, while disclosed techniques are susceptible to various modifications and alternative constructions, certain illustrated embodiments thereof are shown in drawings and have been described above in detail. It should be understood, however, that there is no intention to limit disclosure to specific form or forms disclosed, but on contrary, intention is to cover all modifications, alternative constructions, and equivalents falling within spirit and scope of disclosure, as defined in appended claims.

Use of terms “a” and “an” and “the” and similar referents in context of describing disclosed embodiments (especially in context of following claims) are to be construed to cover both singular and plural, unless otherwise indicated herein or clearly contradicted by context, and not as a definition of a term. Terms “comprising,” “having,” “including,” and “containing” are to be construed as open-ended terms (meaning “including, but not limited to,”) unless otherwise noted. Term “connected,” when unmodified and referring to physical connections, is to be construed as partly or wholly contained within, attached to, or joined together, even if there is something intervening. Recitation of ranges of values herein are merely intended to serve as a shorthand method of referring individually to each separate value falling within range, unless otherwise indicated herein and each separate value is incorporated into specification as if it were individually recited herein. Use of term “set” (e.g., “a set of items”) or “subset,” unless otherwise noted or contradicted by context, is to be construed as a nonempty collection comprising one or more members. Further, unless otherwise noted or contradicted by context, term “subset” of a corresponding set does not necessarily denote a proper subset of corresponding set, but subset and corresponding set may be equal.

Conjunctive language, such as phrases of form “at least one of A, B, and C,” or “at least one of A, B and C,” unless specifically stated otherwise or otherwise clearly contradicted by context, is otherwise understood with context as used in general to present that an item, term, etc., may be either A or B or C, or any nonempty subset of set of A and B and C. For instance, in illustrative example of a set having three members, conjunctive phrases “at least one of A, B, and C” and “at least one of A, B and C” refer to any of following sets: {A}, {B}, {C}, {A, B}, {A, C}, {B, C}, {A, B, C}. Thus, such conjunctive language is not generally intended to imply that certain embodiments require at least one of A, at least one of B, and at least one of C each to be present. In addition, unless otherwise noted or contradicted by context, term “plurality” indicates a state of being plural (e.g., “a plurality of items” indicates multiple items). A plurality is at least two items, but can be more when so indicated either explicitly or by context. Further, unless stated otherwise or otherwise clear from context, phrase “based on” means “based at least in part on” and not “based solely on.”

Operations of processes described herein can be performed in any suitable order unless otherwise indicated herein or otherwise clearly contradicted by context. In at least one embodiment, a process such as those processes described herein (or variations and/or combinations thereof) is performed under control of one or more computer systems configured with executable instructions and is implemented

as code (e.g., executable instructions, one or more computer programs or one or more applications) executing collectively on one or more processors, by hardware or combinations thereof. In at least one embodiment, code is stored on a computer-readable storage medium, for example, in form of a computer program comprising a plurality of instructions executable by one or more processors. In at least one embodiment, a computer-readable storage medium is a non-transitory computer-readable storage medium that excludes transitory signals (e.g., a propagating transient electric or electromagnetic transmission) but includes non-transitory data storage circuitry (e.g., buffers, cache, and queues) within transceivers of transitory signals. In at least one embodiment, code (e.g., executable code or source code) is stored on a set of one or more non-transitory computer-readable storage media having stored thereon executable instructions (or other memory to store executable instructions) that, when executed (i.e., as a result of being executed) by one or more processors of a computer system, cause computer system to perform operations described herein. A set of non-transitory computer-readable storage media, in at least one embodiment, comprises multiple non-transitory computer-readable storage media and one or more of individual non-transitory storage media of multiple non-transitory computer-readable storage media lack all of code while multiple non-transitory computer-readable storage media collectively store all of code. In at least one embodiment, executable instructions are executed such that different instructions are executed by different processors. For example, a non-transitory computer-readable storage medium stores instructions and a main central processing unit (“CPU”) executes some of instructions while a graphics processing unit (“GPU”) executes other instructions. In at least one embodiment, different components of a computer system have separate processors and different processors execute different subsets of instructions.

Accordingly, in at least one embodiment, computer systems are configured to implement one or more services that singly or collectively perform operations of processes described herein and such computer systems are configured with applicable hardware and/or software that enable performance of operations. Further, a computer system that implements at least one embodiment of present disclosure is a single device and, in another embodiment, is a distributed computer system comprising multiple devices that operate differently such that distributed computer system performs operations described herein and such that a single device does not perform all operations.

Use of any and all examples, or exemplary language (e.g., “such as”) provided herein, is intended merely to better illuminate embodiments of disclosure and does not pose a limitation on scope of disclosure unless otherwise claimed. No language in specification should be construed as indicating any non-claimed element as essential to practice of disclosure.

All references, including publications, patent applications, and patents, cited herein are hereby incorporated by reference to same extent as if each reference were individually and specifically indicated to be incorporated by reference and were set forth in its entirety herein.

In description and claims, terms “coupled” and “connected,” along with their derivatives, may be used. It should be understood that these terms may be not intended as synonyms for each other. Rather, in particular examples, “connected” or “coupled” may be used to indicate that two or more elements are in direct or indirect physical or electrical contact with each other. “Coupled” may also mean

that two or more elements are not in direct contact with each other, but yet still co-operate or interact with each other.

Unless specifically stated otherwise, it may be appreciated that throughout specification terms such as “processing,” “computing,” “calculating,” “determining,” or like, refer to action and/or processes of a computer or computing system, or similar electronic computing device, that manipulate and/or transform data represented as physical, such as electronic, quantities within computing system’s registers and/or memories into other data similarly represented as physical quantities within computing system’s memories, registers or other such information storage, transmission or display devices.

In a similar manner, term “processor” may refer to any device or portion of a device that processes electronic data from registers and/or memory and transform that electronic data into other electronic data that may be stored in registers and/or memory. As non-limiting examples, “processor” may be a CPU or a GPU. A “computing platform” may comprise one or more processors. As used herein, “software” processes may include, for example, software and/or hardware entities that perform work over time, such as tasks, threads, and intelligent agents. Also, each process may refer to multiple processes, for carrying out instructions in sequence or in parallel, continuously or intermittently. Terms “system” and “method” are used herein interchangeably insofar as system may embody one or more methods and methods may be considered a system.

In present document, references may be made to obtaining, acquiring, receiving, or inputting analog or digital data into a subsystem, computer system, or computer-implemented machine. Obtaining, acquiring, receiving, or inputting analog and digital data can be accomplished in a variety of ways such as by receiving data as a parameter of a function call or a call to an application programming interface. In some implementations, process of obtaining, acquiring, receiving, or inputting analog or digital data can be accomplished by transferring data via a serial or parallel interface. In another implementation, process of obtaining, acquiring, receiving, or inputting analog or digital data can be accomplished by transferring data via a computer network from providing entity to acquiring entity. References may also be made to providing, outputting, transmitting, sending, or presenting analog or digital data. In various examples, process of providing, outputting, transmitting, sending, or presenting analog or digital data can be accomplished by transferring data as an input or output parameter of a function call, a parameter of an application programming interface or interprocess communication mechanism.

Although discussion above sets forth example implementations of described techniques, other architectures may be used to implement described functionality, and are intended to be within scope of this disclosure. Furthermore, although specific distributions of responsibilities are defined above for purposes of discussion, various functions and responsibilities might be distributed and divided in different ways, depending on circumstances.

Furthermore, although subject matter has been described in language specific to structural features and/or methodological acts, it is to be understood that subject matter claimed in appended claims is not necessarily limited to specific features or acts described. Rather, specific features and acts are disclosed as exemplary forms of implementing the claims.

What is claimed is:

1. A method of image processing comprising:
obtaining initial training images;

generating blurred training images comprising intentionally blurring one or more of the initial training images;
generating low resolution training images or high resolution training images or both and comprising using the blurred training images;

training a neural network to generate blurred high resolution pixel values comprising using the low resolution training images or high resolution training images or both; and

inputting lower resolution images to the neural network while separately generating weights or biases or both of the neural network based on statistics of the low resolution images.

2. The method of claim 1, further comprising:
blurring the initial training images comprising changing image data of the initial training images.

3. The method of claim 1, further comprising:
blurring the initial training images comprising applying a function-related value to image data of the initial training images.

4. The method of claim 1, further comprising:
training the neural network to convert a resolution on input blurred lower resolution images.

5. The method of claim 1, further comprising:
processing a blurred image to generate a desired size of a lower resolution training image or higher resolution training image or both.

6. At least one non-transitory article having at least one computer readable medium comprising a plurality of instructions that in response to being executed on a computing device, cause the computing device to operate by:

obtaining initial training images;
generating blurred training images comprising intentionally blurring one or more of the initial training images;
generating lower resolution training images or high resolution training images or both and comprising using the blurred training images;

training a neural network to generate blurred higher resolution pixel values comprising
using the lower resolution training images or higher resolution training images or both; and

inputting lower resolution images to the neural network while separately generating weights or biases or both of the neural network based on statistics of the low resolution images.

7. The at least one non-transitory article of claim 6, wherein the instructions, in response to being executed, further cause the computing device to operate by generating a factor to compensate for reducing a size of the initial training images.

8. The at least one non-transitory article of claim 7, wherein the factor is used to convolve the initial training image to form the blurred training images.

9. The at least one non-transitory article of claim 6, wherein generating the blurred training images comprises intentionally blurring less than all of one or more of the initial training images.

10. A method, comprising:

generating one or more blurred images by, at least,
blurring pixel values of one or more training images;
generating at least lower resolution training images or high resolution training images or both and comprising using the blurred images;

training a neural network to generate blurred images at a higher resolution using at least the lower resolution training images or higher resolution training images; and

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inputting lower resolution images to the neural network while separately generating weights or biases or both of the neural network based on statistics of the low resolution images.

11. The method of claim 10, further comprising: blurring the training images at least in part by changing image data of the training images. 5

12. The method of claim 10, further comprising: blurring the training images at least in part by applying a function-related value to image data of the training images. 10

13. The method of claim 10, further comprising: training the neural network to convert a resolution on input blurred lower resolution images.

14. The method of claim 10, further comprising: processing a blurred image to generate a desired size of at least one of a lower resolution training image or higher resolution training image. 15

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15. The method of claim 10, wherein a first parameterized function is determined for the one or more images having a resolution less than a first resolution, and wherein a second parameterized function is determined based on the first parameterized function and used to approximate one or more visual effects for the one or more training images having a resolution greater than or equal to the first resolution.

16. The method of claim 15, further comprising: applying one or more enhancements to the one or more training images, having a resolution greater than or equal to the first resolution, after one or more visual effects are approximated.

17. The method of claim 10, further comprising: downscaling the one or more training images, having less than a first resolution, to a resolution of an initial aliased training image generated by a rendering engine.

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